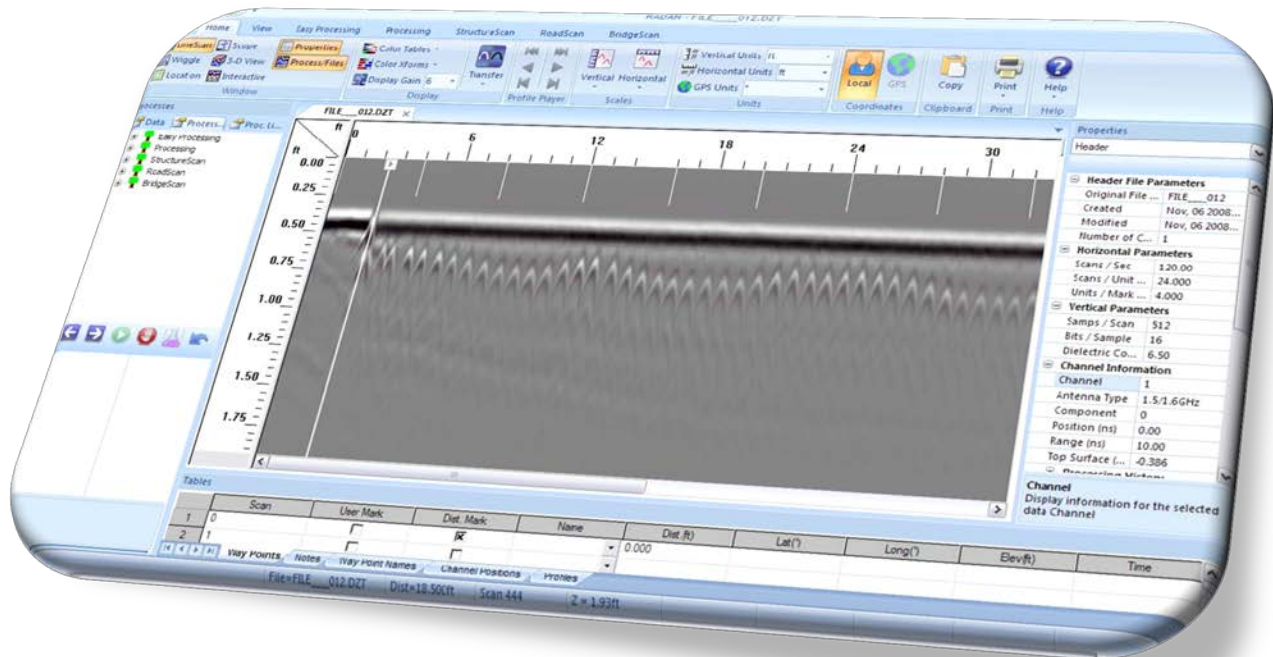




RADAN® 7

Geophysical Survey Systems, Inc.
www.geophysical.com

February, 2012



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Should you have any questions concerning this Agreement, you may contact GSSI by writing Geophysical Survey Systems, 12 Industrial Way, Salem, New Hampshire 03079 U.S.A.

You acknowledge that you have read this agreement, understand it and agree to be bound by its terms and conditions. You further agree that it is the complete and exclusive statement of the Agreement between you and GSSI which supersedes any proposal or prior Agreement, oral or written, and any other communications between us relating to the subject matter of this Agreement.

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How to Use This Manual: Must Read

This manual is designed for both experienced and novice users of RADAN 7. This manual is broken up into five sections to help you find answers you are looking for.

Section 1 is General Description of the software, general requirements, getting started, installing and activating your software

Section 2 is Basic Use of RADAN, launching the software, updating the software, and navigating through the screens

Section 3 is a description of every screen and menu option in RADAN.

Section 4 is basic processing steps/tutorials for your data

Section 5 is basic processing for specific applications.

Note: Some processes may be duplicated in different sections. This is designed that way to assist you to quickly find the help you might need by the question(s) you may be asking.

For example, you might have the software open and wonder what a menu option or icon might do, then Section 3 would best answer your question.

You might have a data file open and wonder what you can do, or what to do next, then Section 4 or Section 5 might answer your questions.

System Requirements and Notes

Recommended System Requirements for RADAN 7

- Microsoft Windows® 7 (32 or 64 bit)
- Intel Core i5 (or better) processor
- 3+ GB system memory
- 500+ GB hard drive with a minimum of 100 GB available space
- 256+ MB dedicated graphics chipset with OpenGL drivers (Note: We only support NVidia and Intel graphics chipsets)

Minimum System Requirements for RADAN 7

- Microsoft Windows® XP
- 1.0+ GHz Pentium 4 (Note: We do not support single core single thread processors)
- 2 GB system memory
- 160 GB hard drive with a minimum of 20 GB available space
- 128 MB graphics chipset with OpenGL drivers (Note: We only support NVidia and Intel graphics chipsets.)

What Data Can Be Processed with RADAN 7:

- RADAN 7 is necessary for viewing of SIR-30 data. It adds capabilities to view and process the new raw data format of SIR-30 and Structure Scan Mini data.
- SIR-3000 (2D dzt, 3D b3d, GPS tmf, plt and gga text from the SDR data logger)
- SIR-20(2D dzt, 3D b3d, GPS tmf and gga text from the SDR data logger)
- StructureScan Optic data
- Individual profiles of older systems such as SIR-10 and SIR-2000
- Files processed by RADAN 4.x RADAN 5.x and RADAN 6.x
- 3D files which have the Microsoft Access (mdb) based database

RADAN 7 is Not Currently Recommended for Users of:

- StructureScan users working with the black pad
- Terravision:
 - Y Gain Equalization is not currently implemented in RADAN 7
 - RADAN will not properly import GPS for Terravision but will read Terravision files that have been opened in RADAN 6
- SIR-20 control units (This should not be confused with SIR-20 post processing systems. RADAN 7 is designed to work with SIR-20 data)
- It does not contain a replacement for controlling SIR-20 systems) All hardware control features of RADAN 6.6 and earlier have been removed from RADAN 7
- CF-29 and earlier toughbooks do not meet the minimum specifications for RADAN 7

Section 1: Getting Started

Thank you for purchasing RADAN® 7. The packing list included with your shipment lists all of the items in your order. The RADAN 7 program and example files are stored on a single CD-ROM disk.

GSSI Activation Policies

Your RADAN 7 license includes one license for installation on one computer.

General Description

RADAN 7 software was designed to process, view, and document data collected with products from GSSI. RADAN 7 module can perform the following functions:

- Display multiple screens of radar data as line scan, wiggle trace, and/or oscilloscope.
- Manipulate color table and color transform parameters to enhance data display.
- Edit file headers and distance markers.
- Process individual files in Process Lists.
- Process multiple files using Project Processing.
- Modify or restore data gains.
- Correct position (shift data scans along the time axis).
- Provide horizontal scaling and distance normalization.
- Incorporate topographic changes with top surface normalization.
- Display the frequency spectrum of data.
- Apply Infinite Impulse Response (IIR) and Finite Impulse Response (FIR) filter
- Spatial Filtering (2 Dimensional/F-K filter).
- Perform migration.
- Perform Predictive Deconvolution.
- Perform Envelope processing functions (Hilbert Transform).
- Velocity analysis.
- Local Peak Interpretation.
- Interactive Interpretation.
- Print to all Windows supported printers.

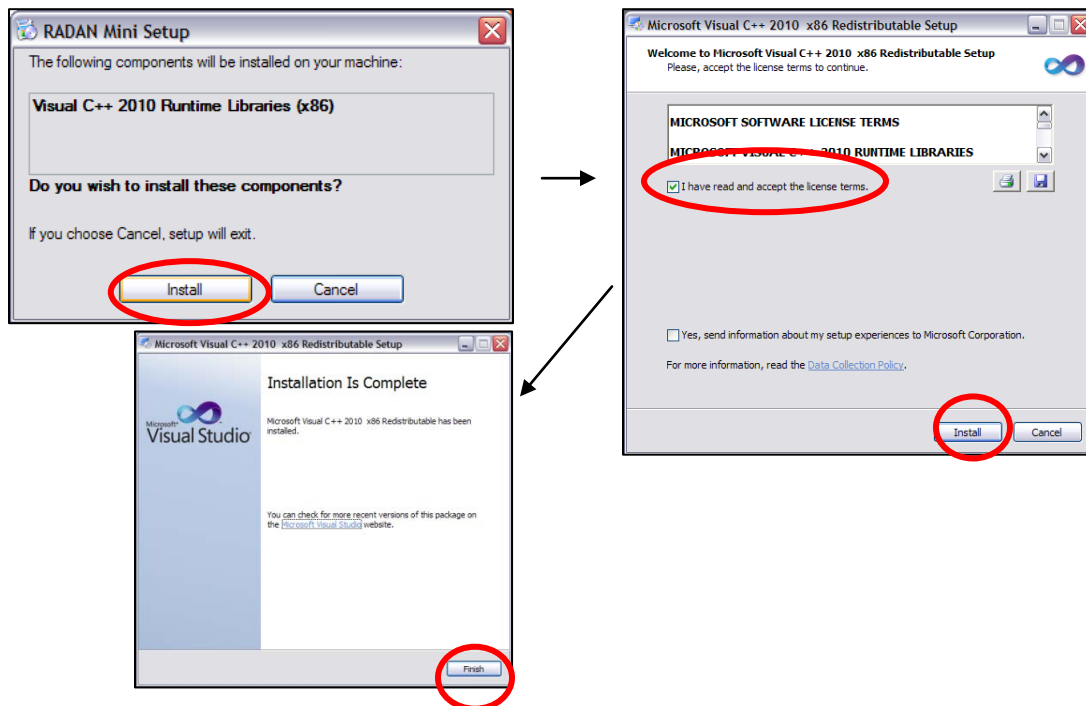
Installing RADAN 7

It is highly recommended that you are connected to the internet when installing the software so that activation can take place. If you are not, you will need to contact GSSI for other installation instructions.

- 1** Insert the RADAN 7 Installation CD into your computer's CD drive.
- 2** The installation program should start automatically. Follow instructions on screen. If the program did start automatically, skip to step 5.
- 3** Using Windows Explorer, double-click the CD drive that contains the RADAN 7 Installation disk. It is probably Drive D: or E:.
- 4** Find the Start Icon and double-click it. 
- 5** Follow on-screen instructions:
 - **Install RADAN 7:** Select this option to install the software onto your computer.
 - **Technical Support Web Site Info:** This will open a PDF file and has excellent information about our support web site and how to use this site. **It is highly recommended that you open, save and print this file.**
 - **GSSI Contacts:** This will open a PDF file on essential contacts for you here at GSSI. **It is highly recommended that you open, save and print this file.**

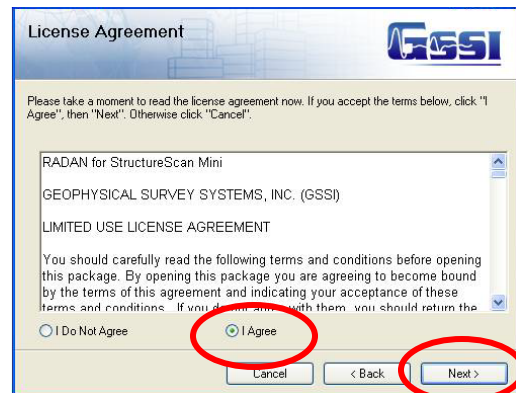
Installation Instructions

Click Install > Accept > Next > OK or Finish through all the screens. After you have selected “Install RADAN 7” from the Main menu, these will be the screens you will see:

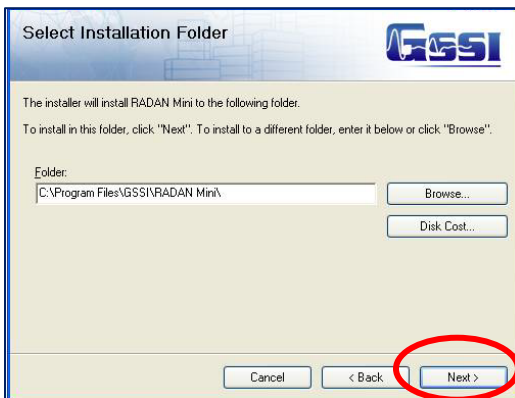




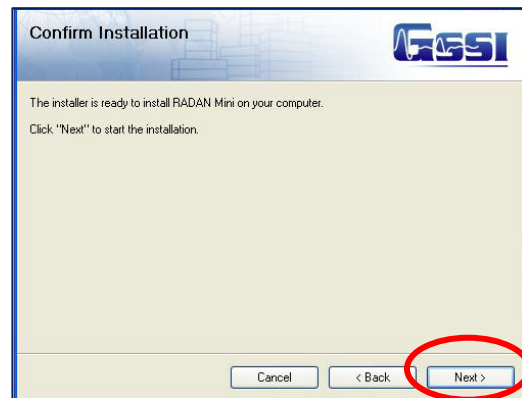
1



2



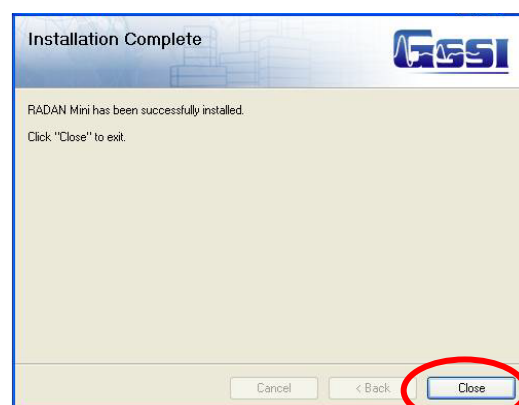
3



4



5



6

Important Note: Depending upon the age of your computer, your operating system (Windows XP, Windows Vista, or Windows 7), and the age of your operating system, you may get other screens asking you to install other software. You should accept these, or if you have questions, contact your IT person or GSSI.

Activating/Validating the RADAN 7 Software

Activating the Software

After installing the software, RADAN 7 will automatically start.

The first time you run the software, you will be asked to activate it. You **MUST** be connected to the internet to activate the software.

- 1** From the label located on the inside cover of the CD case, input the Product Key and the Serial Number into the appropriate fields.
- 2** The Computer Name is pre-filled by GSSI (this is the name of your computer from your computer system).
- 3** Enter a valid email address.
- 4** Click Activate/Validate your Product Key.



Product Key	XXXXXXXXXX
Serial Number	
Computer Name	TRAINING-5
Email Address	
Status	This software can be used 29 more times before activating.
Next License Check	Demo will expire on or before 8/8/2010
Activate/Validate your Product Key	
Continue to RADAN Mini	
Manage your license(s) at support.geophysical.com	

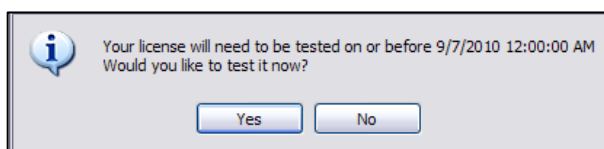
Note: If you choose not to activate your software at this time, the above screen will appear at the launching of the software until you activate the software.

Note: After activating the software, you do NOT have to be connected to the internet to run the software.

Validating the Software

At times, upon launching RADAN 7, you will be asked to test the license. Click “Yes” and a pre-populated screen will appear. Simply click “Continue with RADAN Mini.”

Note: You will need to be connected to the internet to complete this process.

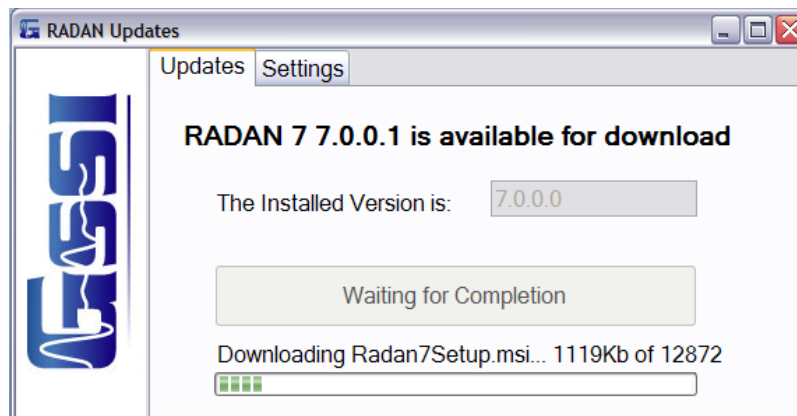
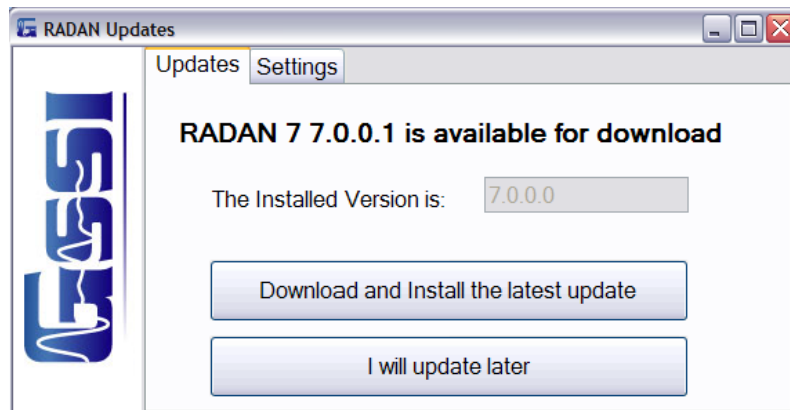


Product Key	098722-561928-393656
Serial Number	12345
Computer Name	SALES-28
Email Address	sommerfeldtr@geophysical.com
Status	Registered and Activated
Next License Check	9/7/2010 12:00:00 AM
Activate/Validate your Product Key	
Continue to RADAN Mini	
Manage your license(s) at support.geophysical.com	

Updating the Software

GSSI can automatically update the software if there is one available. Upon launching RADAN 7 AND if you are connected to the internet, you will be asked if you would like to load the update.

- Click “Download and Install the latest update” if you wish to update your software at this time.
- Click “I will update later” to update the software at a later date.



Section 2: Using RADAN 7

Setting Up Your Laptop/PC and Data Transfer

Create folders on your PC/Laptop. As a suggestion, you should create a different source folder for each of your projects. Then create a “processed” or “output” folder within each of the source folders. The source folder will contain raw, unprocessed data that was collected and saved. The processed or output folder will contain files that were processed from the raw, unprocessed data. After you have created the folders, copy the raw files from your device to the appropriate source folder.

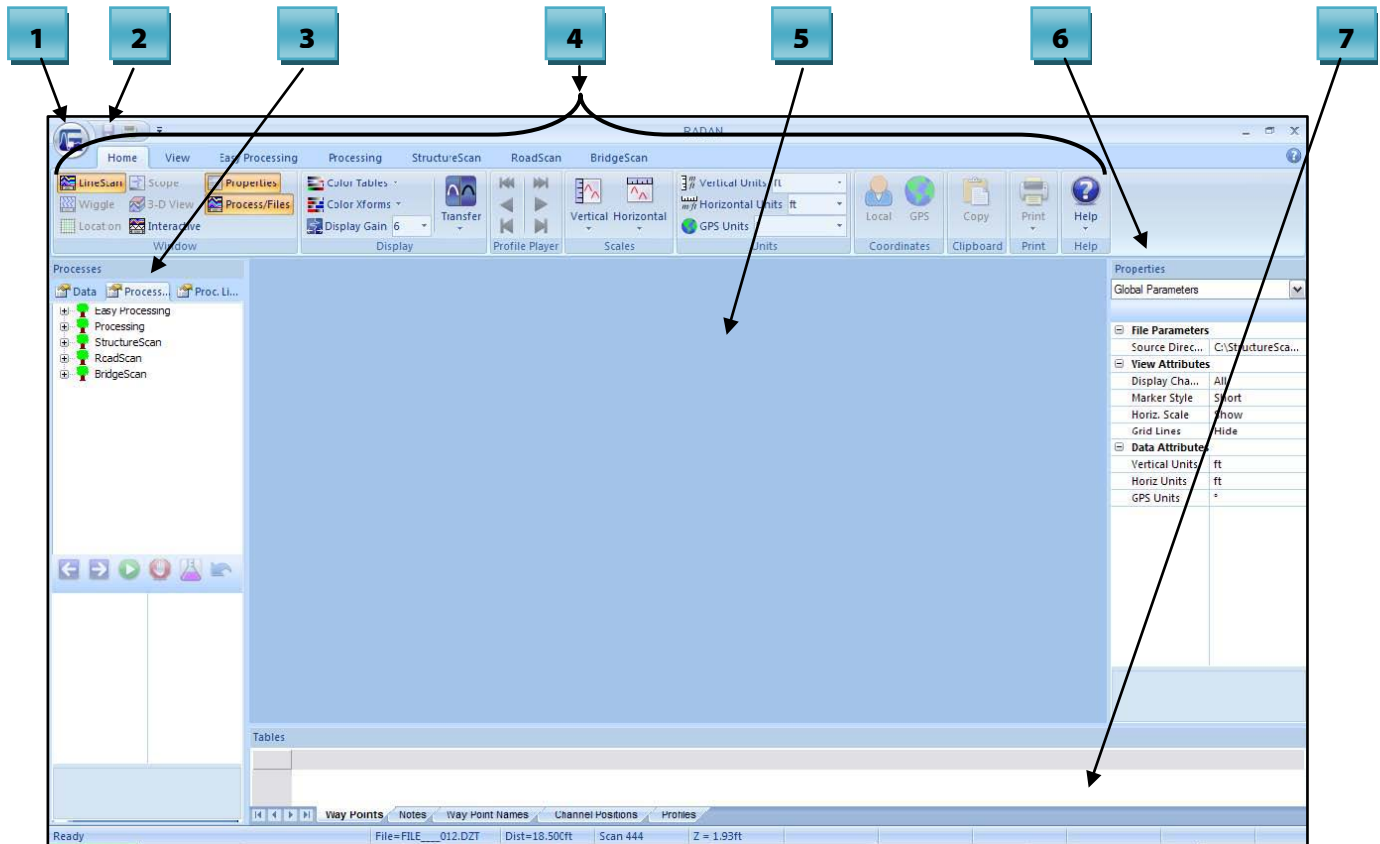
Of course, you may organize your folders using your own set of rules. However, you must know the folder names in order to configure RADAN 7 at the start of each project.

Launching the Software and The Main Screen

Launching the Software

Launch RADAN 7 by double-clicking the RADAN 7 icon

The following screen is displayed:



- 1** GSSI Button
- 2** Quick Access Toolbar
- 3** Data/Processes/Proc. List Pane
- 4** Ribbons
- 5** Data Pane
- 6** Properties Pane
- 7** Tables Pane

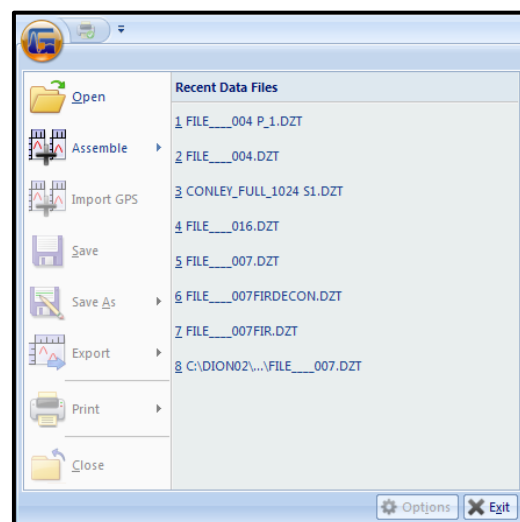
The Main Screen

Note: Many of the options below will be detailed throughout the View and Processing the Data Section of the manual.

1. GSSI Button

Clicking on the GSSI Button allows you to:

- Open a File/Project
- Assemble Files
- Import GPS
- Save a File/Project
- Save As a File/Project under a different name or format
- Export your data
- Print your data
- Close a File/Project
- Open a previously processed File/Project
- Exit RADAN 7 Software



2. Quick Access Toolbar

If you right-click any icon located in the GSSI button or in any Ribbon, you may add that icon to the Quick Access Toolbar. This is for options you use often.

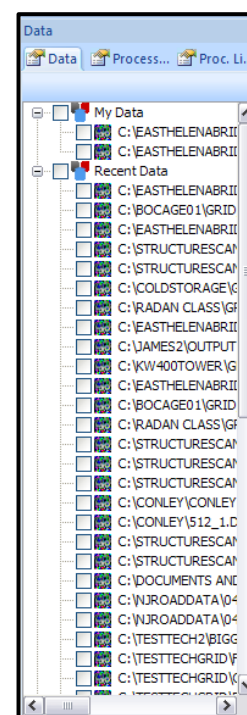
3. Data, Processes & Proc. List Pane

There are three tabs in this section:

Data Tab: This tab will give you quick access to your data, recently processed data and GSSI example data

Processes Tab: Depending upon your application, this tab will give you guided steps to process your data.

Proc. List Tab: The options located in this area will allow you to quickly apply to the data the more commonly used processes. A macro is a series of steps and/or options put together as one option. GSSI has created these commonly used processes as macros. This is detailed more in the Viewing and Processing the Data section



4. Ribbon

This area contains all the options and processes you can do with your data. Each tab in the Ribbon also is broken down by application. This is detailed more in the Viewing and Processing the Data section

5. Data Pane

This is where your data is displayed when you open a file. This screen can contain multiple files, each in its own tab.

6. Properties Pane

This area will display Global Parameters for all your files in a particular project, Header and Display Parameters information about a specific file (when opened), and 3D Display Parameters when a 3D file is opened.

7. Table Pane

Depending upon the type of data that is being processed, this pane will display and allow input different Data Base information about a data file. More information about this is located in Section 4: Basic Processing.

Configuring the Software

Before opening a file, you will need to set up some Global Parameters in the Properties pane, which is located on the right side of the screen.

Note: You may double-click on the left or right column to change/select options or click on either column and click on dots or down arrow to change/select options.

Source Directory: Select the folder where the source (original data is located).

Display Channel: Select the number of channels you would like to display/process.

Marker Style: Select None, Long, or Short for the type of User/Distance marks to be displayed.

Horiz. Scale:

- **Show:** Display horizontal scale.
- **Hide:** Do not display horizontal scale.

Grid Lines:

- **Show:** Display grid lines.
- **Hide:** Do not display grid lines.

Vertical Units: Units for the vertical scale.

Horiz Units: Units for the horizontal scale.

Note: When a file is opened, the scale will default to whatever units were saved with the file.

GPS Units: Units for GPS horizontal scale.

Note: Once you open a file, you cannot change any of the Global Parameters. You must close all files to make changes.

The screenshot shows the 'Properties' window in RADAN 7. At the top is a 'Global Parameters' dropdown menu. Below it are three expandable sections: 'File Parameters', 'View Attributes', and 'Data Attributes'. The 'File Parameters' section shows 'Source Direc...' with the value 'C:\StructureSca...'. The 'View Attributes' section shows 'Display Cha...' set to 'All', 'Marker Style' set to 'Short', 'Horiz. Scale' set to 'Show', and 'Grid Lines' set to 'Hide'. The 'Data Attributes' section shows 'Vertical Units' set to 'ft', 'Horiz Units' set to 'ft', and 'GPS Units' set to '°'. The bottom of the window is a large empty light blue area.

Properties	
Global Parameters ▼	
[-] File Parameters	
Source Direc...	C:\StructureSca...
[-] View Attributes	
Display Cha...	All
Marker Style	Short
Horiz. Scale	Show
Grid Lines	Hide
[-] Data Attributes	
Vertical Units	ft
Horiz Units	ft
GPS Units	°

Section 3: Navigating Through RADAN – The Menus, Ribbons, and Panes

Menus

GSSI Button



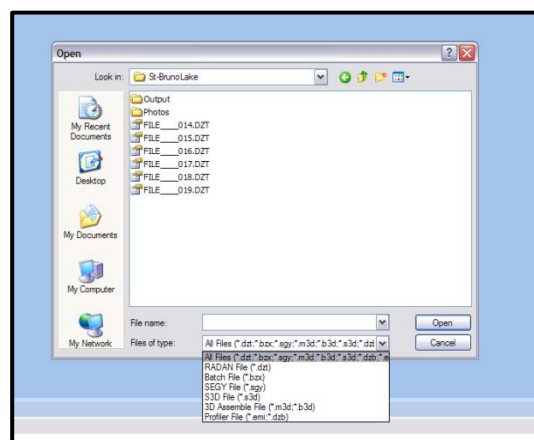
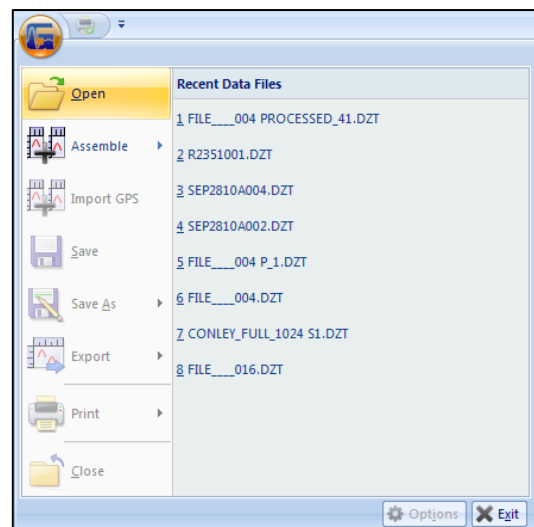
Open

Click the GSSI Button. You may open any previously opened file by selecting a file in the “Recent Data Files” section, or click Open to select a file from your Source or Output folder.

If Open is clicked, click on the down arrow in the Files of Type section to open one of the following file formats:

- **.dzt**:– RADAN file, a single profile file (default)
- **.bzx**: a formally created Batch file, a group of .dzt files to be viewed and processed together.
- **.sgy**: Seismic Files
- **.s3d**: a formally created Super 3D file, which is many grid files put together.
- **.m3d, .b3d**: a formally created 3D file, whether it was created by the system or manually
- **.emi, .dzb**: Files created using the Profiler

Once a file type is selected, click on the file to open and click Open, or double-click on the file to open.



Assemble

Append Files

This option will allow you to append files to create one file.

- 1** After Append Files is selected, browse (if necessary) to the folder where the files you wish to append are located. Enter a name of the file for the newly created file. Select No Merge to keep each file separate, and Yes Merge to combine the files into one file.
- 2** Click Next to continue or Cancel to cancel the process.
- 3** There will be two windows: Available Files in Folder are files available to append; and Batch Files are the files that will be appended.

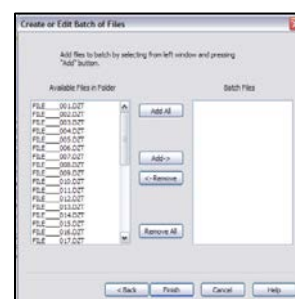
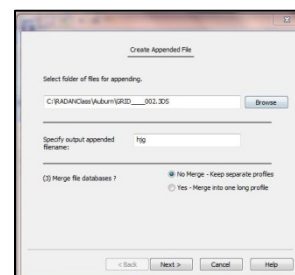
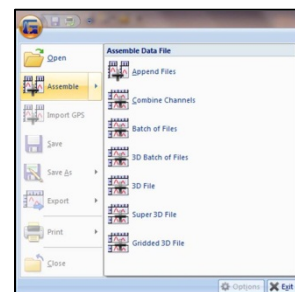
Adding Files To Append Together

- a) Add All:** Click here to add all files from the left window to the right window.
- b)** Click on a file and click Add to add a file to the right window.
- c)** You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
- d)** If the files you want to add files that are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.

Removing Files

- a) Remove All:** Click here to remove all files from the right window to the left window.
- b)** Click on a file and click Remove to remove a file from the right window.
- c)** You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
- d)** If the files you want to remove files that are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.

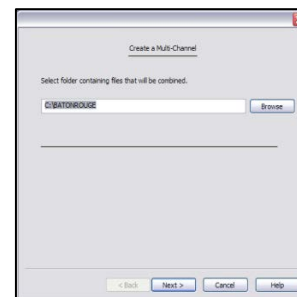
- 4** Click Back to return to the previous screen.
- 5** Click Finish to complete the process.
- 6** Click Cancel to cancel the process.



Combine Channels

If data was collected using multi-channels, you may combine these files for viewing.

- 1** After Combine Channels is selected, browse (if necessary) to the folder where the files you wish to combine for viewing are located.
- 2** Click Next to continue or Cancel to cancel the process.
- 3** There will be two windows: Available Files in Folder are files available to combine channel; and Batch Files are the files that will be added to the combine channel.



Adding Files To Combine Channels

- a) Add All:** Click here to add all files from the left pane to the right pane.
- b)** Click on a file and click Add to add a file to the right pane.
- c)** You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
- d)** If the files you want to add files are that grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.



Removing Files

- a) Remove All:** Click here to remove all files from the right pane to the left pane.
 - b)** Click on a file and click Remove to remove a file from the right pane.
 - c)** You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
 - d)** If the files you want to remove files that are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.
- 4** Click Back to return to the previous screen.
 - 5** Click Finish to complete the process.
 - 6** Click Cancel to cancel the process.

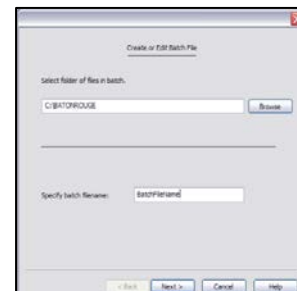
Batch of Files

If you need to run the same process for multiple files, you may create a batch file that contains these files. This allows you to run the process once and the process will run for every file in the batch file.

Note: The batch file created will be a file (.bzx) and contain only the names of the files to batch together. It will NOT be a data file (.dzt) of the files combined together.

- 1** After Batch of Files is selected, browse (if necessary) to the folder where the files you wish to batch are located.
- 2** Enter a name of the batch file for the newly created batch file.
- 3** Click Next to continue or Cancel to cancel the process.

There will be two panes: Available Files in Folder are files available to append; and Batch Files are the files that will be batched together.



Adding Files to Batch Together

- a) Add All:** Click here to add all files from the left pane to the right pane.
- b)** Click on a file and click Add to add a file to the right pane.
- c)** You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
- d)** If the files you want to add files that are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.



Removing Files

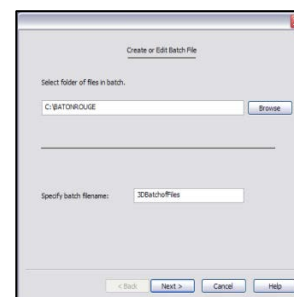
- a) Remove All:** Click here to remove all files from the right pane to the left pane.
 - b)** Click on a file and click Remove to remove a file from the right pane.
 - c)** You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
 - d)** If the files you want to remove files that are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.
- 4** Click Back to return to the previous screen.
 - 5** Click Finish to complete the process.
 - 6** Click Cancel to cancel the process.

3D Batch of Files

This will allow you to combine individual data files (.dzt) that were collected in a grid format and batch them together to create a 3D Grid Batch file. Files collected can be collected in the X direction only, Y direction only, or both X and Y direction.

Note: The batch file created will be a file (.bzx) and contain only the names of the files to batch together. It will NOT be a data file (.dzt) of the files combined together.

- 1 After 3D Batch of Files is selected, browse (if necessary) to the folder where the files you wish to batch are located
- 2 Enter a name of the batch file for the newly created batch file
- 3 Click Next to continue or Cancel to cancel the process.



There will be two panes: Available Files in Folder are files available to append; and Batch Files are the files that will be batched together. Add ALL files that should be included with the 3D grid.

Adding Files to Batch Together

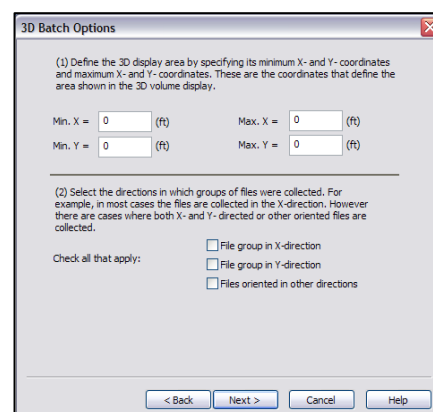
- a) **Add All:** Click here to add all files from the left pane to the right pane.
- b) Click on a file and click Add to add a file to the right pane.
- c) You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
- d) If the files you want to add are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.



Removing Files

- a) **Remove All:** Click here to remove all files from the right pane to the left pane.
 - b) Click on a file and click Remove to remove a file from the right pane.
 - c) You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
 - d) If the files you want to remove are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.
- 4 Click Back to return to the previous screen.
 - 5 Click Finish to complete the process.
 - 6 Click Cancel to cancel the process.

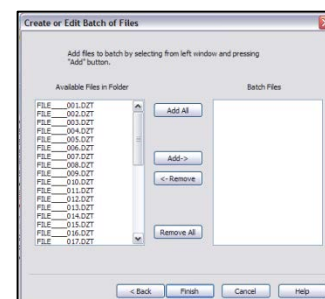
- 7** Enter the Starting and Ending points of your grid.
- 8** Click all that apply:
 - If files were collected in the X direction.
 - If files were collected in the Y direction.
 - If files were collected neither the X nor the Y direction.
- 9** Click Back, Next, or Cancel.
 - If you checked that files were collected in the X direction, you will be asked to add those files as well as the orientation.
 - If you checked that files were collected in the Y direction, you will be asked to add those files as well as the orientation.



There will be two panes: Available Files in Folder are files available to append; and Batch Files are the files that will be batched together. Add ALL files that should be included with the 3D grid.

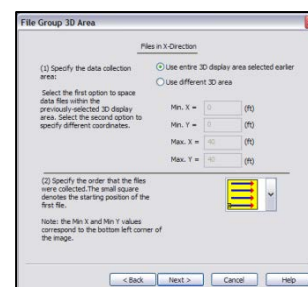
Adding Files

- a) Add All:** Click here to add all files from the left pane to the right pane.
- b)** Click on a file and click Add to add a file to the right pane.
- c)** You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
- d)** If the files you want to add are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.



Removing Files

- e) Remove All:** Click here to remove all files from the right pane to the left pane.
- f)** Click on a file and click Remove to remove a file from the right pane.
- g)** You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
- h)** If the files you want to remove are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.
- 10** Click Back, Next, or Cancel.
- 11** Click on the Down arrow next to the yellow orientation section. Select what direction and orientation was collected.
- 12** Click Back, Next, or Cancel.



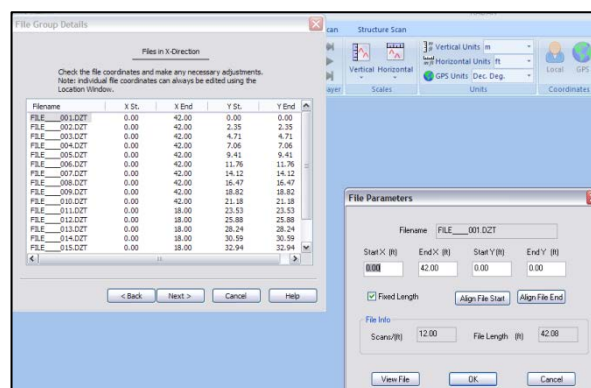
13 You may double-click on each individual file and input the starting and ending XY coordinates of that profile.

- Enter starting and ending XY coordinates.
- View File to view the individual file
- Click OK.

14 Click Back, Next or Cancel,

15 Lastly, you may input GPS coordinates for this grid.

16 Click Back, Finish, or Cancel.

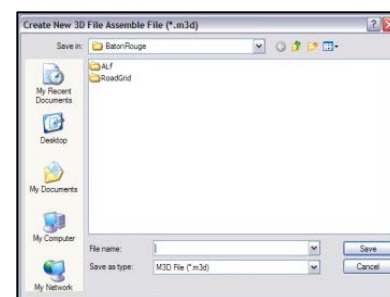


3D File

This will allow you to combine individual data files (.dzt) that were collected in a grid format and combine them together to create a 3D Grid file (.m3d). Files collected can be collected in the X direction only, Y direction only, or both X and Y direction.

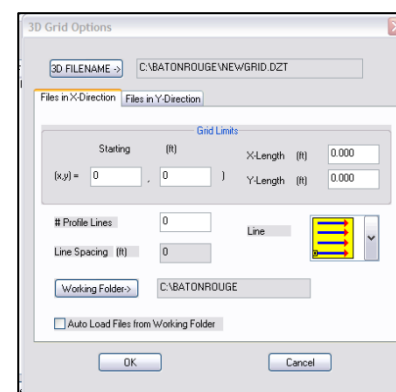
1 After selecting 3D File, enter a filename for the Grid you want to create.

2 Click Save to save and continue.



3D Filename: This is the name of the single .DZT file that RADAN will construct from the individual profiles, and the location where it will be stored. Clicking on this button will allow you to change either the name or the storage directory.

Files in X/Y Direction: Both of these tabs MUST be filled even if you collected the grid in one direction. If you collected the same grid twice, but with perpendicular transects, these tabs allow you to define different input parameters for each direction. For example, if you collected the X transects in zig-zag, but the Y as unidirectional lines, you can input the different line order here. Also if you collect data in the Y direction as with some other geophysical instruments (Geoscan RM15 or FM36, or Geometrics 858), you can just check the Files in Y-Direction tab.



Starting Coords (units): This is the coordinate of the bottom left corner of your grid. If your area is tied into a larger site grid, you can input those coordinates here so that the axis of the resultant 3D file matches with the larger area.

X-Length/Y-Length: These are the maximum coordinates of your grid. For example, if your grid is 100 inches × 100 inches, you would put those values in here. If your profile lines are not all the same length, you should put in the measurement of the longest one. **This must be filled in for BOTH the Files in X-Direction and Files in Y-Direction tabs.**

Profile Lines: Total number of lines in either the X or the Y direction.

Line Spacing: This is the distance between each survey transect. The software figures out this number by dividing the grid size by the number of profile lines. You should use this as error checking. If you collected data with transects placed one foot apart, and you have the correct number of transects for the grid size, than this number should be 1. Anything else and you have a positioning error.

Helpful Hint: Do not forget to count the “0” transect. If you are scanning a 10 × 10 foot area with profiles every one foot, and your first profile is at 0 and your last is at 10, you will have 11 profiles.

Line Order: This is a pull down menu. Visualize your site grid and the order that your files were collected, and choose the orientation that matches your collection method.

Working Folder: This is where your data is stored. Clicking this button will open a browser so you can select a different directory.

Auto Load Files: If you check this box, RADAN will go to the working directory and automatically input the data files in alpha-numerical order. This is the same order that is shown when you sort the data files by name in Windows Explorer (by clicking on the “Name” column header). If the files are not in the correct naming convention, you may find it easier to rename them in Windows Explorer.

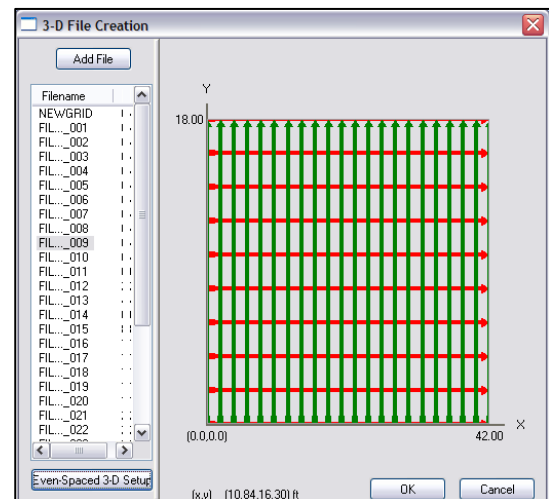
3 After you have worked with all the selections in the dialog, click OK to open the 3-D File Creation window.

4 The window shows the actual locations and orientations of your data profiles. If you clicked “Auto Load Files...” on the previous the left pane will show a list of file names with starting and ending coordinates.

5 Existing filenames and coordinates can be edited two ways. The first way is click on a filename located in the left pane. Then double-click that filename to edit the coordinates. The second way to edit a file is to move the mouse cursor to a line located in the right pane and click on it.

6 You may delete a line by clicking on the file in the left pane and pressing the delete key.

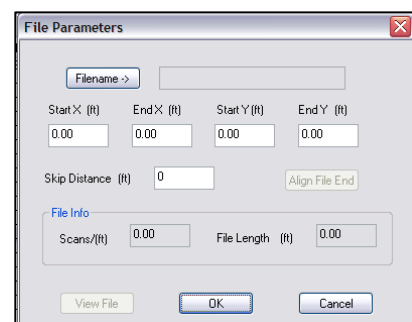
7 When you are satisfied with the look of the grid, click OK and the software will combine all the files and create a single grid file.



Other 3D File Creation Options

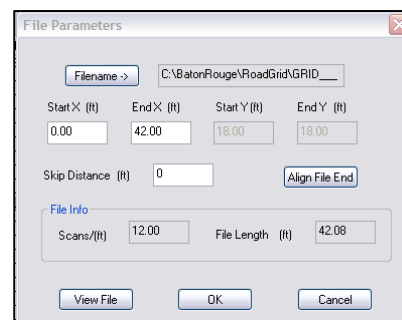
You may, if necessary, add files to your grid. Click Add File. Click Filename to browse and select the appropriate file. Then enter the X and Y starting and ending coordinates for that line. Click OK to save and continue.

The Skip Distance option permits you to skip a certain distance from the start of the file when writing to the 3D output file. This is particularly beneficial for files that were mistakenly started with the antenna in back of the starting point for the grid.



If your files were collected in Zig-Zag, you will see the align file button: to the right of the Skip Distance window. This button will adjust the file so that the last scan is aligned with the end of the grid. This option is typically used in cases where the user is more confident in the ending position of the profile than the starting position. This option is only available for evenly spaced x- or y-directed files.

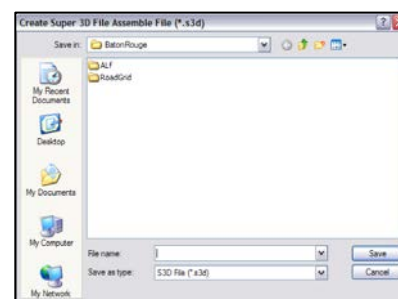
If you did select a line to edit, you may change the coordinates of this line. Click OK when finished.



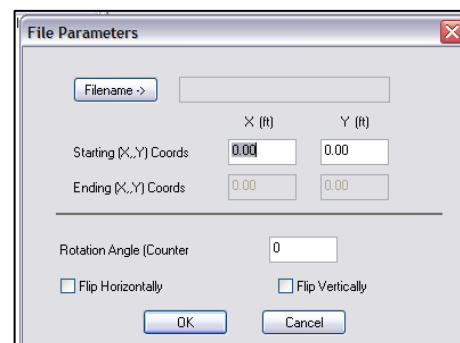
Super 3D File

You may combine multiple processed 3D Grids to create one “super” 3D Grid. This will create a new file with the extension .s3d.

After you select Super 3D File, Enter a filename for the newly created Super 3D file (.s3d). Click Save to save and continue. Follow these next set of steps:



- 1 Add File:** Click Add File to retrieve a grid.
- 2 Populate:** Populate the File Parameters window.
 - a) Filename:** Click Filename >> to browse and retrieve a grid.
 - b) Starting (X,Y) Coords:** Enter the X,Y position for this grid.
 - c)** If this is the first grid you are adding, the Starting (X,Y) coordinate is more likely 0,0.
 - d)** If this is NOT the first grid you are adding, the Starting (X,Y) coordinate is relative to the first grid you added.
 - e)** If necessary, you can Rotate, Flip Horizontally, and/or Flip Vertically, depending on how the grid was collected relative to the coordinates of the first grid you added.
 - f)** Click OK when you are done.

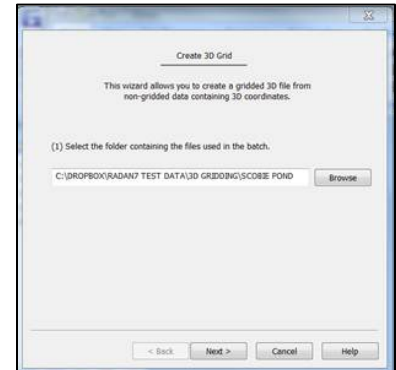


- 3 Repeat:** If you have more grids to add, continue back to step 1.
- 4** Once you have completed adding all the grids, click OK and the system will combine all the grids and create a Super Grid file.

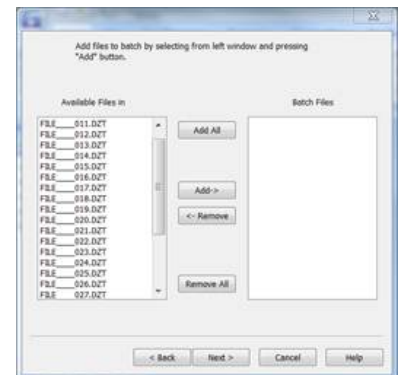
Gridded 3D File

This option takes one or more 3D files or files with GPS coordinates and creates a 3D gridded file.

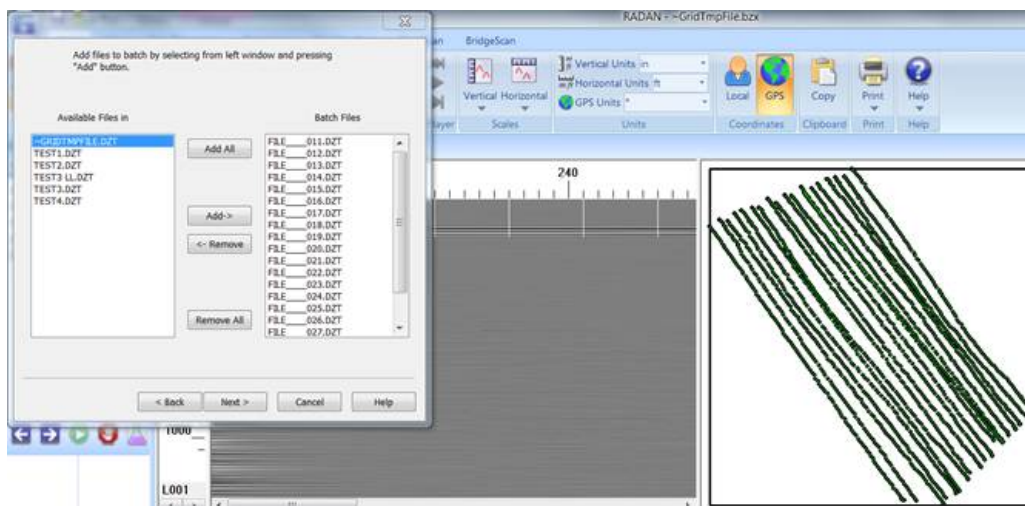
When the Gridded 3D File is selected, it opens a wizard. The first page of the wizard prompts the user to select the folder containing the file(s) to be gridded. All files to be gridded must reside in the same folder.



Once the folder is chosen and the “Next” button is pressed, the list of available files appears. Select the files from the left and add to the “Batch Files” window in the right. In the image below, all the files are profiles with GPS coordinates. But, you can also grid a 3D file containing without GPS coordinates, such as a SS optic file. The gridded is especially useful when gridding 3D files containing data from lines in perpendicular directions. The gridding eliminates the cross-hatching that typically appears in these files.



When files are added to the right window, a temporary file is created and the locations of the files in 3D space are indicated by the location view as seen in the image below. Once all the files have been added, click the “Next” button.



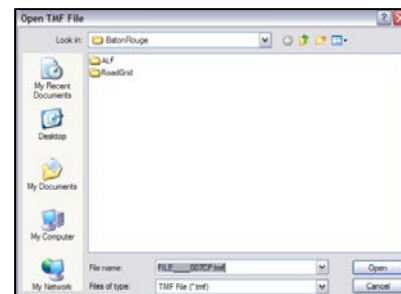
The next window that appears allows you to specify the dimensions of the 3D gridded area. By default, the area that appears encompasses all the data. You can change the area to grid a subset of the data in this window. Then click “Next”. (Note, if don’t have GPS coordinates, the area coordinates will be your horizontal units (e.g. feet or meters)).

The next window that appears allows you to specify the gridding intervals and gridding method. Note how the gridded file size changes when you change your grid interval. This will probably dictate how fine a grid interval you are willing to select. Four is the default number of nearest neighbors. The maximum neighbor distance should be at least the separation distance between profile lines.

After specifying all the info, press “Finish”. A gridded file will be created. While it’s being created, there will be a status bar indicating its progress. Once the 3D gridded file is created, you will be prompted to provide its name. Then it will automatically open up.

Import GPS

If you collected your data using GPS, you may select this option to import GPS data. First, open the data file. Then click the Import GPS menu option. The GPS file name will default to the same file name of the data file, except the GPS file will have a .tmf extension. You may select another GPS file and click Open.



Save

This will save the current active file along with all the parameters of the file.

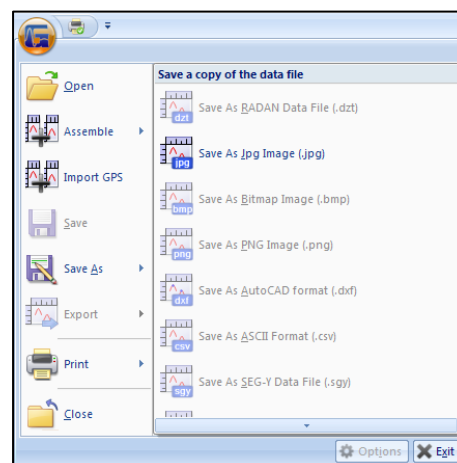
Save As

This will save the current active file as a RADAN file (.dzt) or an SEG-Y file.

Export

You can save the current active file as:

- RADAN file (.dzt)
- Picture as a JPG format (.jpg)
- Picture as a Bitmap format (.bmp)
- Picture as a PNG format (.png)
- AutoCAD file format (.dxf)
- ASCII format, Comma Delimited (.csv)
- SEG-Y format (.sgy)



Print

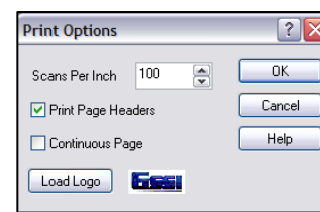
Print the image of the current active file.

Print: Printer setting will display.

Quick Print: Print directly to your default printer.

Print Preview: Preview your output on your screen prior to printing.

Setup: Enter print options such as Scan per Inch, Print Page Headers, Continuous Page, and your own logo (Load Logo) or GSSI logo.



Close

Click Close to close the active file.

Exit

Click Exit to exit the software.

Ribbons: Structure and Functions

The Ribbon contains options to process the data, different ways to display the data, and processing options specific to specialized applications. Each tab in the Ribbon contains these different options.



Home Ribbon

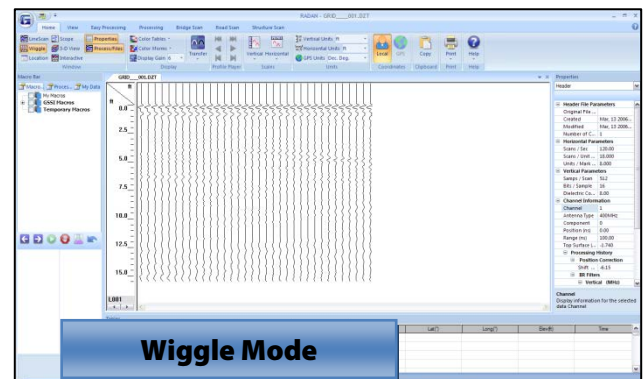
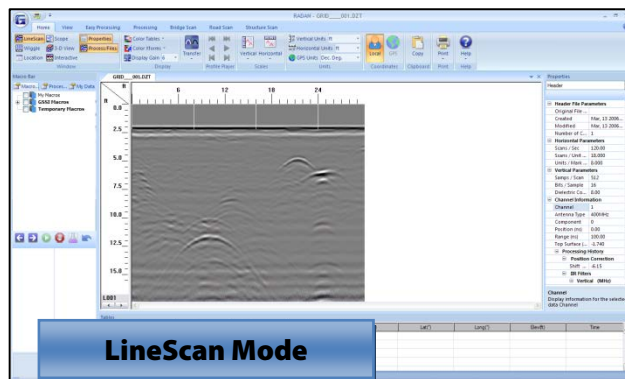
Window Group

Linescan

In the Linescan format your data is displayed in a color-amplitude form, and a color is assigned to a specific positive or negative amplitude value of the recorded signal, depending upon the color table and color transform selected. The vertical scale represents time (or depth) while the horizontal scale represents the horizontal distance traveled by the radar antenna. The Linescan display is the most useful for mapping man-made objects, such as underground storage tanks, pipes, and drums.

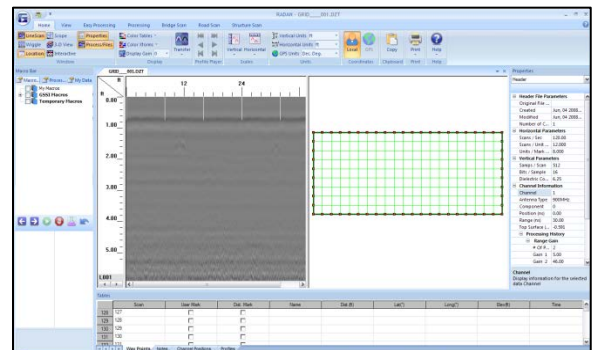
Wiggle

In the Wiggle format the data, consisting of multiple radar scans, are displayed as waveforms or “wiggle traces.” Wiggle plots are more useful for identifying geologic features, such as a clay layer or a water table. In both formats, the whole data file is displayed, with time zero (beginning of each scan) on top and time (or depth) increasing downward.



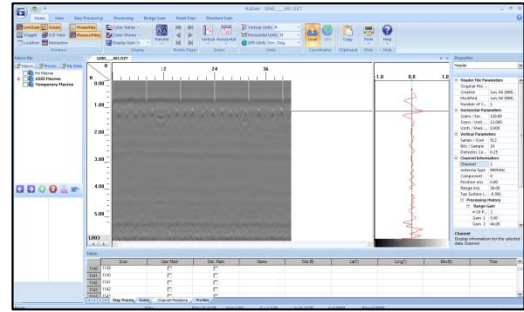
Location

Used primarily for 3D Grid files, this will display a grid where you can place your mouse on your data while holding your left mouse button down to determine the location of that spot on the grid. You may also move the grid around by pointing at the grid, holding the left mouse button down and dragging the mouse.



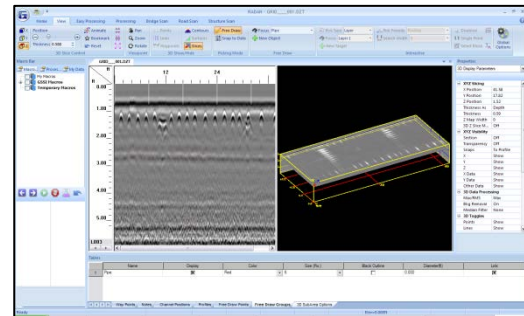
Scope

This will display an O-Scope to the right of the data. You may hold the left mouse button down while pointing to the data and this will correspond with the data with the O-Scope using a horizontal line.



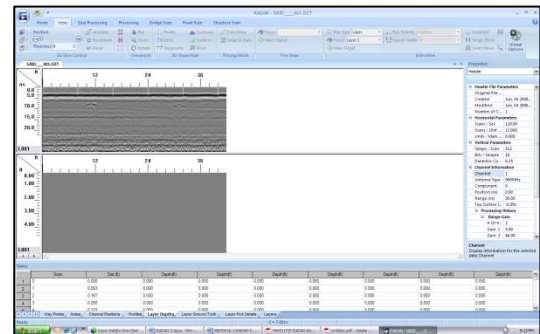
3D View

This will display your data as a 3D Cube. More information and options in this mode will be discussed in Section 4, the 3D View Options section.



Interactive

This view will allow you to “pick” targets and layers from your data for the purpose of exporting the data to a Comma Delimited ASCII file (.csv). More information and options in this mode will be discussed in Section 4, the View Interactive Options section.



Properties

This will toggle the display of the Properties Pane on the right side of the screen.

Process Files

This is toggle the display of the Macro Bar/Process Bar/My Data Pane of the left side of the screen.

Display Group

Color Tables

This is used to code the amplitude of each scan (i.e., the recorded radar signal). You may choose one of the standard display color tables from a list of thirty tables. A color table represents the amplitude of the recorded radar signal mapped to different colors. Colors on the left side of the spectrum represent strong negative amplitudes. As the colors go to the right, this represents the amplitude getting weaker (or closer to 0) and the middle of the spectrum is the area of weak amplitude. As the colors go further right from the middle, this represents the amplitude getting stronger on the positive side.

For instance, white in color table 1 corresponds to the highest positive amplitude pulse; therefore, when it appears on the radar record, it means that there is a strong reflection (or a high dielectric contrast).

Generally, dark means low amplitude signal. Therefore, a large black region on the linescan plot could be indicative of a uniform structure (such as a homogeneous sand deposit) with little or no dielectric contrast.

RADAN defaults to Color Table 17.

Color Xforms

You can also change the Color Transform to enhance weak amplitude or small contrast reflectors. The color transform determines whether the color scale applied to the radar wave's amplitude is linear, logarithmic, exponential, or customized. This function can also be used to de-emphasize certain features.

For example, in a logarithmic map, all low amplitude signals are assigned into a compressed lower color range, and the range of high amplitude signals is extended.

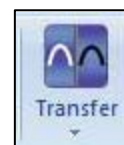
If white represents a high-amplitude signal, then there will be more white area for a given data set than in a linear transform.

Display Gain

To change the Display Gain, click Display Gain or simply right click on the data window and select Display Gain. You can either then select from a preset list of multiples ranging from -6 up to 60, or input a Custom value. This will change all samples, no matter where they are in the pulse, by the same amount. Altering the display gain does not change data values like Range Gain. Trying different values in the Display Gain may make it easier to spot lower amplitude targets.

Transfer

The displayed scan will be replaced with its frequency spectrum plotted in terms of relative amplitude (0 to 1) versus frequency in MHz.



Profile Player Group

Control Scrolling your Data Left and Right

First Row: Scroll all the way to the beginning or end.

Second Row: Play Forward or Reverse.

Third Row: Scroll one screen left or right.

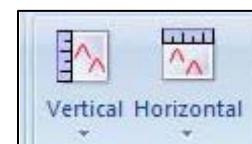


Scales Group

Select Vertical and Horizontal Scales

Vertical Scale: Select between Time (ns), Depth, or Samples.

Horizontal Scale: Select between Distance, Coordinates (GPS), Scans, or None.



Units Group

Vertical and Horizontal Units

Select between English units or Metric units, as well as GPS Coordinates.

GPS Units

Toggle between Local or GPS coordinates display.



Clipboard

Copy

Copy Active Window to Clipboard so that you can paste the image to a Third Party Software.

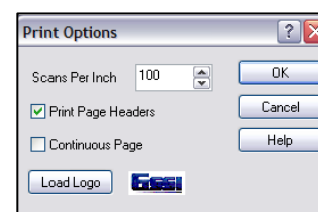
Print

Print: Printer setting will display.

Quick Print: Print directly to your default printer.

Print Preview: Preview your output on your screen prior to printing.

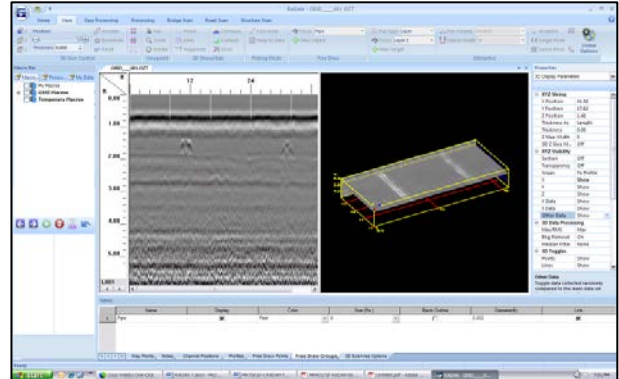
Setup: Enter print options such as Scan per Inch, Print Page Headers, Continuous Page, and your own logo (Load Logo) or GSSI logo.



View Ribbon

This tab contains menu options while displaying 3D data (Home Ribbon->3-D View) and Interactive Mode (Home Ribbon->Interactive).

Note: Information about specific processing for 3D and Interactive 3D is discussed in detail in Section 4.



3D Slice Control Group

You can slice the X wall or the Y wall side to side, or the Z slice up and down.

X: Click X and move the Position Slider bar to move the X wall.

Y: Click Y and move the Position Slider bar to move the Y wall.

Z: Click Z and move the Position Slider bar to move the Z slice up and down.

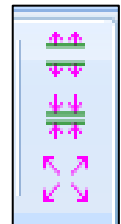
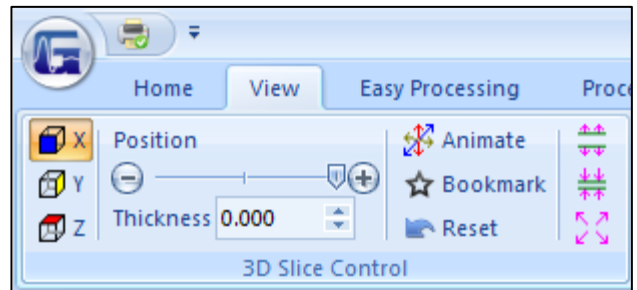
Thickness: Configure the thickness of each slice (X, Y, or Z).

Animate: Click here to have the software automatically automate the Z Slice up and down. Use the Up and Down arrow keys on your keyboard to control the speed of the animation.

Bookmark: Bookmark the slice settings.

Reset: Reset the image to original slice settings.

- Hold down the X, Y, or Z key on your keyboard and click on the first row of arrows to expand the 3D image in the X, Y, or Z direction respectfully.
- Hold down the X, Y, or Z key on your keyboard and click on the second row of arrows to contract the 3D image in the X, Y, or Z direction respectfully.
- Hold down the X, Y, or Z key on your keyboard and click on the third row of arrows to snap back to the original size.



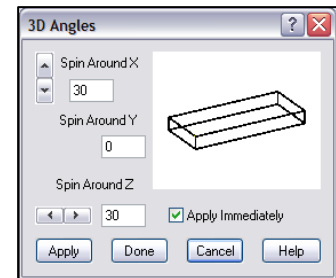
Viewpoint

Pan: Click here, move your mouse pointer on the 3D image. Then while holding down the left mouse button, move the image

Zoom: Click Zoom, move your mouse point on the 3D image. You may control the size of the zoom by move the wheel on your mouse. Double-click to zoom in on the square area. (Click the Reset icon to return to the original image).

Rotate: Click here to select the rotation to view your 3D Cube.

- Use the Up and Down Arrow or enter the numeric angle to rotate you image in the X, Y, and Z axis.
- Toggle with a check mark to Apply Immediately.
- Click Apply without exiting, click Done to apply the changes and exit, or Cancel to exit without applying the changes.



3D Show/Hide

Points: Toggle to show or hide points (targets) in you entered in your 3D Data.

Lines: Toggle to show or hide lines (pipes, rebar, etc) you entered in your 3D Data.

Waypoints: Toggle to show or hide waypoints (User Marks or Computer Generated Marks).

Contours: Toggle to show or hide amplitude contour of your 3D Data.

Surfaces: Toggle to show or hide surfaces of your 3D Data.

Slices: Toggle to show or hide X, Y, and Z slices of your 3D Data.

Picking Mode

Free Draw: Freely draw anywhere in your data rebar, pipes, conduits, etc.

Snap to Data: Have the software “snap” your drawings to a particular target.

Free Draw

Focus: In Free Draw mode, current focus of the object that is being entered.

New Object: Click here to add more objects to focus and draw.

Interactive

Available in Snap to Data Mode or Interactive Mode (Home Ribbon->Interactive).

Pick Type: Pick between Layer or Target.

Focus: Focus between specific layers or specific targets.

New Target: Click here to add more targets.

Polarity: When pick layers or targets, to pick Positive, Negative, Absolute, or None polarity.

Search Width: In pixels, enter the search width for Single Point picking tool.

Interactive Mode ONLY (Home Ribbon->Interactive)

Disabled: Disable Picking Tool.

Single Point: Enter Single Points for target picking.

Select Block: Select a user-defined blocked area to add points.

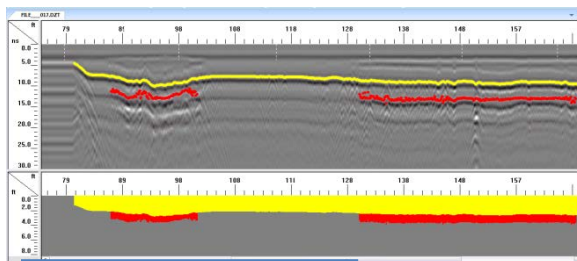
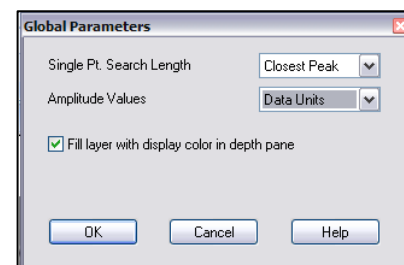
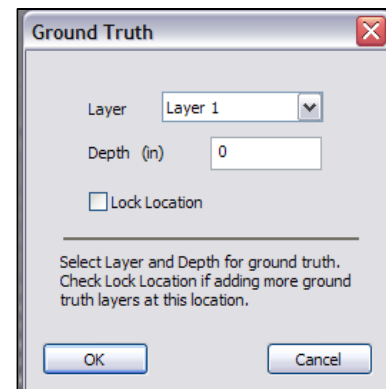
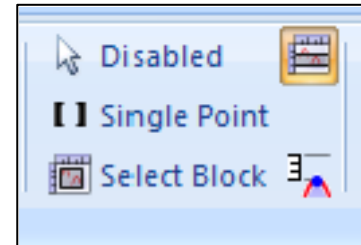
Select Range: Select a range of data from the beginning to the end to add points.

Ground Truth: Enter Ground Truth Information.

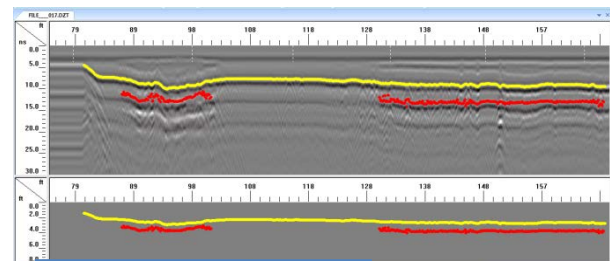
- **Layer:** Select Layer for ground truth.
- **Depth (in):** Enter Depth information.
- **Lock Location:** Input a check mark if adding more ground truth layers at this location.
- Click OK or Cancel.

Global Parameters

- **Single Pt Search Length:** When using Single Point Picking Tool, to enter a “Dot” at the closest peak or use the cursor length.
- **Amplitude Values:** Use Data Unit, dB, or Normalized dB.
- Check to fill layer with display color in depth pane.



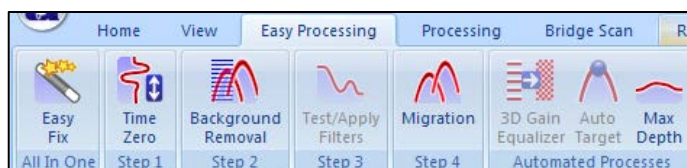
Filled Layers



Non-Filled Layers

Easy Processing Ribbon

This ribbon gives you options to easily process your data. Primarily, this is meant for data collected using Concrete Scan, Utility Scan, and data collected using the StructureScan Mini and Palm Antenna.



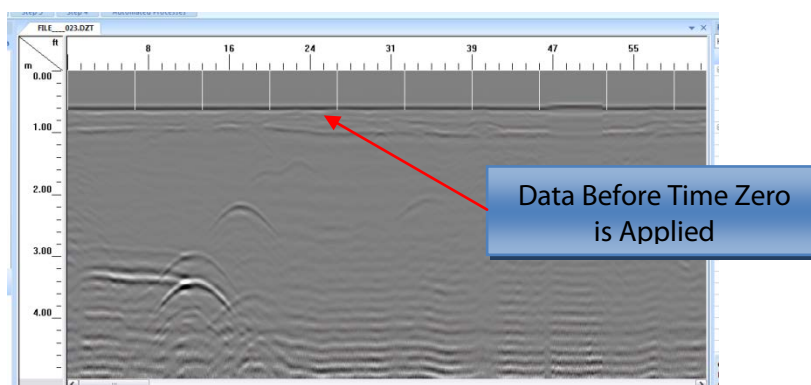
Easy Fix

This option will allow you process your data step by step. The steps will guide you through Time Zero, Background Removal, and Migration. All these steps are customizable.



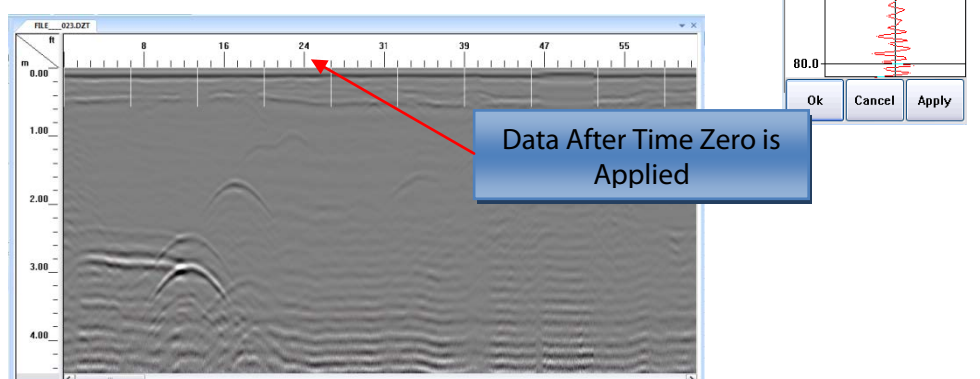
Time Zero

Sometimes it is necessary to vertically adjust the position of the whole profile in the data window (adjust time-zero). You may want the first positive peak of the direct wave from a ground coupled, bistatic antenna to be centered at the top edge of the screen so that you can consider ground surface to be at the top of the window (at Time Zero). A corrected 0-position will give you a more accurate depth calculation because it sets the top of the scan to a close approximation of the ground surface.



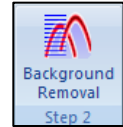
When you click the Time Zero icon, the left pane displays the Time Zero Process Bar. Simply “grab” the positive peak and hold your left mouse button down. Drag the positive peak and adjust it until it lines up with the 0.0 line. Click Apply to Test your position and, if necessary, click Reset and re-adjust.

Once you are satisfied with the Time Zero Position, click OK.



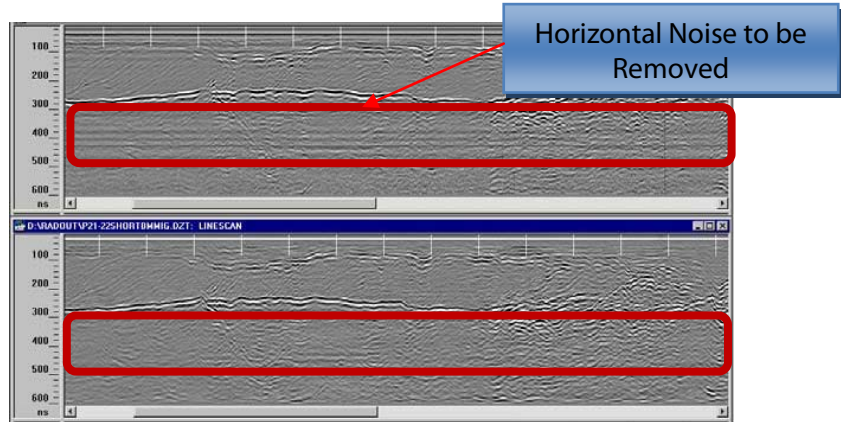
Background Removal

Background Removal Filter is the best way to remove bands of ringing noise. Low frequency features in the data will be removed, such as antenna ringing.



The filter length should be set to the number of scans equal to the feature length you want to remove. In some cases this may or may not be realizable, as the file may be longer than the maximum number of scans that can be implemented for a particular filter. In these cases the user will have to accept some filtering of long i.e. 'real' horizontal reflectors in the data.

The maximum value for a Horizontal Background Removal FIR filter is 1023 scans. Note that any continuous feature within the data, such as a water table reflector or a stratigraphic boundary between two soil types, may also be filtered out to some extent. In this case these features are not present. You must be careful not to filter out actual real data.



- 1 When you click on the Background Removal Icon, the left pane displays the Background Removal Process Bar. Click BR Type to remove specific horizontal type noise.

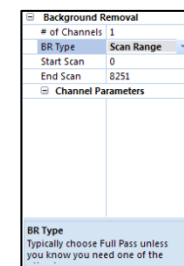
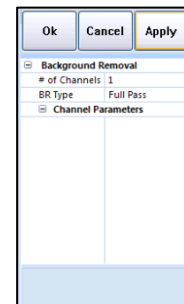
Full Pass: Will remove the horizontal noise throughout the enter profile.

Scan Range: Will remove the horizontal noise within specific scans.

- **Start Scan:** Enter the starting scan number where the horizontal noise begins.
- **End Scan:** Enter the ending scan number where the horizontal noise ends.

- 2 Click Apply to test your position and, if necessary, click Reset and re-adjust.

- 3 Once you are satisfied with the Background Removal, click OK.



Test/Apply Filters

This option will allow you to apply filters to your data. Apply filters can be a trial and error process. You should test the filter, and if the results are desirable, click Apply.

Migration



The radar antenna radiates energy with a wide beamwidth pattern such that objects several feet away may be detected. As a consequence of this, objects of finite dimensions may appear as hyperbolic reflectors on the radar record as the antenna detects the object from far off and is moved over and past it. Deeper objects may be obscured by numerous shallower objects that appear as constructively interfering hyperbolic reflectors. Steeply dipping surfaces will also cause diffracted reflections of radar energy. This diffracted energy can mask other reflections of interest and cause misinterpretation of the size and geometry of subsurface objects. The apparent geometry of steeply dipping layers are an illusion and need to be corrected in many cases. Migration is a technique that moves dipping reflectors to their true subsurface positions and collapses hyperbolic diffractions.

There are two Migration methods available in RADAN: Kirchhoff and Hyperbolic Summation. The Hyperbolic Summation method is faster but less accurate than the Kirchhoff method.

Hyperbolic Summation: Hyperbolic Summation works by summing along a hyperbola placed on the data, and placing the resulting average at the apex of the hyperbola. This process is repeated with the apex on every point of the data.

Kirchhoff (default): The Kirchhoff method is more accurate than the Hyperbolic Summation. An average value is still derived by summing along a hyperbola placed on the data and placed at the apex. However, Kirchhoff Migration also applies a correction factor to this averaged value, based upon the angle of incidence and distance to the feature. It also applies a filter to compensate for the summation process. This filter improves resolution by emphasizing the higher frequencies and applying a phase correction. Generally, because of speed considerations, the Hyperbolic method is used if it provides adequate results. If the Hyperbolic Summation method does not provide good results, the Kirchhoff Migration method should be tried.

Both the Hyperbolic Summation and Kirchhoff Migration require that the hyperbolic width and relative velocity are specified.

The following file header parameters must also be assigned a value before the data can be migrated:

- Samples/scan
- Range (ns)
- Scans/meter

Helpful Hint: You should use a survey wheel during the data collection process in order to migrate the data accurately.

Velocity is the speed at which the radar pulses travel through the material. The relative velocity is the ratio between the length of an object in the distance axis (in number of scans/meter) to its length in the time axis on the screen (number of samples/meter).

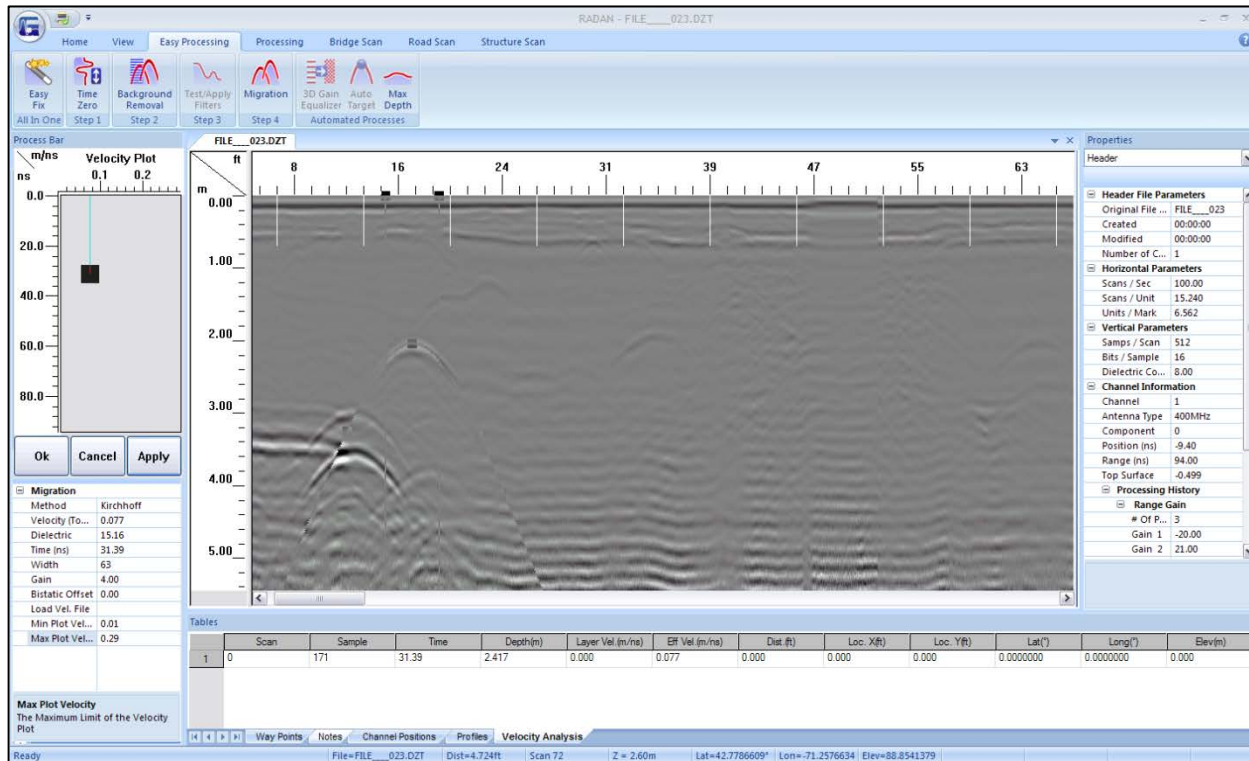
Width (in number of scans) is used to sum across the data file. This value should be set to about the same number of scans as the diffraction hyperbolas in the data. Larger values tend to give more accurate results, but if the value is too large, deterioration will occur.

Gain function is used to increase the data amplitudes after migration, since migration usually reduces the amplitude of the radar signal. The Gain is usually set to a value between 1.5 and 5.

You should adjust the shape of the Ghost Hyperbola to match a real hyperbola in the data. Notice that as you change the shape of the hyperbola, the velocity changes. The Ghost Hyperbola is nothing more than a tool to help you identify the correct velocity of your material. You may do this by grabbing the handles and adjusting the ghost hyperbola, or using the left pane and grabbing the square to adjust the ghost hyperbola.

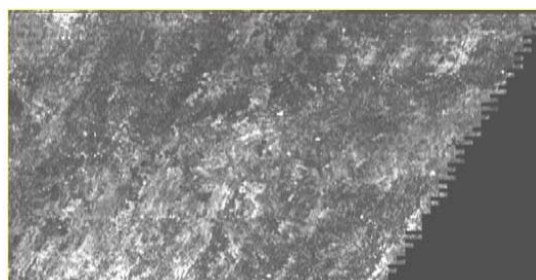
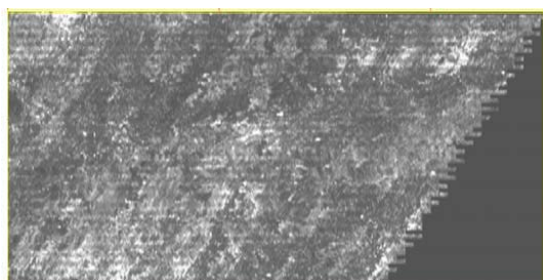
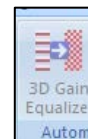
Click Apply to Test your position and, if necessary, click Reset and re-adjust.

Once you are satisfied with the Migration, click OK.



3D Gain Equalizer

Gain Equalization is a very basic, automated processing algorithm that looks for differences in the average gain values of data files in a 3D set. It is ideal for correcting data sets that have been improperly gained, or have had gain differences from profile to profile. RADAN will analyze the relative amplitude values of each profile and alter the gains accordingly. After the process is run, you will be prompted to save the data file with a new name. Below is an example of data before and after gain equalization. Notice the horizontal stripes in the left data example.



Auto Target

If data was collected using GSSI's Structure Scan Mini, this option will automatically pick targets by placing "dots" on the targets.



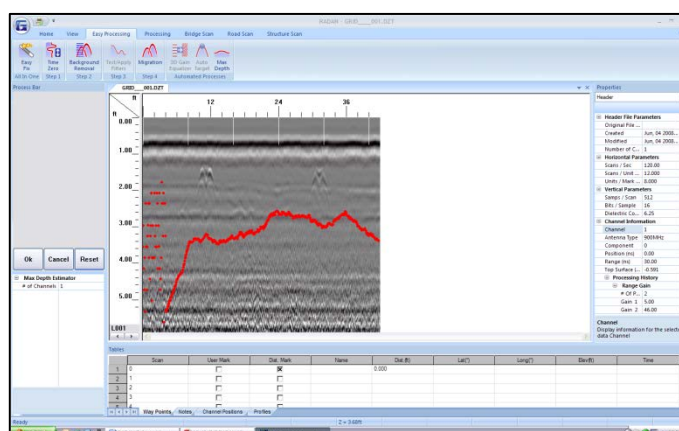
Max Depth

This function analyzes the noise and signal loss (attenuation) from scan to scan and gives you an estimate of your effective depth penetration. The output of this function is shown as a red line on the screen and saved as an ASCII .shp file for review later.



To use this function, follow these steps.

When the Max Depth Icon is selected, the left pane will display the Max Depth Process Bar. Simply click Apply and Reset to test the setting. Click OK to accept the setting.



Processing Ribbon

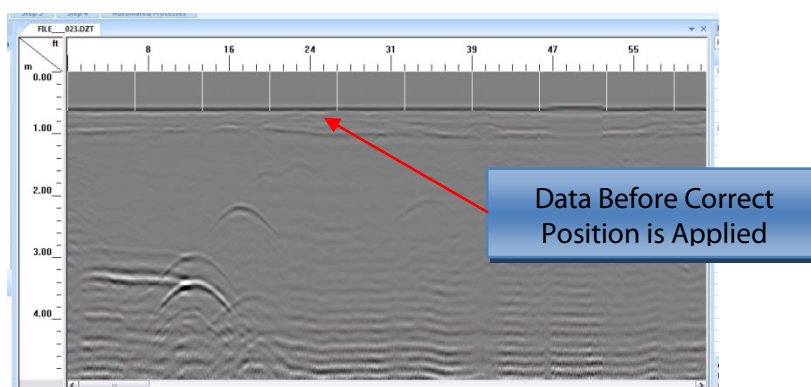
This ribbon will allow you to process (both standard processing and advanced processing) any data and application.



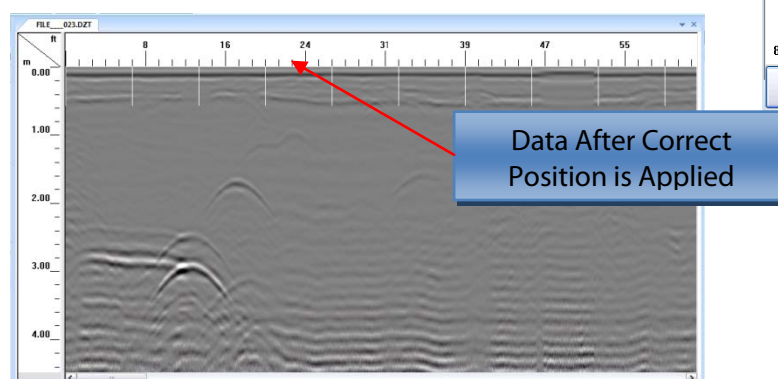
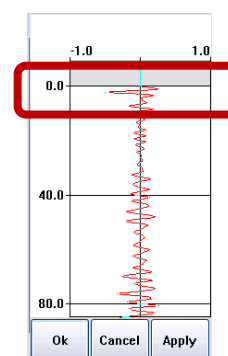
Time Zero

Correct Position

Sometimes it is necessary to vertically adjust the position of the whole profile in the data window (adjust time-zero). You may want the first positive peak of the direct wave from a ground coupled, bistatic antenna to be centered at the top edge of the screen so that you can consider ground surface to be at the top of the window (at Time Zero). A corrected 0-position will give you a more accurate depth calculation because it sets the top of the scan to a close approximation of the ground surface.



When you click on the Time Zero icon, the left pane displays the Time Zero Process Bar. Simply “grab” the positive peak and hold your left mouse button down. Drag the positive peak and adjust it until it lines up with the 0.0 line. Click Apply to Test your position and, if necessary, click Reset and re-adjust. Once you are satisfied with the Time Zero Position, click OK.



Filters Group

FIR Filter



FIR filters have a finite-duration impulse response. FIR filters, when encountering a feature in the data, are guaranteed to output a finite filtered version of that feature. This property makes it possible to design filters that are perfectly symmetrical and have linear phase characteristics.

FIR filters will therefore produce symmetrical results so reflections will not be shifted in time or position. Note, because of the symmetrical nature of FIR filters, FIR filter lengths should always be an odd number. Generally, FIR filters are normally preferred in digital signal post processing; however, users more familiar with IIR filters might choose to use them instead.

There are two types of FIR filters available in RADAN, Boxcar and Triangular Filters.

Boxcar Filter: The Boxcar filter is a rectangular window function that performs a simple running average on the data. A portion of the data, determined by the filter length, is averaged, and the average is output as a single point at the center of the active portion of the filter window. The filter moves on to the next sample and the process is repeated. The Boxcar filter assigns equal weight to the data all along the filter length.

Triangular Filter: The Triangular filter emphasizes the center of the filter more heavily than the ends of the filter. This type of filter is a weighted moving average, with the weighting function shaped like a triangle. A portion of the data, determined by the filter length, is multiplied and summed by this function. The result is output at the center of the triangle. The filter then advances one sample and the process repeats.

When the FIR Filter icon is click, the left pane will display the FIR Filter Process Bar.

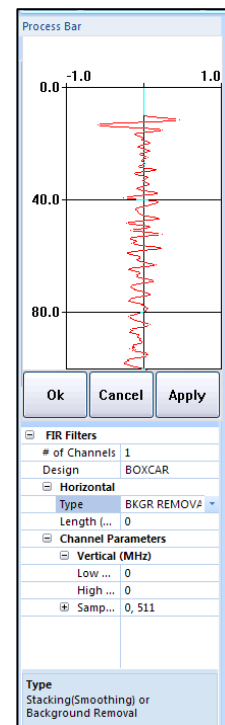
of Channels: If multi-channel was used, which channel to process.

Design: Boxcar or Triangular Filter

Type: Background Removal or Stacking

- Background Removal:** Background Removal Filter is the best way to remove bands of ringing noise. Low frequency features in the data will be removed, such as antenna ringing. Note: The length of the filter should always be a greater number of scans than the length in scans of the longest flat “real” event in the data that you want to keep. This filter will remove the surface reflection (direct coupling) pulse. In many cases, it is important to keep that direct coupling pulse as that is a visual check of your surface. If you want to keep the pulse, change the starting sample number to a value below the surface reflection. This may necessitate overlap filtering later to remove non-linearities created in the data by “window” filtering
- Stacking:** Apply a simple running-average to stack the data. Stacking combines the adjacent selected radar scans and outputs a single scan. When stacking values are used in RADAN, the program will retain the marks in the file. However, the scans per unit distance and marks per unit distance will be changed in the header. For example if you had a raw file with 80 scans per meter and 1 meter per mark and you stacked by a factor of two (2) the output file would have 40 scans per meter written into the header (reduced by a factor of 2).

Length: Enter the number of scans to filter. High number of Background Removal, low odd number for Stacking



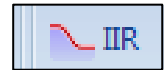
Vertical (MHz)

Low Pass: The Vertical Low Pass Filter will eliminate all the high frequency noise that often is seen in relatively deep radar profiles. You will be asked to input a frequency inside the range, which depends on the data sampling frequency. The cutoff frequency may be estimated by looking at the frequency spectrum of the radar data. We suggest using the frequency spectrum feature of RADAN to look at the frequency content of the radar file. A Vertical Low Pass Filter will reject frequencies above an established threshold. It is a good idea to reject all frequencies above the range where meaningful data are observed.

High Pass: If the desired horizontal features are of higher frequency content, use a FIR or IIR Vertical High Pass Filter with a cut-off frequency corresponding to about $\frac{1}{4}$ the center frequency of the antenna. For example, if you are using a 400 MHz (Model 5103) antenna, set the high pass cut-off to 100 MHz as a starting point, then try changing the frequency of the filter up or down within the antenna bandwidth to find the best fit.

- 1** First measure the frequency of the noise bands you want to eliminate. This can be accomplished by using the O-Scope Display mode in the Home Ribbon.
- 2** Click the right mouse button within the data window and choose Transfer>Spectrum.
- 3** Set the vertical high pass frequency equal to or slightly lower than the highest frequency you wish to eliminate and run the Boxcar filter. The frequency range that you may choose will depend on the data sampling frequency. You should start at low frequencies first. If this does not adequately filter the noise, slowly increase the high pass cut-off frequency. If the frequency of the noise bands is much lower (i.e., wider bands) than the reflections of interest, this approach will work quite well. However, if the noise bands and the reflections of interest are of the same frequency (width) the effectiveness of this approach is minimal.
 - **Samples:** Enter the Starting and Ending Samples where the filters are to be applied.
- 4** Click Apply and Reset, and continue to apply different values until you get the desired result. Once you do, click OK to run the process, or Cancel to exit the process without running the process.

IIR Filter



IIR filters were introduced before the advent of computers, when simple LRC circuits were used as analog filters. When an IIR filter encounters a feature in the radar data, it produces an output that decays exponentially towards zero but never reaches it, hence the name “infinite.” IIR filters are not necessarily symmetrical and while they achieve excellent amplitude response, their phase response is non-linear and so they can cause slight phase shifts in the data.

The IIR Filter Parameters Box lets you define the high and low pass horizontal and vertical filters as well as the time interval that will be processed, defined by the start and end sample number (RADAN automatically defaults to the whole time window).

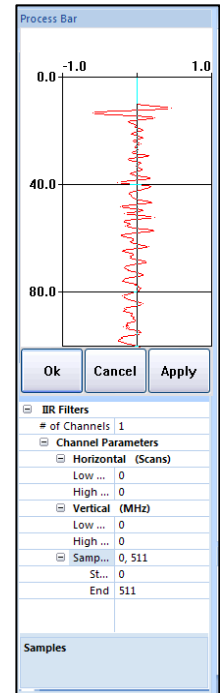
The cutoff frequencies are defined in number of scans horizontally and in MHz vertically.

You can apply any combination of vertical filters simultaneously and see the result using the Apply button.

Horizontal filters, due to their implementation, will not work under Apply. Select OK to see the horizontal filter's effect on the file.

RADAN uses IIR filters having only one pole so that there is not a sharp break at the cutoff frequency, which may provide limited noise reduction. As a consequence of this, it may be beneficial to run the same filter more than once. Or, another approach is to modify the Color Transform (under the Display Parameters Dialog Box) to hide what little noise remains in your data.

Click Apply and Reset, and continue to apply different values until you get the desired result. Once you do, click OK to run the process, or Cancel to exit the process without running the process.



FK Filter



The spatial FFT filter, which is a two-dimensional frequency filter, takes place in the time-space domain. It is often called a frequency–wave number, or F-K, domain. This approach generates a two-dimensional matrix, the complex elements of which represent the phase and amplitude of various spatial waves present in the radar data. It allows the user to develop a two-dimensional filter to attenuate the noise.

Performing the inverse Fourier transformation of the product matrix from the transformed data and the filter yields a data with reduced noise. Technically, at this stage the frequency domain data is reconstructed back to the time domain.

The advantage of F-K filtering over successive vertical and horizontal one-dimensional frequency filtering is that it enables a better distinction to be made between the signal and the noise. The signal and noise spectra may overlap in one dimension, which makes their separation impossible, but this is less likely in the F-K domain.

The 2-D filter is defined by four parameters: Radius, Delta Radius (dRadius), Alpha and Delta Alpha (dAlpha). In a two-dimensional F-K domain their analytical expression in polar coordinates is as follows:

$$F(r,a) = k * F_0(r,a)$$

$$\text{RADIUS } r_0 = (r_1 + r_2)/2; \text{ Alpha } a_0 = (a_1 + a_2)/2;$$

$$\text{dRADIUS } dr = r_2 - r_0; \text{ dAlpha } da = a_2 - a_0;$$

Four different filters are available in the RADAN Spatial FFT module: High-cut Horizontal, High-cut Vertical and the two corresponding symmetrical filters. They are described in the Spatial FFT section further in this manual.

Each filter can be used with filtering windows of different shapes. This determines the type of function necessary to obtain a finite-length impulse response by truncating an infinite-duration impulse response (frequency domain analog of time domain window functions).

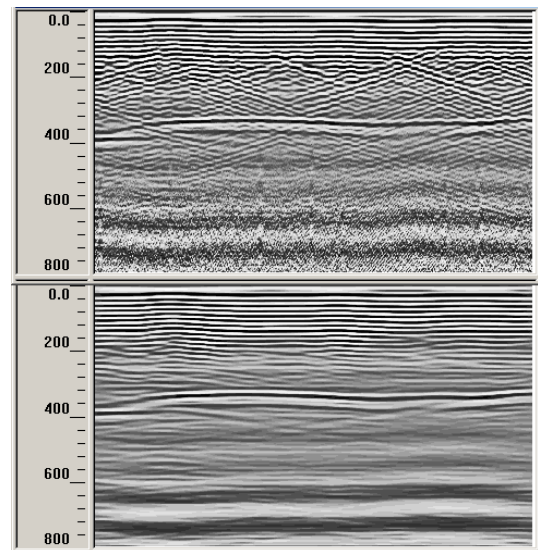
The linear window parameters, which are the radius parameters, represent fractions of the maximum data frequency. The following selection of filtering windows is available: Rectangle, Bartlett, Hanning, Hamming, and Blackman. The rectangular window cuts off at the specified limits, when the other windows allow for a smoother, more gradual transition. Their analytical expression, based on the parameters defined above, is as follows (assuming $r_t = r - r_0$; $a_t = a - a_0$, $PI=3.14..$):

Rectangular	$k = 1$, for $-dr \leq r_t \leq dr$ and $-da \leq a_t \leq da$;
Bartlett	$k = (1 - r_t/2dr) * (1 - a_t/2da)$, for $-2dr \leq r_t \leq 2dr$ and $-2da \leq a_t \leq 2da$;
Hanning	$k = 0.5 * (1 + \cos(2 * PI * r_t / 4dr)) * 0.5 * (1 + \cos(2 * PI * a_t / 4da))$, for $-2dr \leq r_t \leq 2dr$ and $-2da \leq a_t \leq 2da$;
Hamming	$k = (0.54 + 0.46 * \cos(2 * PI * r_t / 4dr)) * (0.54 + 0.46 * \cos(2 * PI * a_t / 4da))$, for $-2dr \leq r_t \leq 2dr$ and $-2da \leq a_t \leq 2da$;
Blackman	$k = (0.34 + 0.5 * \cos(2 * PI * r_t / 4dr) + 0.16 * (\cos(2 * PI * r_t / 4dr) ** 2)) * (0.34 + 0.5 * \cos(2 * PI * a_t / 4da) + 0.16 * (\cos(2 * PI * a_t / 4da) ** 2))$, for $-2dr \leq r_t \leq 2dr$ and $-2da \leq a_t \leq 2da$;
otherwise	always $k = 0$.

Using The 2D Spatial Filter To Enhance Target Reflections In GPR Data

The 2D spatial filter (i.e., frequency/wave-number (F-K)) filtering technique is a robust tool for isolating sloping features with known frequency components in GPR data. This technique was first implemented by the seismic industry to remove ground roll from seismic data. In the case of GPR, many of the important targets contain reflections with characteristic slopes. At right is an example of noise removal.

Note the lattice-like noise pattern in the top window of data. A spatial filter can be adjusted to remove features with a specific dip angle, while leaving other features intact (bottom window).



- 1** Click the FK Filter icon.
- 2** Click the box near Target Parameters. Approximate values for the ground propagation velocity, depth, and diameter of the pipe should be entered in the dialog.

- 3** The F-K spectrum of the highlighted section of the data file is shown on the left side of the dialog with the outline of the F-K target filter super-imposed.
- 4** When the highlighted section of the data file contains a strong hyperbolic reflection and the user-specified target parameters are accurate, the outline of the F-K target filter should extend around the highest amplitude portion of the spectrum.
- 5** To activate the F-K target filter, click OK. Features in the data with slopes outside the filter range are removed from the processed data.

This function is entered by selecting Spatial Filter in the Process menu or by using the button. A dialog box appears. It shows spectrum display and filter parameters on the right and a 2D FFT diagram on the left. The diagram is a graphical representation of the 2D filter in a time-space domain. Its vertical axis represents signal frequency F (MHz) in the vertical direction (in time); the horizontal axis shows the wave number K (cycle/cm) in the horizontal direction (in space, i.e., along the profile).

The recommended processing sequence starts with a spectrum display. The Recalc button transforms the file into the two dimensional frequency domain and displays the spectrum of the active file in the 2D FFT Diagram

The Display group of parameters on the right side of the dialog box controls the spectrum display. Once a spectrum is displayed, use the Gain and Zoom parameters to enhance the display as desired. By default, the spectrum of 512 scans by 512 samples data section is calculated. You can select a different block of data by changing the start and end values under Scans and Samples, or you may select a portion of the file using the overlay box which will appear on the data display when the 2D FFT function is selected. Click the Recalc button again to display the spectrum of the selected section.

A 2-D filter for the inverse spatial FFT can now be set up. This can be done either by setting the filter parameters directly or by using target parameters.

Filter Parameters Setup

- 1** To set filter parameters directly, make sure the Use Target Parameters box is blank (this is the default).
- 2** Check and adjust the Geometrical Parameters (Min and Max Frequencies, Alpha and Delta Alpha).

These parameters can be entered in the appropriate boxes or adjusted by dragging the four square boxes in the 2D diagram.

Minimal and maximal frequencies are represented by the arcs in the diagram.

Alpha and Delta Alpha that control the horizontal components are represented by two straight lines (radiuses) in the diagram.

Alpha expresses the filter symmetry and remains close to zero whenever the radiuses are placed symmetrically on both sides of the imaginary vertical centerline (zero wave number). In this configuration, horizontal and vertical features will be emphasized.

An asymmetric setting (high Alpha) will emphasize slant reflections. Default is a zero Alpha setting, but it is recommended to experiment with combinations of asymmetric radiuses and symmetric filters (see below) that can, in some cases, make weak reflections clearly visible.

Delta Alpha controls the angle between the radiuses that physically corresponds to the linear dimension of reflections.

- 3** Once the Geometrical Parameters of the filter have been set, the 2D diagram can be printed out using the Print button. The arc and radius positions will be drawn with the four parameters printed below: Radius, dRadius, Alpha, and dAlpha (see the "About Filters" section for the definitions).

Filter: This parameter determines the part of the 2D diagram that will be processed with inverse spatial FFT. The five possible options are:

- **None:** The FFT file will be reconstructed without any changes.
- **High-Cut Horizontal:** Reconstruction will use only the wide arc domain between the radiuses.
- **High-Cut Vertical:** Reconstruction will use only the wide arc domain between horizontal zero-frequency line and radiuses (outside the radiuses).
- **High-Cut Vert Symm:** Two equal sectors outside the radiuses are used for reconstruction.
- **High-Cut Horz Symm:** Two equal sectors between the radiuses are used.
- The symmetrical filters are useful in cases where the radiuses are positioned asymmetrically relative to the imaginary vertical centerline.

Window: Each filter can be used with any of the filtering windows available: Rectangle, Bartlett, Hanning, Hamming, and Blackman. The rectangular window cuts off at the specified limits, whereas the other windows allow for a smoother, more gradual transition. The analytical expression for these windows is given earlier in this manual (see section, About Filters).

Overlap Scans: Determines the overlap between consecutive passes of the filtering window. More overlap produces a smoother result; in most cases, the default value of 8 is adequate.

Using the Target Parameters

The Geometric Parameters setup is replaced with the Point Target Parameters setup, if the Use Target Parameters option is activated (the corresponding box is checked). The following parameters are required:

- **Diameter (m):** Enter the typical size of an expected target.
- **Depth (m):** Enter the expected target depth.
- **Ground Relative Permittivity:** Enter an estimate of the relative dielectric constant of the ground material.
- **Low K removal number:** Cutoff wave number. Wave numbers less than this value are removed by the filter.

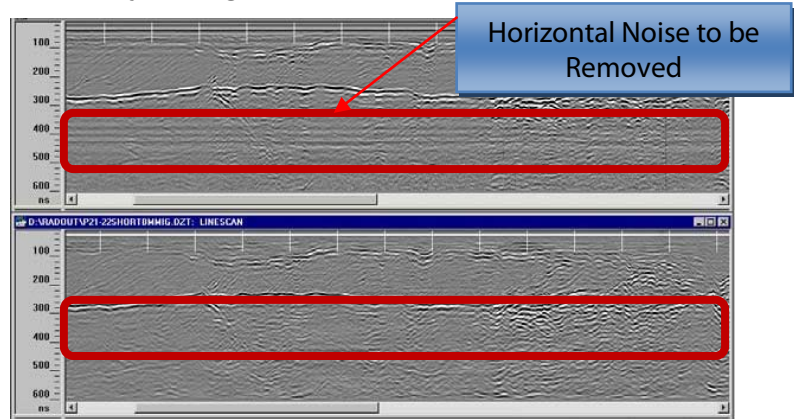
- 1** RADAN will adjust the filter parameters according to the specified target parameters. In the Use Target Parameters mode, no arcs and radiuses are displayed in the 2D diagram, but a spectrum can still be displayed using the Recalc button.
- 2** When the filter parameter setup is complete, click OK and the result of the F-K filtering will be displayed in the data window.
- 3** You will be prompted to save the file. Do so if you're satisfied with the result. Otherwise, cancel the Save operation, go back to the Spatial FFT dialog, readjust parameters and try filtering again. A trial and error approach may be necessary due to the complexity of this procedure.

Background Removal

Background Removal Filter is the best way to remove bands of ringing noise. Low frequency features in the data will be removed, such as antenna ringing.

The filter length should be set to the number of scans equal to the feature length you want to remove. In some cases this may or may not be realizable, as the file may be longer than the maximum number of scans that can be implemented for a particular filter. In these cases the user will have to accept some filtering of long i.e. 'real' horizontal reflectors in the data.

The maximum value for a Horizontal Background Removal FIR filter is 1023 scans. Note that any continuous feature within the data, such as a water table reflector or a stratigraphic boundary between two soil types, may also be filtered out to some extent. In this case these features are not present. You must be careful not to filter out actual real data.



When you click on the Background Removal Icon, the left pane displays the Background Removal Process Bar. Click BR Type to remove specific horizontal type noise.

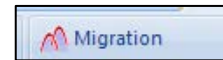
- **Full Pass:** Will remove the horizontal noise throughout the entire profile
- **Scan Range:** Will remove the horizontal noise within specific scans.
 - **Start Scan:** Enter the starting scan number where the horizontal noise begins
 - **End Scan:** Enter the ending scan number where the horizontal noise ends

Click Apply to Test your position and, if necessary, click Reset and re-adjust.

Once you are satisfied with the Background Removal, click OK.

Focus Group

Migration



The radar antenna radiates energy with a wide beamwidth pattern such that objects several feet away may be detected. As a consequence of this, objects of finite dimensions may appear as hyperbolic reflectors on the radar record as the antenna detects the object from far off and is moved over and past it. Deeper objects may be obscured by numerous shallower objects that appear as constructively interfering hyperbolic reflectors. Steeply dipping surfaces will also cause diffracted reflections of radar energy. This diffracted energy can mask other reflections of interest and cause misinterpretation of the size and geometry of subsurface objects. The apparent geometry of steeply dipping layers are an illusion and need to be corrected in many cases. Migration is a technique that moves dipping reflectors to their true subsurface positions and collapses hyperbolic diffractions.

There are two Migration methods available in RADAN: Kirchhoff and Hyperbolic Summation. The Hyperbolic Summation method is faster but less accurate than the Kirchhoff method.

Hyperbolic Summation: Hyperbolic Summation works by summing along a hyperbola placed on the data, and placing the resulting average at the apex of the hyperbola. This process is repeated with the apex on every point of the data.

Kirchhoff (default): The Kirchhoff method is more accurate than the Hyperbolic Summation. An average value is still derived by summing along a hyperbola placed on the data and placed at the apex. However, Kirchhoff Migration also applies a correction factor to this averaged value, based upon the angle of incidence and distance to the feature. It also applies a filter to compensate for the summation process. This filter improves resolution by emphasizing the higher frequencies and applying a phase correction. Generally, because of speed considerations, the Hyperbolic method is used if it provides adequate results. If the Hyperbolic Summation method does not provide good results, the Kirchhoff Migration method should be tried.

Both the Hyperbolic Summation and Kirchhoff Migration require that the hyperbolic width and relative velocity are specified.

The following file header parameters must also be assigned a value before the data can be migrated:

- Samples/scan
- Range (ns)
- Scans/meter

Helpful Hint: You should use a survey wheel during the data collection process in order to migrate the data accurately.

Velocity is the speed at which the radar pulses travel through the material. The relative velocity is the ratio between the length of an object in the distance axis (in number of scans/meter) to its length in the time axis on the screen (number of samples/meter).

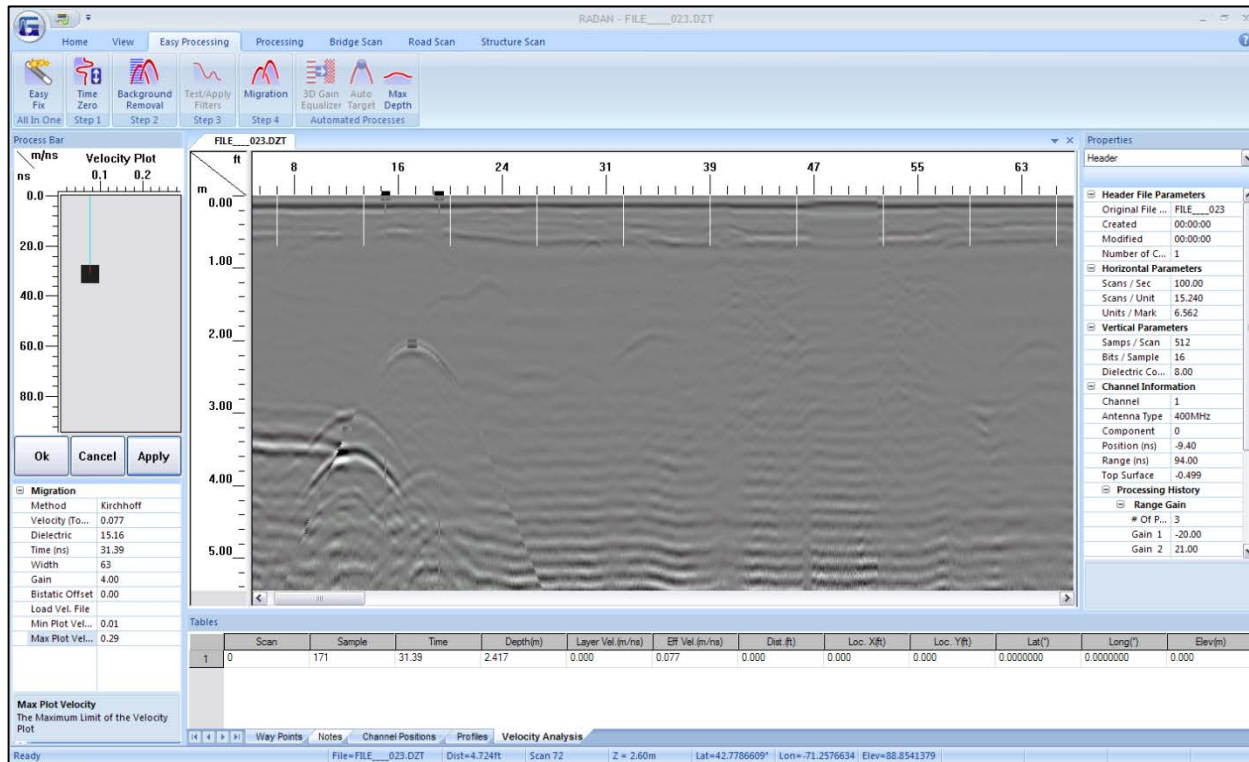
Width (in number of scans) is used to sum across the data file. This value should be set to about the same number of scans as the diffraction hyperbolas in the data. Larger values tend to give more accurate results, but if the value is too large, deterioration will occur.

Gain function is used to increase the data amplitudes after migration, since migration usually reduces the amplitude of the radar signal. The Gain is usually set to a value between 1.5 and 5.

You should adjust the shape of the Ghost Hyperbola to match a real hyperbola in the data. Notice that as you change the shape of the hyperbola, the velocity changes. The Ghost Hyperbola is nothing more than a tool to help you identify the correct velocity of your material. You may do this by grabbing the handles and adjusting the ghost hyperbola, or using the left pane and grabbing the square to adjust the ghost hyperbola.

Click Apply to Test your position and, if necessary, click Reset and re-adjust.

Once you are satisfied with the Migration, click OK.



Velocity Analysis



Data Preparation

To use the velocity analysis program in the RADAN Velocity Analysis module, you must first have a data set recorded at multiple offsets using a bistatic antenna configuration using the common-midpoint (CMP) method (described below) or wide angle refraction reflection (WARR) method.

GSSI makes two bistatic antenna models, which can be used for CMP acquisition: the Model 3207 (100 MHz), and the Model 3200 MLF (16-80 MHz). Using a radar system with a multi-channel capability (e.g., the SIR 20, 10A, 10A+, 10B, or 10H systems), any combination of antennas can be used to gather CMP data. Additionally, a dual-antenna adapter is available for bistatic operation with a single-channel system, such as GSSI's SIR-2, 2000, or 3000.

To estimate velocity from a CMP “gather,” you will need to collect GPR data using a bistatic antenna configuration in the Static Stack mode. Initially, transmitting and receiving antennas start off adjacent to one another at the “zero offset” position. (Actually, because of the size of the antennas' housings, transmitting and receiving antennas are never at the true zero offset position. Rather, the initial offset is the distance from the symmetrical reference - i.e., the zero offset position- to the antenna's centerline.) Data are then collected at multiple offsets, as the antennas are moved symmetrically apart from the zero offset position. The output data quality improves significantly using a large number of offsets and stacks. However, by using a lot of offsets and stacking, acquiring data may be very time consuming. The operator will have to decide the best utilization of time when establishing the best offset interval, the number of stacks per measurement, and number of offsets, versus the desired quality of data output.

With estimated velocities acquired in this way, we can more accurately estimate the depth to certain targets or stratigraphic features. Also, using a DOS subroutine supplied by GSSI with the Velocity Analysis module, estimated velocities can be used to construct a CMP “stack.” Using derived velocities, offset traces are corrected for nonzero offset, and then stacked to form a single trace at the zero offset position. Many CMP gathers can be likewise compressed to a single trace, resulting in a corrected stacked section with a significantly enhanced signal-to-noise ratio.

Theoretical Overview

For a single constant-velocity horizontal layer, the travel time curve as a function of offset is a hyperbola. The time difference between travel time at a given offset and at zero offset is called normal moveout (NMO). The velocity required to correct for normal moveout is called the normal moveout velocity. For a single horizontal reflector, the NMO velocity is equal to the velocity of the medium above the reflector. For a dipping reflector, the NMO velocity is equal to the medium velocity divided by the cosine of the dip angle:

$$(0), \text{ or } V_{\text{nmo}} = V / \cos \theta$$

Travel time, as a function of offset from a series of planar horizontal layers, is approximated by a hyperbola. This approximation is better at small offsets than large offsets. For short offsets, the NMO velocity for a horizontally layered earth is equal to the root mean square (rms) velocity down to the layer boundary under consideration. In media composed of layers with arbitrary dips, the travel time equation gets complicated. However, in practice, as long as dips are gentle and the spread (offset interval) is small (less than the reflector depth), the hyperbolic assumption can be made. Conventional velocity analysis is based on the hyperbolic assumption. For a detailed discussion of velocity analysis, see Seismic Data

Processing, by Ozdogan Yilmaz, published by the Society of Exploration Geophysicists in 1987, an excellent reference from which this brief description has been extracted and adapted.

The velocity analysis method used in this module is based on computing the velocity spectrum. The principle is to display some measure of signal coherency on a graph of velocity versus two-way zero-

offset time. Signal coherency is computed on the CMP gather in small time gates that follow a hyperbolic trajectory. Stacking velocities are interpreted from velocity spectra by choosing the velocity function that produces the highest coherency at times with significant event amplitudes.

Using the Velocity Analysis Function

Once you have gathered some CMP data sets,

- 1** Transfer them to your computer (into a separate directory for Velocity Analysis).
- 2** Next, run RADAN and Open the file.
- 3** Select the desired CMP file having the extension (*.dzt). Note that GSSI has provided a sample CMP data set for your use, should you not have one available to you.
- 4** Next, check the file headers (Header Information in the Right Pane) of the CMP gathers and make sure that the Scans/Unit parameter is entered

Note: The Scans/Unit parameter must have a non-zero value before the velocity analysis can begin.

- 5** With a CMP file open, initiate the Velocity Analysis module by clicking on the appropriate icon on the toolbar, or choosing Process > Velocity Analysis.
- 6** An Open Velocity File Message Box will appear and prompt you to input a *.vlc file.
 - A *.vlc file is a velocity data file created and saved from a previous Velocity Analysis session.
 - If this is the first time activating the Velocity Analysis, click Cancel to continue to the Velocity Diagram screen.
- 7** Upon activating the Velocity Analysis function, you will see the following window on your screen.
- 8** The program will default to the Configuration menu, as indicated when the Configuration button at the right center portion of the screen is selected. In order for the Velocity Analysis program to operate properly, the following information **MUST** be input:
 - In the Configuration section of the menu, select the proper Distance Unit (cm, inch, or foot).
 - Next, set the Input Step. This refers to the step interval distance the antennas are moved apart each time data is stacked during the CMP gather.
 - Input the Minimum Offset. This is the distance between the antennas center to center when they are at their closest during the CMP gather. Note that because of the antenna housing, this value should always be a non-zero number.
 - In the Process section of the menu, input the minimum and maximum velocity values you expect at the site you are surveying in cm/ns.
 - Next, choose the type of analysis you want the program to use: Stacked Amplitude or Cross Correlation. As the name implies, the latter method relies on a cross-correlation of traces in a gather, and not on the lateral continuity of traces. Hence, the Cross Correlation method produces less scatter in the data, but at the cost of using up significantly more processing time. The Stacked Amplitude method generally produces adequate results, but requires much less time for processing. For a detailed description of these analytical methods, see Yilmaz, (1987).

- 9** Last, select whether you want the Zero Time Position derived automatically, or if you want to input a value manually.
- 10** Once you have entered the required information, click the Recalc button and the program will calculate the estimated velocities of the layers present in the data.

The velocity spectrum shown clearly exhibits three layers. The bold elliptical to circular areas in each layer represent the most probable velocity of that layer, based on the correlation or lateral continuity of data.

The Diagram Parameters menu allows the user to locate the maximum velocity within the data set and establish the velocity within other layers. The program can also display isolines, or contours having the same velocity.

To access the Diagram Parameters window, click the Diagram button. The following features will allow you to modify the display in ways described below:

Mouse Cursor Function: When the cursor is placed on the left side of the window, the program shows the user the velocity, two-way travel time, and amplitude at that point.

MAX Function: The Max function returns the cursor to the maximum velocity of the dataset. By clicking Max, the program will automatically pick the maximum velocity layer, place crosshairs at the maximum point, and display the velocity of the layer and the time in nanoseconds (ns) at which it occurs.

Amplitude %: Amplitude % is displayed in another window. The Amplitude % value can be used as an indication of the “coherency” of the layer. The closer to 100%, the greater the correlation (or lateral continuity) of that value with adjacent values.

Isolines Function: The Isolines function enables the user to see contours of areas having the same velocity. This function is active when the Isolines box is selected with the mouse. When selected, a check mark will show next to the word isolines and the display will show velocity contours.

Left/Right Arrow: Clicking on the left or right arrow buttons will cause the program to select other layers to display their velocities, times and amplitudes.

The Velocity menu is accessed by selecting the Velocity button. This function allows the user to make a velocity log of significant layers in the data by clicking on the Velocity button. The velocity log consists of the velocity from each layer, the two-way travel time to the maximum velocity, the effective layer velocity, and the true layer velocity. To set up a velocity log:

- 1** Starting at the upper-most layer, move the mouse cursor to the maximum % Amplitude value.
- 2** Double-click the left mouse button, and the layer information will appear at the top right of the screen in the Velocity Dialogue box.
- 3** Move the mouse cursor to the next velocity layer and repeat the steps above.

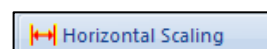
Velocity log information will be displayed in the Velocity window. A line will also be plotted from layer to layer, showing the best correlation of velocity.

Results of the velocity analysis can be saved in a file, or printed. To save these results in a velocity file (*.vlc), select the Save button at the bottom of the screen. To print the velocity log:

- 1** Select the Diagram button.
- 2** Click the Print button. A Print Dialogue box will appear.
- 3** Select Setup to make sure that the printer is setup properly, having the appropriate page quality and orientation.
- 4** OR, to save the velocity log to a file, using the mouse, select the Print to File box at the bottom left of the Printer Dialogue box.
- 5** Click OK to Print to the printer (or file).

Adjust Scans Group

Horizontal Scaling



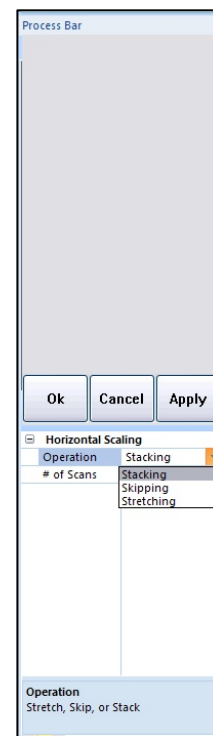
Data may be modified by adjusting the Horizontal Scale using the Stacking, Skipping and Stretching functions. To do this, use the button or go to the Process menu and select Horizontal Scale.

Note that you can only run one of the three choices at a time.

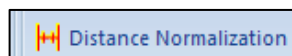
- **Stacking:** Select Stacking to apply a simple running-average to stack the data. Stacking combines the adjacent selected radar scans and outputs a single scan. When stacking values are used in RADAN, the program will retain the marks in the file. However, the scans per unit distance and marks per unit distance will be changed in the header. For example if you had a raw file with 80 scans per meter and 1 meter per mark and you stacked by a factor of two (2) the output file would have 40 scans per meter written into the header (reduced by a factor of 2).
- **Skipping:** Select Skipping to specify the number of scans you wish to skip over from the previous scan. For instance, enter 1 if you wish to omit every other scan from the file (hence compressing it by a factor of 2).
- **Stretching:** Select Stretching to expand the horizontal scale. The Stretching function will calculate the simple average of two adjacent scans (or the specified number of scans) and place the averaged scan in between the existing scans.
- **# of Scans:** Select number of scans to Stack, Skip, or Stretch.

Click Apply to Test the process and, if necessary, click Reset and re-adjust.

Once you are satisfied with the process, click OK.



Distance Normalization



Distance Normalization allows you to establish a constant horizontal scale between marks. This function performs rubber-sheeting so that there is an equal distance, or equal number of scans per unit distance, between markers.

This is generally required when data is collected in continuous mode without a survey wheel and is due to unavoidable inconsistencies in antenna towing speed. When Combo or Distance Marks are set and, Distance Normalization will correct the number of scans between markers by stretching and skipping, thereby correcting for variations in survey speed.

Note: Distance Normalization requires that marker information be stored in the Microsoft Access Database; otherwise it will be inactive.

Note: Distance Normalization will be “grayed out” and inactive if the file is a 3D file or has GPS information.

- Distance Normalization requires a minimum of TWO markers to work.
- Make sure that the markers are correct (no double or missing markers, first and last markers are present, all User markers are converted to Distance or Combo) before running Distance Normalization.
- The Scans per Unit and the Units per Mark must be assigned a value in the input file header in order for Distance Normalization to work.
- Units per Mark are determined at the time of the survey.
- Scans per Unit may be estimated by using the mouse cursor to count the number of scans between markers or by taking the Scans/Sec value in the header and estimating the number of seconds it took to traverse the distance between marker stations.
- Both Scans per Unit and Units per Mark parameters can be changed for the output file.
- Horizontal and vertical units (meters, feet, etc.) must be set before opening the file. Use the Linear Units tab under the Customize command in View menu.

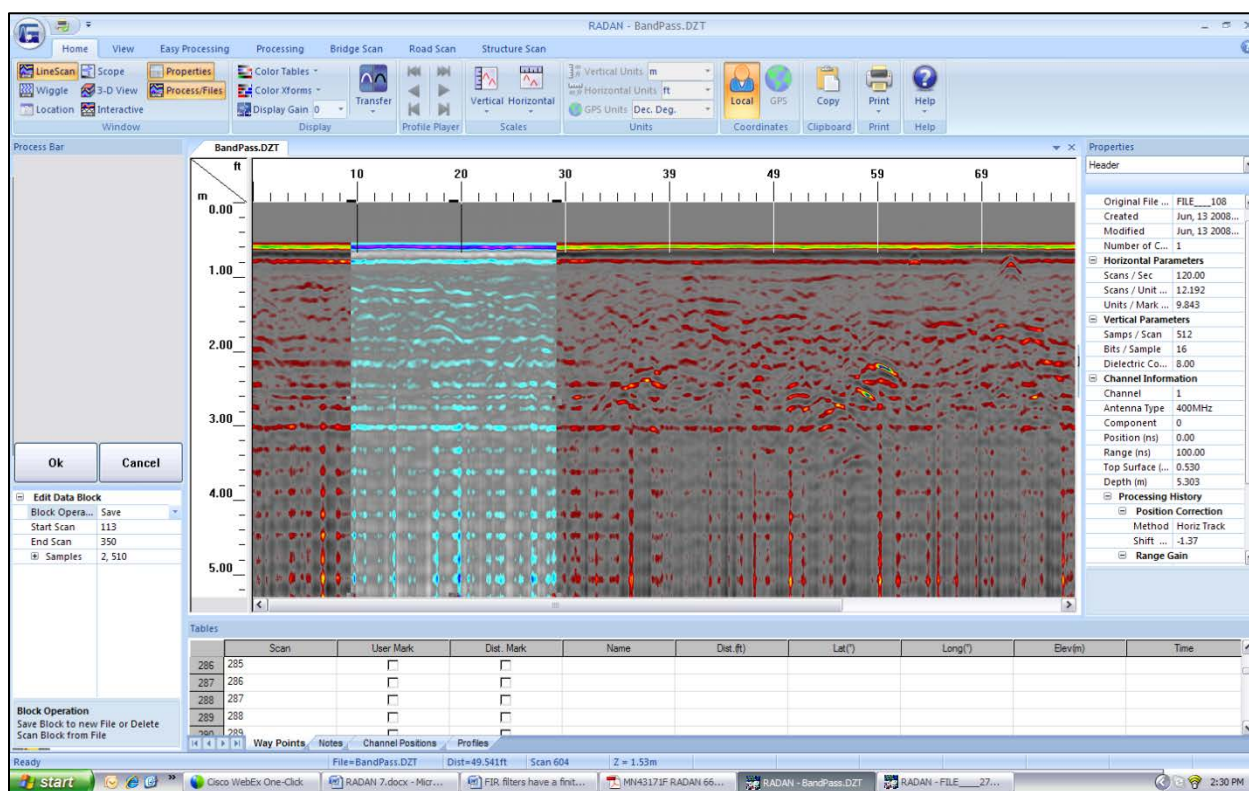
To use, correctly set the scans/unit distance and the units/mark in the Header Information located in the Right Pane. Then click Apply to User Marks, True or False. Click Apply and Reset until you get the desired results. Click Ok to accept the desired reports.

Distance Normalization	
Scans / Unit...	0.000
Units / Mark...	1.000
Apply to Us...	False

Edit Block



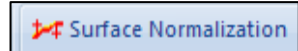
You may save pieces of your data to another file, or delete parts of your data and save the remaining data to another file (the original file never get modified)



- 1 Adjust the Picking tool overlay to highlight the part of the data you want to save or delete. To do this, simply “grab” the handles of the Picking tool overlay and drag the mouse.
- 2 Click Block Operation and select Save to save the block to another file, or select Delete to delete the block from the original data. A new file will be save without the deleted area.
- 3 Click Ok to process, or Cancel to cancel the operation.

Adjust Surface Group

Surface Normalization



Surface Normalization allows you to assign an elevation to the markers in the database and correct for elevation changes. In other words, Surface Normalization adjusts the vertical scale to remove topographic effects, which results in horizontal or near horizontal reflectors appearing as they are in reality. For instance, the water table may appear to have significantly greater relief in radar data than in actuality. Surface Normalization corrects for the topography and displays the water table as a flat reflector.

Vertical Scale: The Vertical Scale may be set to 1:1 (normal view), 1:2, or 1:4. Using the latter two scales compresses the vertical axis (and adjusts the vertical scale accordingly) by factors of 2 and 4, respectively.

This compression of the vertical scale may be necessary if the relative change in elevation for the file is greater than the display range of the monitor being used. If data is not compressed, the file may 'run-off' the top or the bottom of the display window.

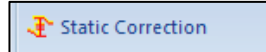
Normal Level: Normal Level can be manually input if the Auto Level option is deactivated. Surface Normalization requires that Combo (or Distance) Markers be used.

Note: Elevation values have to be manually assigned to the markers using Edit Markers. When entered, elevation data will be stored in the Microsoft Access Database file.

Click Apply and Reset and make necessary adjustments. Click Ok when you have your desired results.

Surface Normalization	
Normal Level	2.00
Vertical Scale	1 : 4
Auto Level	True

Static Correction

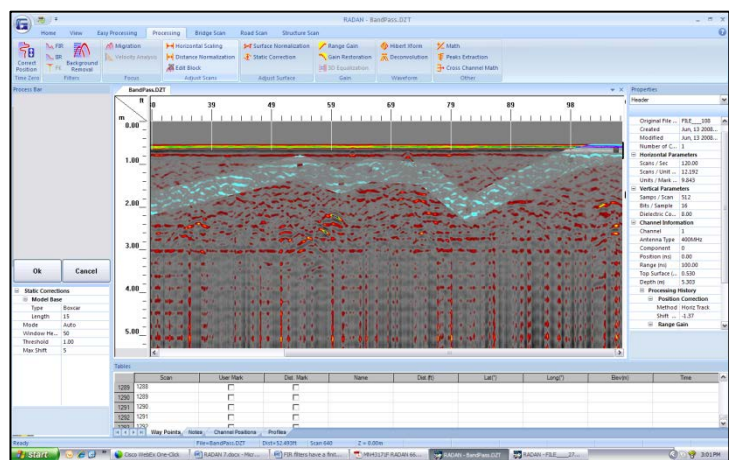


Static Corrections compensates for variations in elevation, phase shifts, and high frequency noise served in the horizontal direction and is generally one of the last processing steps undertaken.

Static Corrections assume that near-horizontal discontinuous due to poor antenna coupling, time zero tracking problems, or some localized changes in the velocity. Sometimes after a lot of processing, once horizontal (or near-horizontal) and continuous layer appear discontinuous and slightly shifted in time, making the reflector difficult to trace from scan to scan. Static Corrections can correct for this.

Static Corrections compensates for noise that is introduced by shifting the reflector within a specified time window (specified by the number of samples) so that it is realigned. Another function of Static Corrections is that it allows you to filter horizontally without influencing the vertical frequency of the data, unlike Horizontal High and Low Pass Filtering. Static Correction proves the horizontal traceability of layers using cross correlation.

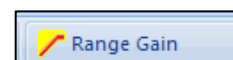
- 1** A rectangular overlay will appear superimposed upon your data. You can use the mouse to shape the rectangle to the desired width and move it so it is superimposed upon the reflector of interest.
- 2** After adjusting the first segment on the reflector, move the cursor with the mouse to the next segment, and click on it. • In this way, create a multi-segmented window that the Static function will use to trace a reflector.
- 3** A dialog box will appear in which you will be able to choose the window height (in number of samples), the filter length, and the type of model (Boxcar or Triangle) you wish to use to filter.



RADAN correlates this model scan of a specified filter length and performs a horizontal boxcar or triangle filter size of the filter length of the number of samples in the window and compares it to the highlighted layer. The correlation threshold is the value used to cross-correlate the model data with the actual data. It is this parameter that tells you how well a layer can be traced from scan to scan. The correlation threshold is usually from 0.5 to 1.0.

Gain Group

Range Gain



There are three Range Gain options available in RADAN: Automatic Gain, Linear Gain and Exponential Gain. Linear and Exponential Gain functions are applied in the Manual Mode and allow the user to manipulate the gain between gain points. The gain value can be entered in the Value box or adjusted by dragging the gain points (small squares) up and down.

When Linear and Exponential Gain functions are selected, gain corrections are applied to the entire data set. These curves will not only amplify the low amplitude signal but also the high amplitude areas.

The Linear function applies a linear gain between gain nodes.

The Exponential function applies an exponential gain curve between gain nodes. The vertical scale of the Exponential Gain function is displayed in terms of decibels (dB) so that it is linear. This is designed to compensate for attenuation.

The Automatic Gain feature attempts to balance the gains over each scan.

A value for the horizontal time constant is required to run the Automatic Gain function. This time constant determines how many scans on the left side of the current scan will be used to shape the automatic gain curve. It applies a weighted filter. A small value giving more weight to scans adjacent to the current scan, a large value giving weight to those further away.

When using both the Automatic Gain and the Manual Gain we recommend setting the number of breakpoints to between 4 and 8.

For the Automatic Gain the overall gain factor is normally set between 2 and 5 and the horizontal time constant is normally set between 11 and 21.

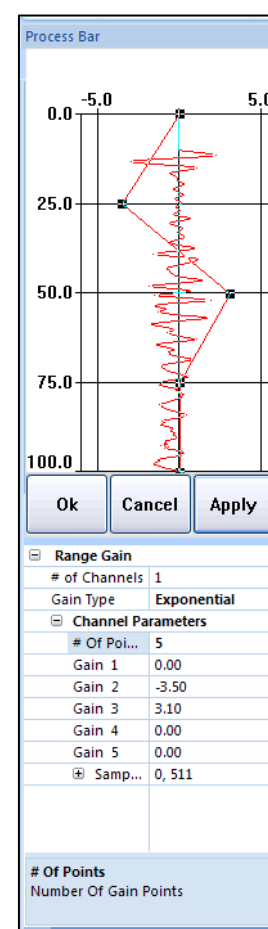
Gain Type

- Automatic
- Linear
- Exponential

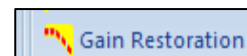
Number of Points: Number of sections the data is divide equally vertically

After the Number of Points is entered, grab the handles of the gain line located in the top pane and adjust the gain.

Click Apply and Reset, and make any necessary adjustments. Click Ok when you have the desired results.



Gain Restoration

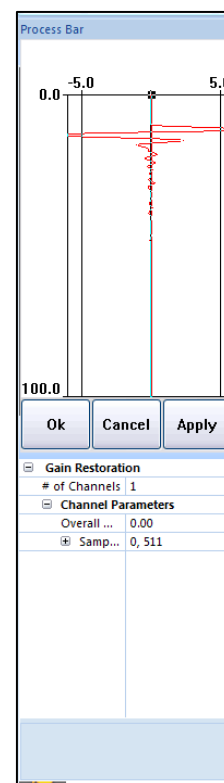


The Restore Gain function removes the gain applied to the data during acquisition. Restoring gain is an important option should you wish to export your data to a forward modeling program, or determine the dielectric permittivity, conductivity, and dispersion (approximate attenuation) of layers. The Restore Gain function uses gain information found in the file header to remove the gain function and normalize the gains.

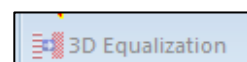
Note: Gain differences exceeding 30 dB may result in a significant loss in data resolution. Saturated data will result in distorted wavelets after Gain Restoration.

- **Overall Gain:** What the overall gain you wish
- **Sample:** Enter Starting and Ending Samples to apply Gain Restoration

Click Apply and Reset, and adjust if necessary. Click Ok when you have your desired results.



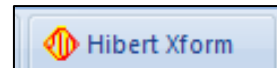
3D Equalization



Gain Equalization is a very basic, automated processing algorithm that looks for differences in the average gain values of data files in a 3D set. It is ideal for correcting data sets that have been improperly gained, or have had gain differences from profile to profile. RADAN will analyze the relative amplitude values of each profile and alter the gains accordingly.

Waveform Group

Hilbert XForm



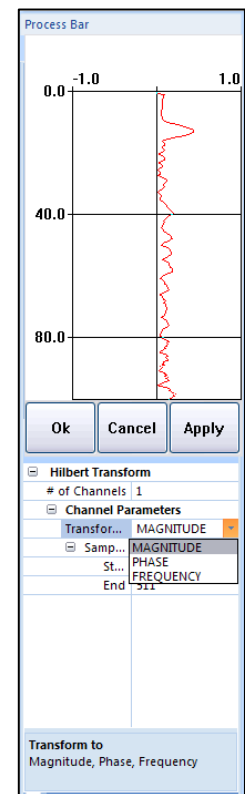
Reflector amplitude and geometry are the primary types of information in GPR data used to make interpretations. The time domain radar data is defined as time and amplitude of the reflected pulses.

Another way of defining the data is to transform it into frequency and phase information is sometimes more sensitive to important subsurface (dielectric) changes in the amplitude or geometric information (e.g., contaminants). A Hilbert Transform, accessed by clicking on the Hilbert icon on the tool bar or by the Processing menu, will change a radar signal represented as a time series into its magnitude (via envelope detect ion instantaneous phase, or instantaneous frequency components (the derivative of phase). The Hilbert Transform expresses the relationship between the phase and magnitude of the signal, or between its real and imaginary parts. It allows the phase of a signal to be reconstructed from its amplitude. The Hilbert Transform is used to display subtle properties of the earth. The magnitude display is useful for indicating the raw energy reflected from an object or layer. The radar wavelet itself may not always be a clear indicator of energy levels because it consists of several cycles. The instantaneous frequency indicates how the earth is filtering the radar signal.

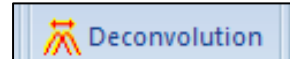
Certain input parameters are required to use the Hilbert Transform. They include:

- Transform To
 - Magnitude
 - Phase
 - Frequency
- Sample Number
 - Starting
 - Ending

Click Apply and Reset, and adjust if necessary. Click Ok when you have your desired results.



Deconvolution



Multiples or “ringing” occur when the radar signal bounces back and forth between an object (such as a piece of metal or layer of wet clay) and the antenna, causing repetitive reflection patterns throughout the data and obscuring information at lower depths. Multiple reflections may also be observed when mapping water bottom, bedrock (or till), or voids. Deconvolution is the filtering method used to remove this type of noise.

RADAN uses a method called Predictive Deconvolution. It is a general technique that includes “spiking” deconvolution as a special case. Predictive deconvolution is aptly named because the method tries to approximate the shape of the transmitted pulse as the antenna is coupled to the ground. Assuming a source wavelet of a specified length, called the operator length, this filter will predict what the data will look like a certain distance away, called the prediction lag, when the source wavelet is subtracted (or deconvolved) from it. This results in the compression of the reflected wavelet. Predictable phenomena, such as antenna ringing and multiples, are moved to distances greater than the prediction lag and are effectively removed from the data.

Deconvolution Parameter Selection: In order for Deconvolution to work properly, certain parameters, such as operator length, prediction lag, prewhitening, gain, start sample, and end sample, must be supplied as input.

Operator Length: The operator length specifies the size of the filter used in terms of the number of samples making up 1 pulse length. Longer filters give a better approximation of the radar wavelet and generally give better results, but take longer to process. A good rule to start with is that the operator length should be about one full cycle of the radar antenna wavelet. A value less than this gives poor results. To remove reverberation, first measure the width of a reverb packet in number of samples. Then set the operator length to that value. Increase the operator length slightly for more effect. Tutorial two in the "Advanced Processing Tutorials" section will detail how to measure the width of a feature and input an appropriate operator length.

Prediction Lag: The Prediction Lag should be set to the desired length of the output pulses (about one-half cycle of the antenna wavelet). Smaller values than this will produce more noise.

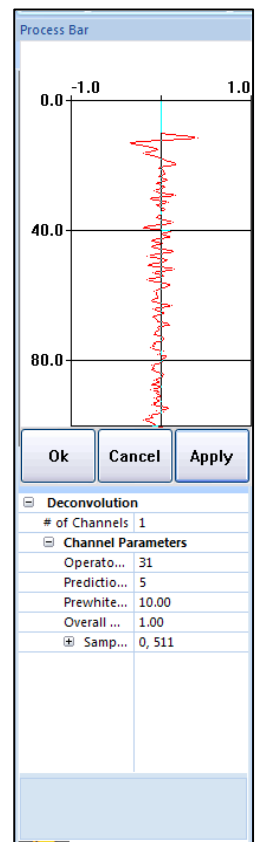
When using Deconvolution to remove multiple reflections, the lag should be equal to or less than the spacing between multiples.

A prediction lag between 5 and 1 is used to approximate “spiking deconvolution.” However, this introduces significantly more noise into the data.

Prewhitening: Prewhitening modifies the autocorrelation function by boosting the white noise (zero delay) component. Mathematically, prewhitening stabilizes the filter, thereby smoothing the output and reducing noise. Values between 0.1 and 1 percent are common, 0.8 is a good value to start with.

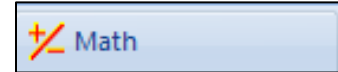
Additional Gain: Additional Gain may be needed because the deconvolution process attenuates the signal, especially when the prediction lag is short. Gain values of 3 to 5 are common, but use whatever achieves an amplitude level equal to the original data.

Start/End Sample: The starting sample and ending sample should be set to establish the “time gate,” specified in terms of sample number, in which the Deconvolution filter is active. For instance, a start sample and end sample of 256 and 1024 respectively may be used to remove multiples beneath a reflector located at sample number 240 in a 1024 Samples/Scan data set.



Other Group

Math



Simple arithmetic functions may be performed on the data by using the Math Functions feature of RADAN. It allows you to add or subtract another file, add or multiply by another constant, or express data in terms of its absolute value, square root, integrate the data. When the math functions log and square root are applied to GPR data, they will emphasize the low amplitude data relative to the high amplitude data.

Note: When using Log or Square Root functions, data amplitudes must be expressed as positive numbers (i.e., data cannot be zero or negative). Otherwise, an error will occur resulting from an improper mathematical procedure. Therefore, you must first take the absolute value of your data.

Using the Exponential or Squared function, you can emphasize strong (high amplitude) signals and reduce or remove weak (low amplitude) signals or noise. All operations are performed on the binary 16-bit data, therefore the constants used will always be in the 32767 to -32767 range and have an integer value.

These operations are performed in the following order:

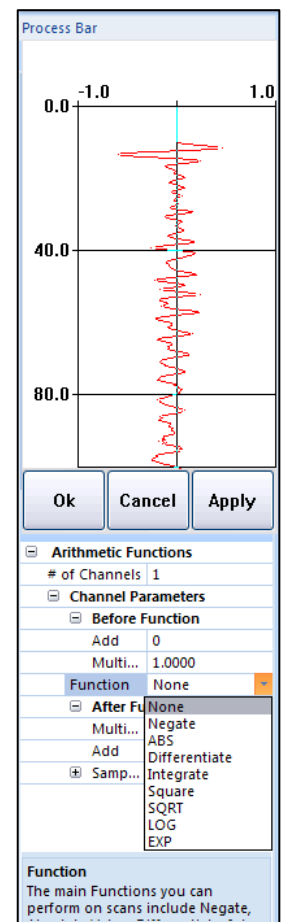
- Add Constant
- Multiply Constant
- Function
- Multiply 2nd Constant
- Add 2nd Constant
- Truncate to Range (-32768 to 32767)

It is possible to Add and Multiply by a constant before and/or after a nonlinear function.

When designing an operation, be aware that the original data is only 8 bits, it will range from -128 to 127 before the math operation. After the operation, the output file will still be 8 bits, any overflows as a result of the operation will be clipped.

Some operations, such as square root or log, will not operate on zero or negative numbers. However they can operate on data by first taking the absolute value of the amplitude. Therefore, the Log, Exponent, Square, or Square Root functions may need to be combined with constant multiplication and addition in order to obtain output within the proper range. This is why multiplication and addition are allowed before (1st add constant and 1st multiply constant, respectively) and after (2nd add constant and 2nd multiply constant, respectively) a function. The Arithmetic Function option allows you to select from a number of linear and nonlinear functions as follows:

- Negate: $y(t) = -x(t)$
- Absolute Value: $y(t) = |x(t)|$
- Differentiate: $y(t) = x(t) - x(t-1)$
- Integrate: $y(t) = x(t) + x(t-1) + x(t-2) + \dots$
- Square: $y(t) = (x(t))^2$
- Square Root: $y(t) = (x(t))^{1/2}$



- Log: $y(t) = \ln [x(t)]$, where $\ln[x(t)]$ is the natural logarithm
- Exponent(Exp): $y(t) = \text{Exp} [x(t)]$ or $e^{[x(t)]}$

Peaks Extraction

Peak Extraction function will automatically trace continuous reflections in the data window of the active file according to the user specified selection parameters.

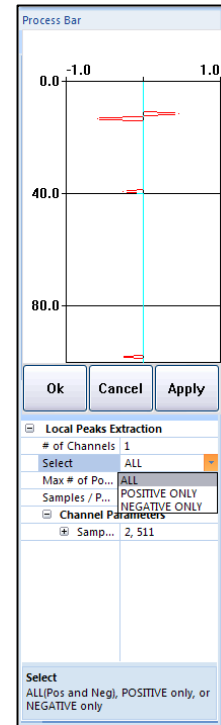
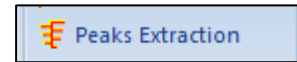
The Peaks Extraction Parameters dialog box has the following entries:

Select: Defines if the function will trace ALL, only the POSITIVE or only the NEGATIVE reflections.

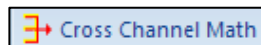
Max # of Points: Defines the number of peaks to trace. The indicated number of strong reflections will be displayed. The reflections may vary along the profile, but their total number in any trace will not exceed the Max #.

Samples/Point: Defines the width of the peaks in the vertical directions. The peaks are displayed as bands of the indicated width, with the amplitude values preserved with

Start and End Samples: By default the entire trace is processed, but the user may choose any horizontal layer, expressed in sample numbers. Only the selected interval will be processed, the rest of the file being left unchanged.



Cross Channel Math



The MC Combo function creates combinations of channels and assigns them to a selected channel in a multi-channel file. A selection of arithmetic operations is available for this operation.

For a single-channel file, Combo function means an arithmetic operation performed on that channel. The Compensate Original Gain option allows you to restore the range gain.

The following list of functions is available:

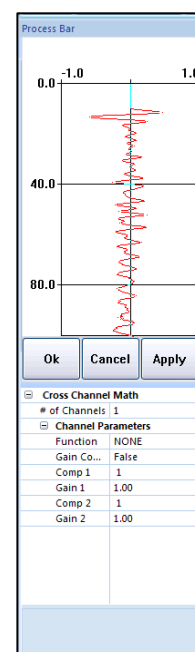
- NONE - no transformation
- ADD - adds the channel
- SUB - subtracts the channel
- ADD3 - add 3 numbers
- SQADD - square root of the absolute value of the sum of the channels
- SQSUB - square root of the absolute value of the difference of the channels
- SQDIV - square root of the absolute value of the division of the channels
- SQMUL - square root of the absolute value of the product of the channels
- VSUMM - square root of the sum of the squares of the two channels
- VSUMM3 - square root of the sum of the squares of the three channels

The Apply button enables the user to see the results of the Combo operation without being prompted to save the file.

The button label changes to Reset; clicking it without changing parameters will redisplay the original data.

If different parameters are selected, clicking Reset will display the results of the new extraction.

Click Reset again, and the original data will be redisplayed. When you're satisfied with the results, click OK to save the processed file.



StructureScan Ribbon



Processing

Structure

Select this option to automatically process the data as follows (same steps as you will see in the Easy Processing Ribbon):

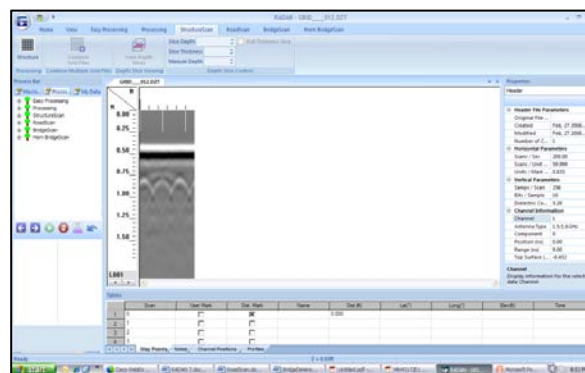
- Time Correction
- Background Removal
- Migration

Combine Grid Files

This option will allow you to combine multiple processed grids into one 'Super Grid.'

View Depth Slices

View a top-down view of your data.



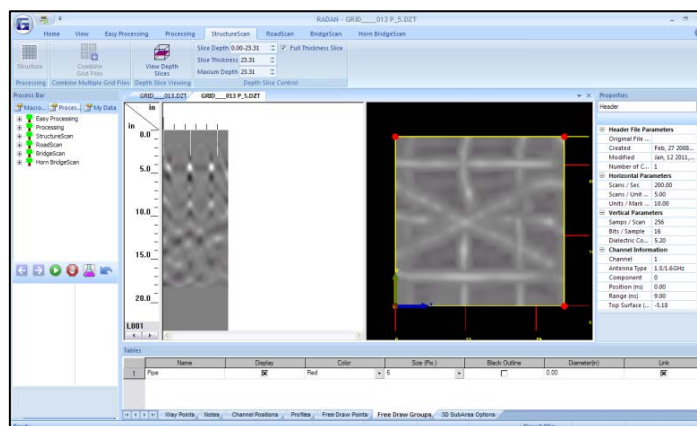
Depth Slice Control

Slice Depth: Adjust the depth of the slice for viewing.

Slice Thickness: Adjust the thickness of the slice.

Maximum Depth: Adjust the maximum depth of your data.

Full Thickness Slice: Check this to see full view from top to bottom, uncheck to be able to view data in slices.



RoadScan Ribbon



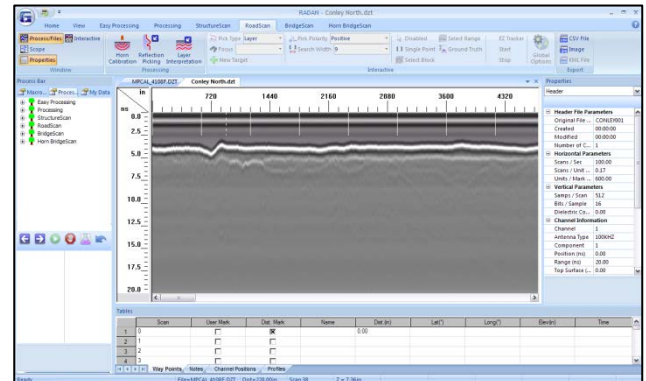
Window Group

Process Files: Toggle on/off the Process Bar/Macro Bar/My Data Window Pane.

Scope: Toggle on/off the O-Scope pane.

Properties: Toggle on/off the Properties Window Pane.

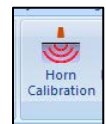
Interactive: Switch to Interactive View for target (rebar) picking.



Processing Group

Horn Calibration

After clicking the Horn Calibration File icon, there are a few things we need to do here. The information shown on the dialog is stored with the processed data file header and is accessed when the calibration file is used in the reflection picking tool. The parameters selected in the dialog determine how the data file is processed in reflection picking.



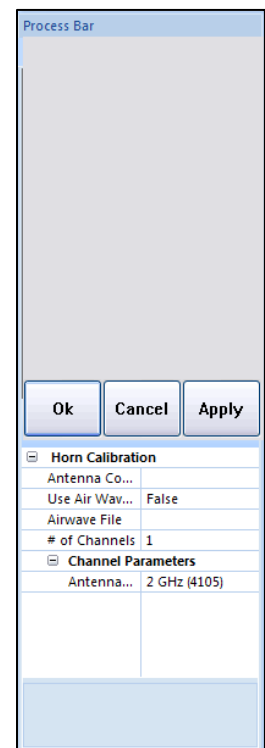
Each one of the parameters in the Calibration File Generation Dialog is discussed below.

Antenna Config: This is the antenna setup for data collection. RSA Profile must be selected in the Antenna Configuration dialog box. The other option “RadBridge Project” is used with the Bridge Assessment module.

Use Air Wave File: The air wave file is optional and normally not used. Select True if an air wave file has been collected. Then enter the air wave filename in the space to the right. The airwave file is most commonly collected by turning the antennas upside down so that they radiate upward. This file is obtained using the same SIR-10/20 settings as the metal plate calibration file.

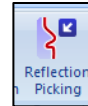
Antenna Type: Select the antenna used for data collection. There is a choice of the Model 4205, Model 4208, Model 4108, Model 4105HR or Model 4108HR antennas. The “HR” options implement an algorithm designed to improve thickness measurement accuracy for pavements 6” (10 cm) or less in thickness. Choose 4105HR for the 2 GHz and 4108HR for the 1 GHz.

Once all of the parameters are set in the Calibration Menu, click OK to begin execution. It is a good idea to save the .CZT file with an easily remembered name to be able to quickly identify it during processing. RADAN also has a special folder called Horn Cal. If you like, you can save your CZT file to that folder. When you run the Reflection Picking process in the next step, RADAN will ask if you would like to search for a CZT file in this folder. It may help to keep things organized.



Reflection Picking

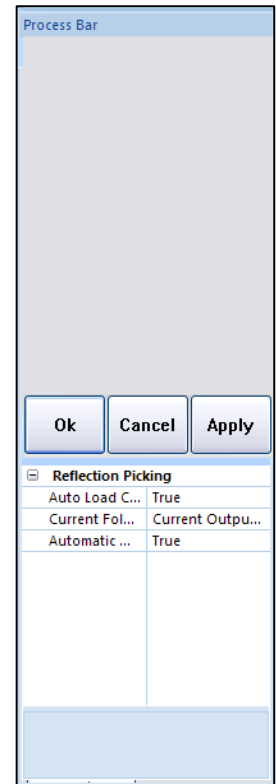
Before beginning your survey processing, you must have created an antenna calibration file.



- 1** Open the survey data file and add display gain if necessary. Display gain can be added by right-clicking on the actual GPR data and selecting a gain value from the display gain pop-up window. This will not permanently alter your data, it will simply make things easier to see.
- 2** Click the Reflection Picking icon. This will:
 - Eliminates unwanted signals from the radar data, so the signals from the pavement layers are clearer and easier to track.
 - Tracks the reflections from the bottoms of the various layers comprising the pavement.
 - Measures the return times of the reflections and calculates the two-way travel time for each of the pavement layers from the data.
- 3** After clicking the icon, the process will prompt you to specify an antenna calibration file.

Auto Load Calibration File: RADAN will search for a CZT file in the Current folder selection. This selection will be either the current output folder you selected in Global Parameters of the Properties Pane, or the Horn Cal folder in the RADAN program directory. If RADAN finds a CZT file, you will see it displayed in the Calibration File window. If you have more than one CZT in the particular, RADAN will select the first one in alphanumeric order.

Check to make sure that the calibration file displayed is correct and then most users can just click Apply (and Reset if necessary). Click OK once you have the desired results. The automatic mode of the processing will work just fine for most people. For advanced users looking to have more control over this process, change Automatic Mode to False.



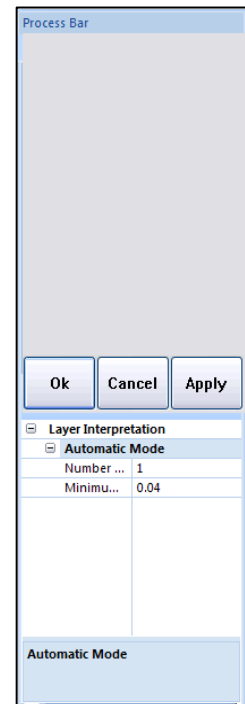
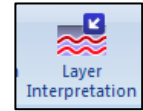
Layer Interpretation

The Layer Interpretation option is designed to help automatically track layers in road structures that exhibit clear and consistent layer reflections. The output is identical to the manually derived output of EZ Tracker, but under certain controlled conditions LI can bypass that manual step. However, even under the most ideal conditions, the output from LI may still require significant manual editing. The Layer Interpretation Process is accessed by clicking the button on the toolbar or by selecting Horn Layer Interpretation in the Infrastructure submenu under Process Menu.

Note: Layer Interpretation can only be performed on files that have been output from the Reflection Picking process.

Enter the Number of Layers you would like to process, and enter the minimum depth of the first layer.

Simply click Apply/Reset and adjust as necessary. When the desired results are displayed, click Apply. In the Automatic Mode, just click finish and LI will attempt to locate the specified number of layers. The minimum depth allows you to set the search range outside of the ground surface reflection. This is to prevent LI from picking the ground surface reflection as a layer by mistake. Automatic Mode works very well for clearly visible single layers. It is especially appropriate for picking overall pavement thickness.



Interactive Group

After you process the Road data, click on Interactive to begin picking your targets. In this view, you can begin inserting and deleting picks to create the ASCII comma delimited file. The Interactive Pane in the RoadScan Ribbon gives you many options to do this.



Pick Type: Select whether you will be picking Layers or Targets.

Focus: Select the type of layer or target you will be picking.

New Target: Add more targets to pick.

Pick Polarity: Select whether the picks are Positive, Negative, Absolute, or None.

Search Width: Enter the number of pixels your cursor will be when adding or deleting picks.

The layers or targets to which all edit operations currently apply. The choice is any of the layers defined in the *.CSV file (1, 2, etc.), All, Targets, or Filtered.

When a layer number is selected, picks in that layer can be added or deleted.

The All and Filtered options only allow the user to delete picks.

The Targets option allows the user to add single independent picks to mark objects such as pipes, culverts, voids, or events such as road intersections, starting and ending points of new pavement, etc.

Filtered appears as an option when the filtering option is enabled (Layer Tab of the Tables Pane).

Picking Tools

These are the modes in which the picks are selected.

Single Point

Layer picks are added each time the user presses the left mouse button and deleted each time the user presses the right mouse button. Only one pick can be added at a time with this option, but multiple picks can be deleted depending on the Search Width.

Adding Picks in Single Point Mode

- 1** Position the mouse cursor over the area where you would like a pick to be added.
- 2** Click the left mouse button. A search will be performed on all of the scans between the left and right inside edges of the mouse cursor to locate the maximum amplitude. If the search is successful, a pick will appear on the data. The type of search used to locate the new pick can be customized in the Layer Options menu choice (Layer Tab of the Tables Pane).
 - A pick will only be successfully added if a reflection can be located over the cursor search width.
 - If no pick is added after pressing the left mouse button, reposition the mouse cursor and click the left mouse button again.
 - For cases where the reflection peak is reversed polarity, the user would first need to select Neg. Peak for the Layer Properties under Layer Options in the Interactive Interpretation main menu, then place the mouse cursor over a negative polarity reflection.

Deleting Picks in Single Point Mode

- 1** To delete poor picks, a procedure similar to adding picks is followed. Position the mouse cursor so that the poor pick is located within the search area of the cursor
 - Click the RIGHT mouse button in Delete Picks mode
 - Click the LEFT mouse button in Add Picks mode
- 2** Then, click the right mouse button. If a portion of the pick circle bisects the vertical midpoint of the cursor, then the pick will be deleted from the database and from the screen.

Select Block and Select Range

The Select Block and Select Range picking tools are designed to operate over a large number of scans.

Select Block

When Select Block is activated, a multi-colored, translucent square will appear over the data when the user clicks the left mouse button. The select block contains tiny squares on each face and corner. These squares act as handles that can be used to resize the select block.

To Resize Select Block

- 1** Position the mouse cursor over one of the handles and press the left mouse button.
- 2** Move the mouse cursor to the desired location with the left mouse button still pressed down.
- 3** When the desired size is reached, simply release the left mouse button.

To Move Select Block

- 1** Place the mouse cursor within the select block and press the left mouse button down.
- 2** Move the mouse cursor with the left button still pressed down until the block is in the desired position. Finally, release the left mouse button.

Select Range

When Select Range is activated, a multicolored translucent overlay appears over the data, extending from the beginning to the end of the file. It operates similarly to the Select Block, except that all operations performed using the Select Range picking tool are performed within the time interval (slice width) of the selected area on *all* of the scans in the file.

The slice width can be changed by clicking the left mouse button on one of the handles (located at the top and bottom at the horizontal midpoint of the slice) and with the left mouse button still pressed, dragging the handle to the desired location.

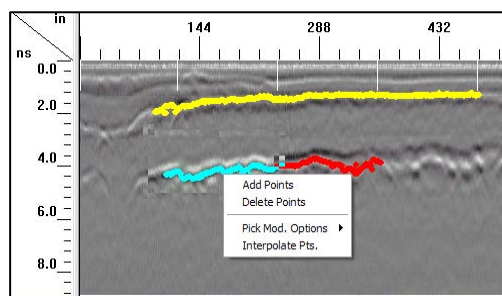
Adding and deleting points are performed within the Select Block and Select Range areas. These options are activated by pressing the right mouse button down while the mouse cursor is positioned within the multicolored Select Block or Select Range area. This action activates a pop-up menu

Add Points: Clicking on Add Points will activate the program to begin a smart search for reflection peaks within the selected region. Circles will overlay the data where reflection peaks are identified by the search algorithm

Delete Points: Clicking on Delete Points will activate the program to start deleting the picks of the Current Layer located within the selected region.

Layer Modification Options:

- **Change Velocity** changes the velocity of the currently selected layer points located in the selected region. Clicking on Change Velocity opens a dialog box for entering the desired velocity. The new velocity is used for all of the currently selected layer picks in the selected region.
- **Change Pick ID** allows the user to change the layer number assigned to the picks located within the selected region. For example, the user desires to change the layer # of a group of layer 3 picks to layer 2. The user must select layer 3 as the Current Layer, position the Select Block (or Select Range) over the group of points and click the right mouse button to access the pick modification options shown in. Clicking on Change Pick ID will open a dialog box for entering the desired layer number. After changing the layer number, the color of the selected picks will change to the color of the new layer.



Note: A selected point is not changed to the new layer if a pick from the new layer is already present in the scan, or if the new layer overlaps another layer pick (i.e., changing a layer 3 pick to a layer 1 pick with a layer 2 pick present in the scan).

Ground Truth

Selecting the Ground Truth menu item from the Interactive Interpretation main menu (see next section) opens a spreadsheet that allows the user to enter ground truth data that can automatically be used to calculate layer velocities.



Ground truth data can be entered on the left half of the spreadsheet and the calculated velocity and nearest scan location are automatically updated and shown in the right side of the spreadsheet.

For example, let's say a core was drilled 104.5 m from the starting location of the file and the measured depths of the layer interfaces were 12 cm, 22 cm, and 33 cm. Each of the first three columns in the Ground Truth spreadsheet should be filled with this information.

The 5th, 6th and 7th columns of the spreadsheet were automatically calculated by RADAN. The Data X shows the location of the nearest scan containing the layer pick, Time is its arrival time, and V is the calculated velocity for the layer. If no picks are located within 20 scans of the core location, then zeros will appear in the last 3 columns.

Once core data have been entered, the spreadsheet can be closed or left open. When the user saves the data as an ASCII *.lay file, the core data are stored in a separate ASCII file with the same name as the *.dzt file and the extension .gtr. Once this file has been created it will automatically open and its contents read by RADAN the next time the user opens the ASCII file providing it is located in the same directory as the ASCII file. Alternatively, if the user saves the pick data in a *.bii file, all of the ground truth information will be stored with the picks in the *.bii file.

The velocities calculated from core data can be used to calculate layer depths. To change the velocities of existing picks, use the select block or select range picking tool and move the selected region over the picks that need to be changed. Then, click the right mouse button to bring up the pick options and select the Change Pick Velocity option. Proceed to change the selected pick velocities as directed in the dialog.

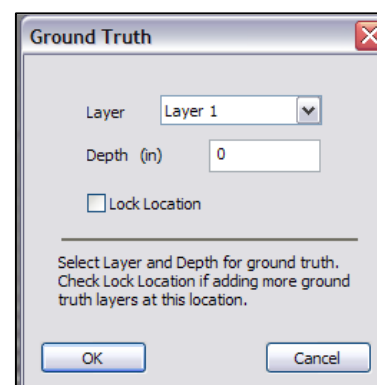
To specify all new layer picks to have velocities calculated from the ground truth data, the velocity option must be changed from the Layer Properties Dialog page accessed through the Layer Options menu item.

When ground truth data from more than one core are entered in the Ground Truth spreadsheet and the user selects core data as the velocity option for picks, the velocity used for each pick is obtained from the nearest core.

Any changes in core data are automatically updated in the picks that have core data calculated velocities.

Ground Truth: Enter Ground Truth Information.

- 1** Click in the data the spot where the ground truth was collected.
- 2** Layer: Select Layer for ground truth.
- 3** Depth (): Enter Depth information.
- 4** Lock Location: Input a check mark if adding more ground truth layers at this location.
- 5** Click OK or Cancel.



Interpolate Points: Will interpolate layer picks (add new picks between existing ones) using the interpolation method (Linear or Nearest Neighbor) specified in the Properties dialog under Layer Options.

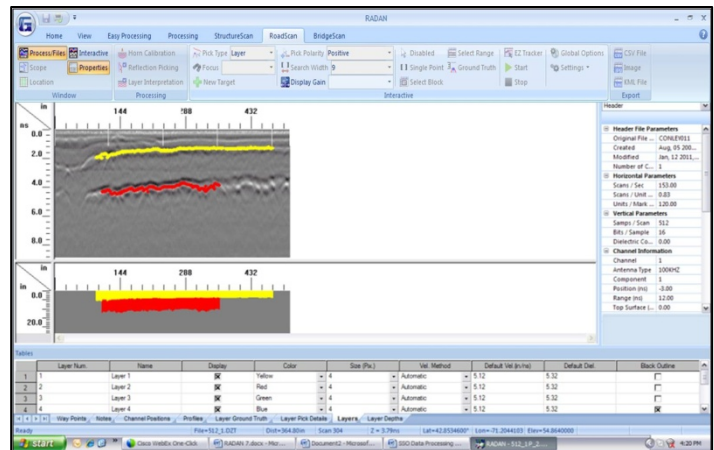
EZ Tracker

EZ Tracker will attempt to trace a layer by following similar conditions in successive scans. You should note the desired layer's phase information and select Positive, Negative, or Both under layer options.

To use EZ Tracker, make sure that the current focus is set to the layer that you want to track and click Start. Then click on the layer. Move the cursor further down the layer and click again. EZ Tracker will attempt to trace that layer between the two points. A closer click spacing will produce better results. If you make a mistake, you can right click to undo the previous section. You should click Save every few picks to save changes. If you come to a gap in the layer, click Stop and then Start to restart the layer at the other side of the gap.

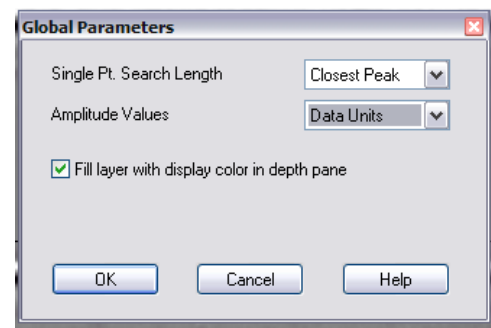
Start by clicking on EZ Tracker. Then click Start. Click, in the Data Area, the beginning of a layer. Then at small intervals, click along the layer. The software will fill the layer in between the dots.

If you wish to begin at a new area of the layer, simply click Stop, then Start again. You may Right Click the mouse to delete the previous click.



Global Options

- **Single Pt Search Length:** When using Single Point Picking Tool, to enter a "Dot" at the closest peak or use the cursor length.
- **Amplitude Values:** Use Data Unit, dB, or Normalized dB.
- Check to fill layer with display color in depth pane.



Save Settings

- Load Default Settings for Interactive Options
- Save as Default Settings will save your Interactive Options as default.

Export Group



You may export your picks to a comma delimited ASCII (CSV) file, save your image to a JPG file, or export the layer information to a KML file (if GPS was used with your data) for Google Earth.

CSV File

Click CSV File to export your picks to a comma delimited file. You may select pick what fields you would like to export to the CSV file. You may:

- **Use Existing:** Use the existing custom fields what you created in Create Custom.
- **Create Custom:** You may pick and choose which fields you want.
- **Modify Existing:** Modify an existing field list to create a new one.

Use Existing: Select an existing customize field list you created from Create Custom or the GSSI standard.

Create Custom: Select from Categories and a list of fields from each of the Categories you want for your CSV file. You should give this Category a user-friendly name for future use.

Modify Existing: Modify an existing category to create a new category.

Image

Save the image as a JPG file.

KML File

Export each layer to a KML file for Google Earth.

A KML file is a file type used to display geographic data in an earth browser, such as Google Earth and Google Maps. KML files have a tag-based structure with names and attributes used for specific display purposes. In RADAN you can export certain kinds of information as a KML file and easily view that information in Google Earth. This section discusses exporting layers identified in Interactive Interpretation.

Note: In order to create a KML file, you must have collected the data with GPS.

Layer: Selects the Layer that you want to export. If you have multiple layers, you must export them one at a time.

Channel: Selects the data channel to export. You can export different channels if you have collected a multi-channel file.

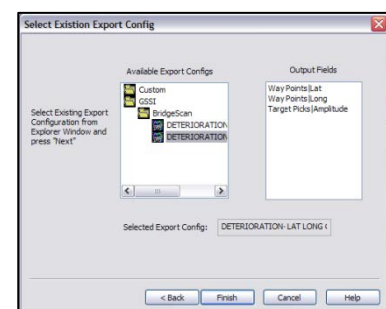
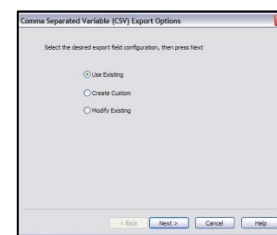
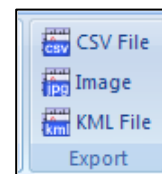
Output Parameter: This can be pick depth, amplitude, velocity, or time depending on the type of data that you want to depict.

Distance Placemarks/Distance Interval: This option outputs a notation at a prescribed distance.

Line width: The KML output line width in pixels.

Minimum/Maximum: This sets the range of output values specified in the Output Parameter option.

Number of Intervals: The number of evenly spaced breakpoints for the output data.



Range/Colors: The range will be set by the number of output intervals. You can decide to code different intervals with colors.

The output of this is a KML file with the same root name as the ASCII pick LAY file. To import into Google Earth, simply drag this file onto Planet Earth and it will zoom to the given location of the data.

Layer Options

In the Layers Tab located in the Table Pane, you can define each layer, how they are displayed and how they behave.

Tables										
	Layer Num	Name	Display	Color	Size (Pic)	Vel. Method	Default Vel (in/ft)	Default Vel	Default Diel	Black Outline
1	1	Layer 1	☒	Yellow	4	Default Vel	5.12	5.32		☐
2	2	Layer 2	☒	Red	4	Default Vel	5.12	5.32		☒
3	3	Layer 3	☒	Green	4	Default Vel	5.12	5.32		☒
4	4	Layer 4	☒	Blue	4	Default Vel	5.12	5.32		☒
5	5	Layer 5	☒	Brown	4	Default Vel	5.12	5.32		☒
6	6	Layer 6	☒	Black	4	Default Vel	5.12	5.32		☒
7	7	Layer 7	☒	Black	4	Default Vel	5.12	5.32		☒

Name: The Layer Names page permits the user to specify a name for each layer. For example, for a typical road structure, the user may want to rename Layer 1 as Asphalt, Layer 2 as Base and Layer 3 as Subbase. Layer name changes are shown in all dialogs and will be output to the ASCII file when the user performs the next ASCII file save

Display: Display the layer or not.

Color: Pick the color for the layer.

Size: Select the pixel size of each pick.

Velocity Method: Sets the propagation velocity used to calculate the thickness of each layer. If the Auto box is checked, the velocity will be automatically calculated for each radar scan. The other options are: (1) User-specified, (2) Core Data, and (3) Velocity Analysis File. The Velocity analysis option is most commonly used for geological investigations where a common-midpoint file has been collected and processed using the Velocity Analysis Option in RADAN. The output file from the Velocity Analysis must be entered in the Global page before this option is activated here.

Default Vel (ins/ns): The velocity of RADAR at the interface of that layer.

Default Diel: The dielectric of the layer.

Black Outline: Toggle On/Off to have a black outline for each pick.

BridgeScan Ribbon



The Bridge Scan Ribbon is designed to process data specifically collected to determine bridge deterioration.

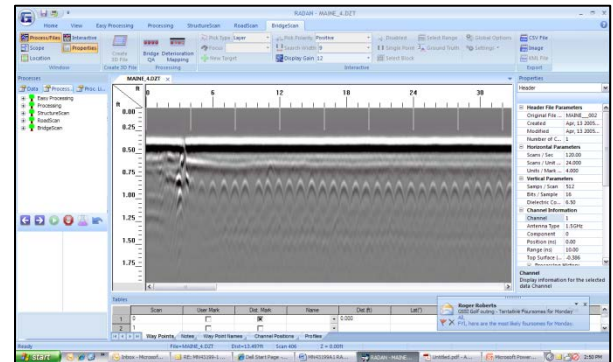
Window Group

Process Files: Toggle on/off the Process Bar/Macro Bar/My Data Window Pane.

Scope: Toggle on/off the O-Scope pane.

Properties: Toggle on/off the Properties Window Pane.

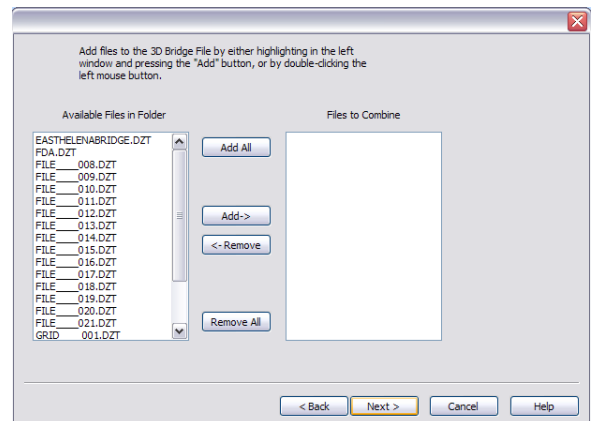
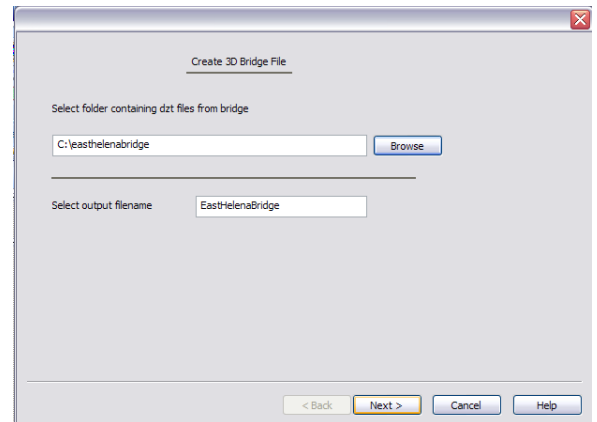
Interactive: Switch to Interactive View for target (rebar) picking.



Create 3D File

This will guide you to combine the files you have collected, process the files, and prepare the newly created “bridge” file for further processing.

- 1 Click on Create 3D File.
- 2 Browse to the appropriate folder where the bridge files that were collected are located.
- 3 Enter an appropriate name of the newly created Bridge File.
- 4 Click Next.
- 5 Add the bridge files from the left pane to the right pane by:
 - Double-clicking on the File, or
 - Click on the File and click Add->, or
 - If Appropriate, click Add All, or
 - Highlight a group of files and click Add->
- 6 You may remove files from the right pane by doing the same procedures in the right pane.



7 Click Next.

8 Input:

- Length of the bridge.
- The starting distance of the FIRST line collected from the Curb.
- The spacing between each line.

9 Click on the down arrow and select how the data was collected.

10 Click Next.

11 Column Information

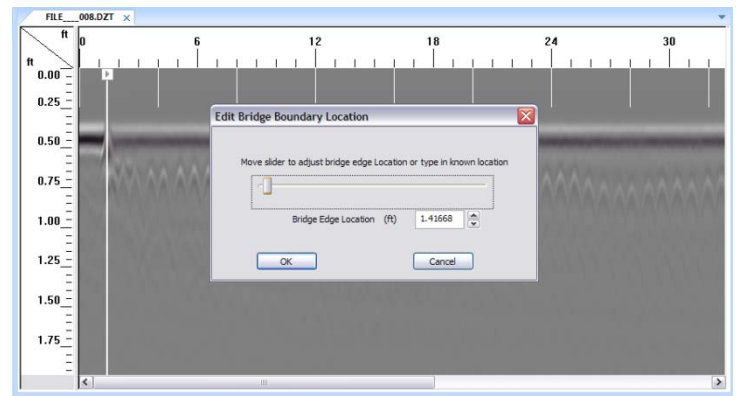
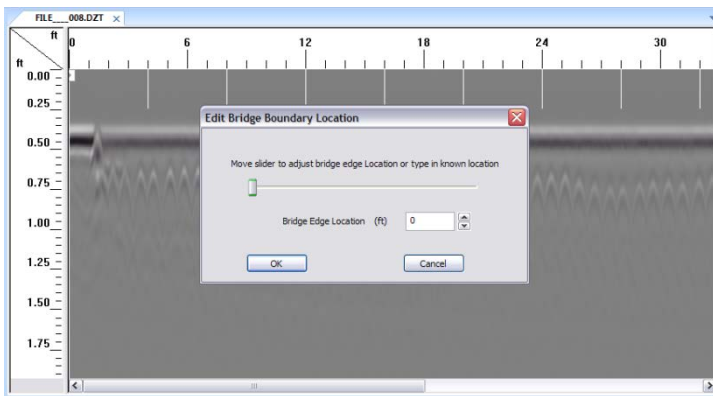
- **Filename:** The name of each file collected.
- **Across Dist.:** The distance from the curb of each of the files collected.
- **Rev. Dir.:** If you collected the files in a zig-zag way, an “X” will automatically be entered that the file will need to be reversed.
- **St. Loc:** This is the starting location of the file
- **Edit St. Loc.:** This will allow you to correct the starting location of the file.
- **End Loc.:** This is the ending location of the file.
- **Edit End Loc.:** This will allow you to correct the ending location of the file.

12 Edit the Starting and Ending Locations of each file.

- Click Edit St. Loc. And Edit End Loc. for EACH file, one at a time.
- Move the slider bar until the vertical line on the data matches the beginning of the bridge, more than likely the middle of the bridge joint. You may use the left and right arrow keys on your keyboard to make minor adjustments, as well as clicking on the up and down arrows on the screen
- Click OK.
- Repeat this process for the Starting and Ending locations for ALL files.

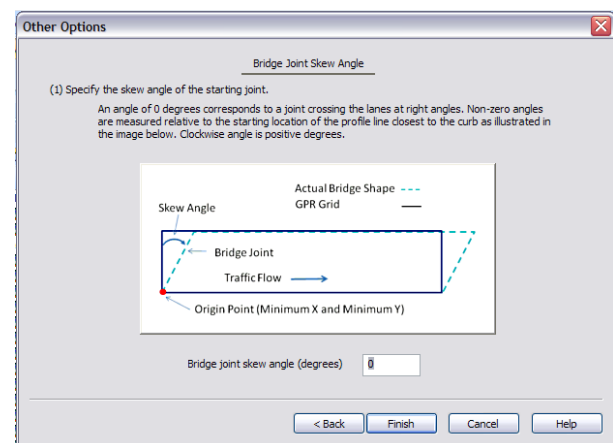
13 Click Next.

	Filename	Across Dist.	Rev. Dir.	St. Loc.	Edit St. Loc.	End Loc.	Edit End Loc.
1	FILE_008.1	2.00		1.416678	Edit St. Loc.	57.875465	Edit End Loc.
2	FILE_009.1	4.00	☒	1.500012	Edit St. Loc.	57.875465	Edit End Loc.
3	FILE_010.1	6.00		0.833340	Edit St. Loc.	57.333794	Edit End Loc.
4	FILE_011.1	8.00	☒	2.041683	Edit St. Loc.	58.500470	Edit End Loc.
5	FILE_012.1	10.00		2.375019	Edit St. Loc.	58.833806	Edit End Loc.
6	FILE_013.1	12.00	☒	1.791681	Edit St. Loc.	58.333802	Edit End Loc.
7	FILE_014.1	14.00		1.250010	Edit St. Loc.	57.792131	Edit End Loc.
8	FILE_015.1	16.00	☒	2.416686	Edit St. Loc.	59.000474	Edit End Loc.
9	FILE_016.1	18.00		2.041683	Edit St. Loc.	58.542137	Edit End Loc.
10	FILE_017.1	20.00	☒	1.083342	Edit St. Loc.	57.625463	Edit End Loc.
11	FILE_018.1	22.00		1.583346	Edit St. Loc.	58.125467	Edit End Loc.
12	FILE_019.1	24.00	☒	2.833356	Edit St. Loc.	59.375477	Edit End Loc.

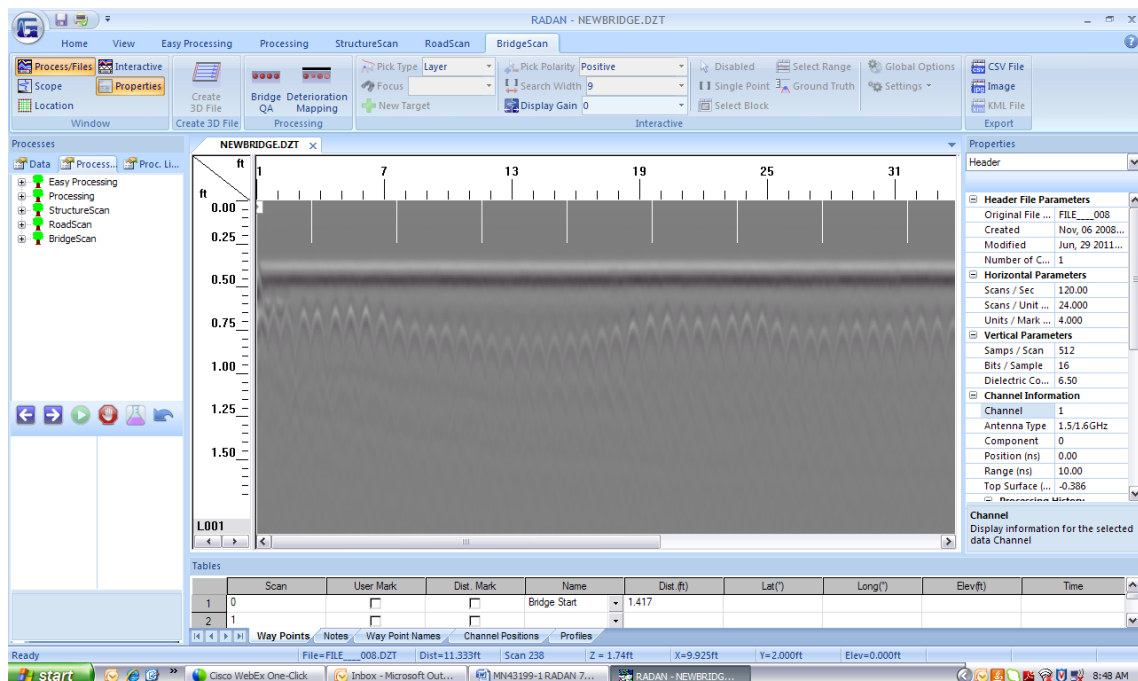


- 14** Enter the Skew Angle of the bridge. A positive number if the bridge angled to the right, and negative number if the bridge angled to the left.

- 15** Click Finish.



The system will then process the files and combine them to create one “Bridge” file. The name of the file is what you named it on the first screen.



Bridge QA

This section describes data collection for quality assurance (QA) studies on new concrete cover bridge decks. This technique is not applicable to asphalt overlaid decks. The goal of this application is to ensure that there is adequate concrete cover over the top layer of reinforcing steel.

Unlike the Bridge Deterioration Mapping application, a Bridge QA project does not require a 3D collection. A single representative profile per travel lane is all that is needed. Typically, this profile is along the wheel path of the lane opposed to the shoulder. There is one important caveat to this: the data must be collected perpendicular to the trend of the top layer of rebar. This means that if the rebar are longitudinal, then you must take profiles across the bridge, not along it.

If a complete map of cover thickness is required, then a 3D file can be collected following the instructions given in your SIR System manual. Be sure to configure the data collection settings (Gain, Position, Range, Scan and Sample density) as they are described in this guide. Also be sure to orient profile lines perpendicular to the trend of the top bars.

Velocity Calculation: The first decision is how the radar wave velocity will be calculated. The velocity is a critical parameter because it directly influences the calculated rebar depths.

The only recommended method is to use calibration hole data obtained from measuring the depth to one of the rebar that the antenna passed directly over during data collection.

To implement calibration hole data, select the Use Calibration Hole Data option, then enter the calibration hole location relative to the start of the profile line and the measured rebar depth.

If the decision was made not to use calibration hole data, then the user should select Specify Concrete Velocity (if it is not already selected) and then enter the estimated or previously known concrete velocity.

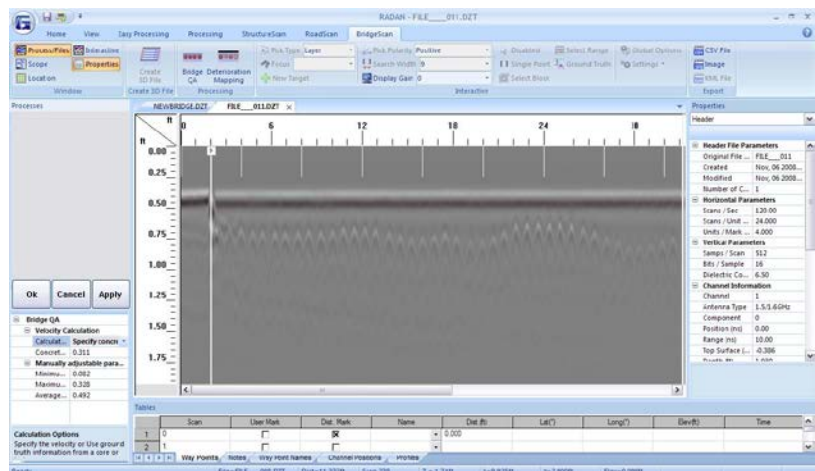
The default velocity is 3.74 inches/nanosecond (9.5 cm/ns).

Manually Adjustable Parameters: The most common use for the Bridge QA module has been on newly constructed concrete bridge decks containing rebar between 1-4 inches (2.5-10 cm) in depth and at a spacing of 6 inches (15 cm) between rebar.

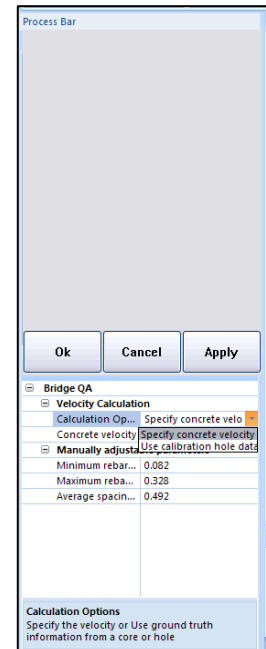
The data processing algorithm uses these values to locate rebar unless the user changes these settings.

The other parameter in the Processing Options dialog box is the Antenna ID option, which only applies to users with more than one Model 5100.

Each Model 5100 used for the bridge QA module should be calibrated at GSSI with the calibration settings entered into the dialog that opens when the user clicks Antenna Calibration Settings. See below for more details. **Note:** This function is not typically used anymore and is only important if you are collecting with multiple antennas.



- 1 In the Processes Pane on the left side of the screen, under the Velocity Calculation section, select Calculation Options:
 - **Concrete Velocity:** where you will specify a velocity in Concrete Velocity
 - **Use Calibration Hole Data:** where you will specify Calibration Hole Location and Calibration Hole Depth
- 2 Manually Adjustable Parameters, specify
 - Minimum Rebar Depth
 - Maximum Rebar Depth
 - Average Spacing between Rebar
- 3 Click Apply to see the results, Cancel to cancel the process, or OK to see and save the results.



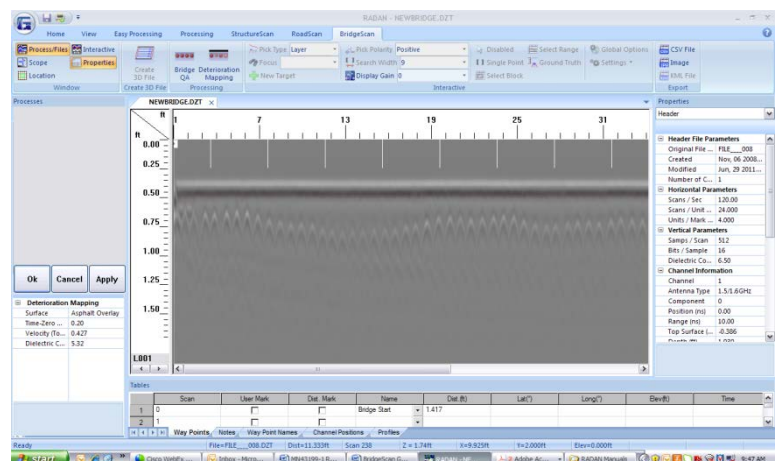
Deterioration Mapping

This technique allows you to locate areas of probable concrete deterioration in a bare concrete or asphalt overlaid concrete deck. This is done by graphing the relative reflection amplitudes across the survey area and assigning threshold values to the amplitude range. The basic premise is that areas of deterioration will attenuate (weaken) the radar signal. It is important to note that this technique does not indicate the type or cause of the deterioration, merely its presence.

This application does have some limitations. Because the technique shows relative change across the single deck, data from different decks cannot be compared. Each deck must be understood in a vacuum. Furthermore since the application depends on mapping differences, this technique is not appropriate for a deck with no deterioration or a deck with near total deterioration.

Any meaningful deterioration map is created by a fusion of the radar data with a comprehensive visual inspection of the deck. The analyst should take careful notes of the location of any surface defect on the top or the bottom of the deck. This allows the analyst to identify areas of extreme deterioration and thus “calibrate” the map output. The presence of any patch or crack should be noted on the top surface and any spalling, staining, or efflorescence should be noted on the bottom. If possible, obtain any maintenance records.

Time-Zero Amplitude Threshold is important for locating the time-zero in the scan. The program will use the peak immediately following the first portion of the scan that exceeds the time-zero amplitude threshold.



The value of 3000 generally works well. If the peak amplitude of the direct coupling is less than 3000, you will need to lower the time-zero amplitude threshold.

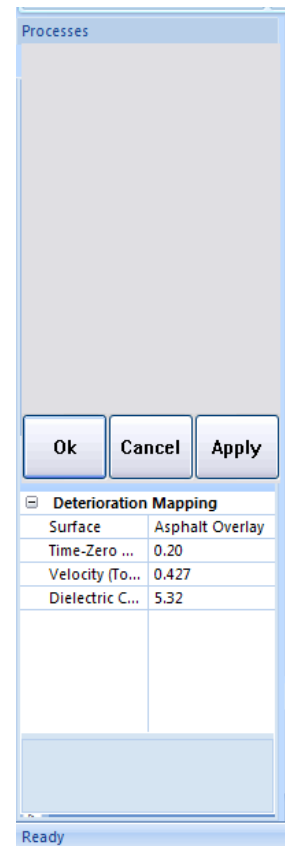
Migration Velocity, the second important parameter, is the radar wave propagation velocity used in the migration process. Migration is basically an imaging process that collapses the arches (or hyperbolas) associated with rebar reflections.

For this process we need an estimate of how fast the radar wave propagates through the asphalt and concrete. The typical ranges of velocities are 4-6 in/ns (10-15 cm/ns). It is up to the user to choose the correct velocity.

You may not pick the correct migration velocity the first time, so observe the processed data (after clicking OK to close the dialogs) and click the Stop button to stop the process, and restart the process with a different migration velocity.

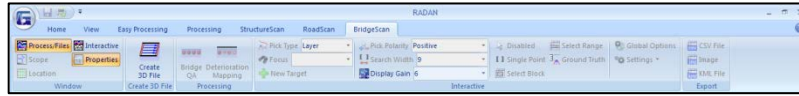
The migration velocity of 4 in/ns (10 cm/ns) was too low. Note that the hyperbolas weren't totally collapsed. The migration velocity of 6 in/ns (15 cm/ns) was too high. The hyperbolas were reversed – they now look like the letter “V.” The best migration velocity for this dataset was 5 in/ns (12.5 cm/ns).

- 1** Click on Deterioration Mapping.
- 2** In the Processes Pane on the left side of the screen, under the Deterioration Mapping section, select:
 - Surface – Asphalt or Concrete Overlay
 - Time Zero – System will default to the location of the surface. You may change this is the system calculates this incorrectly
 - Velocity Top Layer – System will default the velocity. You may change this is the system calculates this incorrectly
 - Dielectric Constant - System will default the dielectric constant. You may change this is the system calculates this incorrectly
- 3** Click Apply to see the results, Cancel to cancel the process, or OK to see and save the results.



Interactive Group

After you process the Bridge data, click on Interactive to begin picking your targets. In this view, you can begin inserting and deleting picks to create the ASCII comma delimited file. The Interactive Pane in the BridgeScan Ribbon gives you many options to do this.



Pick Type

Select whether you will be picking Layers or Targets.

Focus

Select the type of layer or target you will be picking.

New Target

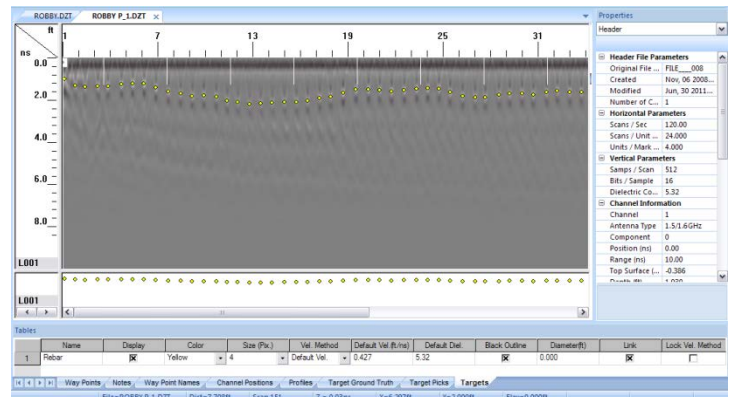
Add more targets to pick.

Pick Polarity

Select whether the picks are Positive, Negative, Absolute, or None.

Search Width

Enter the number of pixels your cursor will be when adding or deleting picks.



The layers or targets to which all edit operations currently apply. The choice is any of the layers defined in the *.CSV file (1, 2, etc.), All, Targets, or Filtered.

When a layer number is selected, picks in that layer can be added or deleted.

The All and Filtered options only allow the user to delete picks.

The Targets option allows the user to add single independent picks to mark objects such as pipes, culverts, voids, or events such as road intersections, starting and ending points of new pavement, etc.

Filtered appears as an option when the filtering option is enabled (Layer Tab of the Tables Pane).

Picking Tools

These are the modes in which the picks are selected.

Single Point

Layer picks are added each time the user presses the left mouse button and deleted each time the user presses the right mouse button. Only one pick can be added at a time with this option, but multiple picks can be deleted depending on the Search Width.

Adding Picks in Single Point Mode

- 1 Position the mouse cursor over the area where you would like a pick to be added.
- 2 Click the left mouse button. A search will be performed on all of the scans between the left and right inside edges of the mouse cursor to locate the maximum amplitude. If the search is successful, a pick will appear on the data. The type of search used to locate the new pick can be customized in the Layer Options menu choice (Layer Tab of the Tables Pane).

- A pick will only be successfully added if a reflection can be located over the cursor search width.
- If no pick is added after pressing the left mouse button, reposition the mouse cursor and click the left mouse button again.
- For cases where the reflection peak is reversed polarity, the user would first need to select Neg. Peak for the Layer Properties under Layer Options in the Interactive Interpretation main menu, then place the mouse cursor over a negative polarity reflection.

Deleting Picks in Single Point Mode

- 1** To delete poor picks, a procedure similar to adding picks is followed. Position the mouse cursor so that the poor pick is located within the search area of the cursor
 - Click the RIGHT mouse button in Delete Picks mode
 - Click the LEFT mouse button in Add Picks mode
- 2** Then, click the right mouse button. If a portion of the pick circle bisects the vertical midpoint of the cursor, then the pick will be deleted from the database and from the screen.

Select Block and Select Range

The Select Block and Select Range picking tools are designed to operate over a large number of scans.

Select Block

When Select Block is activated, a multi-colored, translucent square will appear over the data when the user clicks the left mouse button. The select block contains tiny squares on each face and corner. These squares act as handles that can be used to resize the select block.

To Resize Select Block

- 1** Position the mouse cursor over one of the handles and press the left mouse button.
- 2** Move the mouse cursor to the desired location with the left mouse button still pressed down.
- 3** When the desired size is reached, simply release the left mouse button.

To Move Select Block

- 1** Place the mouse cursor within the select block and press the left mouse button down.
- 2** Move the mouse cursor with the left button still pressed down until the block is in the desired position. Finally, release the left mouse button.

Select Range

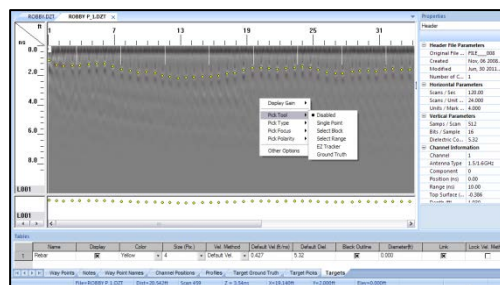
When Select Range is activated, a multicolored translucent overlay appears over the data, extending from the beginning to the end of the file. It operates similarly to the Select Block, except that all operations performed using the Select Range picking tool are performed within the time interval (slice width) of the selected area on *all* of the scans in the file.

The slice width can be changed by clicking the left mouse button on one of the handles (located at the top and bottom at the horizontal midpoint of the slice) and with the left mouse button still pressed, dragging the handle to the desired location.

Adding and deleting points are performed within the Select Block and Select Range areas. These options are activated by pressing the right mouse button down while the mouse cursor is positioned within the multicolored Select Block or Select Range area. This action activates a pop-up menu

Add Points: Clicking on Add Points will activate the program to begin a smart search for reflection peaks within the selected region. Circles will overlay the data where reflection peaks are identified by the search algorithm

Delete Points: Clicking on Delete Points will activate the program to start deleting the picks of the Current Layer located within the selected region.



Layer Modification Options:

- **Change Velocity** changes the velocity of the currently selected layer points located in the selected region. Clicking on Change Velocity opens a dialog box for entering the desired velocity. The new velocity is used for all of the currently selected layer picks in the selected region.
- **Change Pick ID** allows the user to change the layer number assigned to the picks located within the selected region. For example, the user desires to change the layer # of a group of layer 3 picks to layer 2. The user must select layer 3 as the Current Layer, position the Select Block (or Select Range) over the group of points and click the right mouse button to access the pick modification options shown in. Clicking on Change Pick ID will open a dialog box for entering the desired layer number. After changing the layer number, the color of the selected picks will change to the color of the new layer.

Note: A selected point is not changed to the new layer if a pick from the new layer is already present in the scan, or if the new layer overlaps another layer pick (i.e., changing a layer 3 pick to a layer 1 pick with a layer 2 pick present in the scan).

Ground Truth

Selecting the Ground Truth menu item from the Interactive Interpretation main menu (see next section) opens a spreadsheet that allows the user to enter ground truth data that can automatically be used to calculate layer velocities.



Ground truth data can be entered on the left half of the spreadsheet and the calculated velocity and nearest scan location are automatically updated and shown in the right side of the spreadsheet.

For example, let's say a core was drilled 104.5 m from the starting location of the file and the measured depths of the layer interfaces were 12 cm, 22 cm, and 33 cm. Each of the first three columns in the Ground Truth spreadsheet should be filled with this information.

The 5th, 6th and 7th columns of the spreadsheet were automatically calculated by RADAN. The Data X shows the location of the nearest scan containing the layer pick, Time is its arrival time, and V is the calculated velocity for the layer. If no picks are located within 20 scans of the core location, then zeros will appear in the last 3 columns.

Once core data have been entered, the spreadsheet can be closed or left open. When the user saves the data as an ASCII *.lay file, the core data are stored in a separate ASCII file with the same name as the *.dzt file and the extension .gtr. Once this file has been created it will automatically open and its contents read by RADAN the next time the user opens the ASCII file providing it is located in the same directory as the ASCII file. Alternatively, if the user saves the pick data in a *.bii file, all of the ground truth information will be stored with the picks in the *.bii file.

The velocities calculated from core data can be used to calculate layer depths. To change the velocities of existing picks, use the select block or select range picking tool and move the selected region over the picks that need to be changed. Then, click the right mouse button to bring up the pick options and select the Change Pick Velocity option. Proceed to change the selected pick velocities as directed in the dialog.

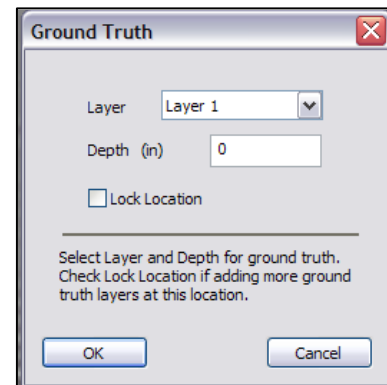
To specify all new layer picks to have velocities calculated from the ground truth data, the velocity option must be changed from the Layer Properties Dialog page accessed through the Layer Options menu item.

When ground truth data from more than one core are entered in the Ground Truth spreadsheet and the user selects core data as the velocity option for picks, the velocity used for each pick is obtained from the nearest core.

Any changes in core data are automatically updated in the picks that have core data calculated velocities.

Ground Truth: Enter Ground Truth Information.

- 1** Click in the data the spot where the ground truth was collected.
- 2** Layer: Select Layer for ground truth.
- 3** Depth (): Enter Depth information.
- 4** Lock Location: Input a check mark if adding more ground truth layers at this location.
- 5** Click OK or Cancel.



The Ground Truth dialog box contains the following fields and controls:

- Layer:** A dropdown menu currently set to "Layer 1".
- Depth (in):** A text input field containing the value "0".
- Lock Location:** An unchecked checkbox.
- Instructions:** "Select Layer and Depth for ground truth. Check Lock Location if adding more ground truth layers at this location."
- Buttons:** "OK" and "Cancel".

Interpolate Points: Will interpolate layer picks (add new picks between existing ones) using the interpolation method (Linear or Nearest Neighbor) specified in the Properties dialog under Layer Options.

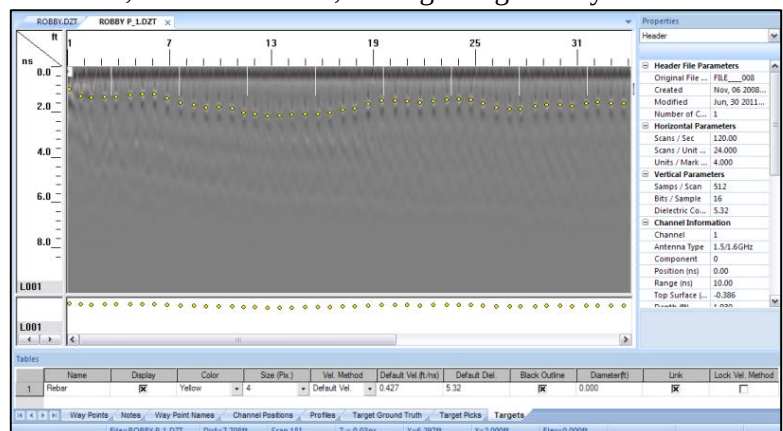
EZ Tracker

EZ Tracker will attempt to trace a layer by following similar conditions in successive scans. You should note the desired layer's phase information and select Positive, Negative, or Both under layer options.

To use EZ Tracker, make sure that the current focus is set to the layer that you want to track and click Start. Then click on the layer. Move the cursor further down the layer and click again. EZ Tracker will attempt to trace that layer between the two points. A closer click spacing will produce better results. If you make a mistake, you can right click to undo the previous section. You should click Save every few picks to save changes. If you come to a gap in the layer, click Stop and then Start to restart the layer at the other side of the gap.

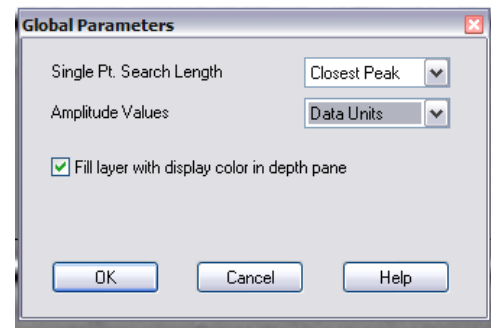
Start by clicking on EZ Tracker. Then click Start. Click, in the Data Area, the beginning of a layer. Then at small intervals, click along the layer. The software will fill the layer in between the dots.

If you wish to begin at a new area of the layer, simply click Stop, then Start again. You may Right Click the mouse to delete the previous click.



Global Options

- **Single Pt Search Length:** When using Single Point Picking Tool, to enter a “Dot” at the closest peak or use the cursor length.
- **Amplitude Values:** Use Data Unit, dB, or Normalized dB.
- Check to fill layer with display color in depth pane.



Save Settings

- Load Default Settings for Interactive Options
- Save as Default Settings will save your Interactive Options as default.

Export Group

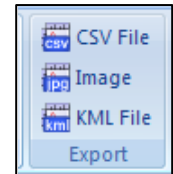


You may export your picks to a comma delimited ASCII (CSV) file, save your image to a JPG file, or export the layer information to a KML file (if GPS was used with your data) for Google Earth.

CSV File

Click CSV File to export your picks to a comma delimited file. You may select pick what fields you would like to export to the CSV file. You may:

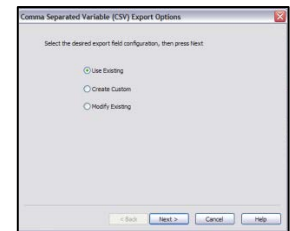
- **Use Existing:** Use the existing custom fields what you created in Create Custom.
- **Create Custom:** You may pick and choose which fields you want.
- **Modify Existing:** Modify an existing field list to create a new one.



Use Existing: Select an existing customize field list you created from Create Custom or the GSSI Standard.

Create Custom: Select from Categories and a list of fields from each of the Categories you want for your CSV file. You should give this Category a user-friendly name for future use.

Modify Existing: Modify an existing category to create a new category.



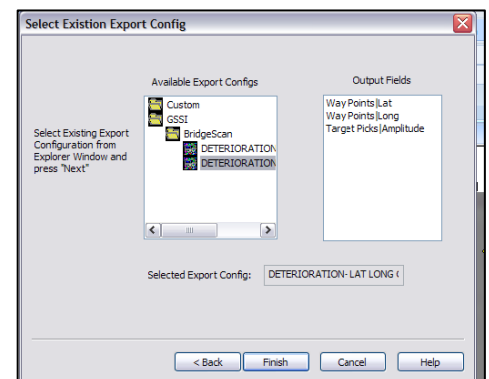
Image

Save the image as a JPG file.

KML File

Export each layer to a KML file for Google Earth.

A KML file is a file type used to display geographic data in an earth browser, such as Google Earth and Google Maps. KML files have a tag-based structure with names and attributes used for specific display purposes. In RADAN you can export certain kinds of information as a



KML file and easily view that information in Google Earth. This section discusses exporting layers identified in Interactive Interpretation.

Note: In order to create a KML file, you must have collected the data with GPS.

Layer: Selects the Layer that you want to export. If you have multiple layers, you must export them one at a time.

Channel: Selects the data channel to export. You can export different channels if you have collected a multi-channel file.

Output Parameter: This can be pick depth, amplitude, velocity, or time depending on the type of data that you want to depict.

Distance Placemarks/Distance Interval: This option outputs a notation at a prescribed distance.

Line width: The KML output line width in pixels.

Minimum/Maximum: This sets the range of output values specified in the Output Parameter option.

Number of Intervals: The number of evenly spaced breakpoints for the output data.

Range/Colors: The range will be set by the number of output intervals. You can decide to code different intervals with colors.

The output of this is a KML file with the same root name as the ASCII pick LAY file. To import into Google Earth, simply drag this file onto Planet Earth and it will zoom to the given location of the data.

Layer Options

In the Layers Tab located in the Table Pane, you can define each layer, how they are displayed and how they behave.

Layer Num	Name	Display	Color	Size (Px.)	Vel. Method	Default Vel (in/ft)	Default Vel (m/s)	Default Die	Black Outline
1	Layer 1	☒	Yellow	4	Default Vel	5.12	5.12		☐
2	Layer 2	☒	Red	4	Default Vel	5.12	5.12		☒
3	Layer 3	☒	Green	4	Default Vel	5.12	5.12		☒
4	Layer 4	☒	Blue	4	Default Vel	5.12	5.12		☒
5	Layer 5	☒	Brown	4	Default Vel	5.12	5.12		☒
6	Layer 6	☒	Black	4	Default Vel	5.12	5.12		☒
7	Layer 7	☒	Black	4	Default Vel	5.12	5.12		☒

Name: The Layer Names page permits the user to specify a name for each layer. For example, for a typical road structure, the user may want to rename Layer 1 as Asphalt, Layer 2 as Base and Layer 3 as Subbase. Layer name changes are shown in all dialogs and will be output to the ASCII file when the user performs the next ASCII file save

Display: Display the layer or not.

Color: Pick the color for the layer.

Size: Select the pixel size of each pick.

Velocity Method: Sets the propagation velocity used to calculate the thickness of each layer. If the Auto box is checked, the velocity will be automatically calculated for each radar scan. The other options are: (1) User-specified, (2) Core Data, and (3) Velocity Analysis File. The Velocity analysis option is most commonly used for geological investigations where a common-midpoint file has been collected and processed using the Velocity Analysis Option in RADAN. The output file from the Velocity Analysis must be entered in the Global page before this option is activated here.

Default Vel (ins/ns): The velocity of RADAR at the interface of that layer.

Default Die: The dielectric of the layer.

Black Outline: Toggle On/Off to have a black outline for each pick.

Tables and Panes

Data Tab/Processes/Proc. List Tab Pane

Data Tab

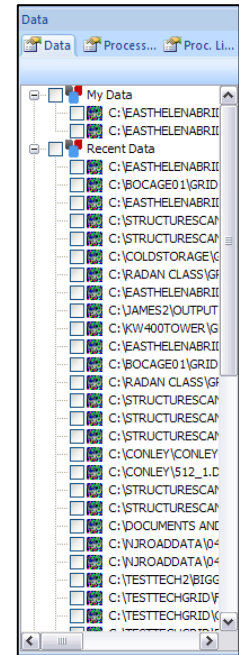
This will contain a list of My Data, My Recent Data, and GSSI Example Data for quick reference and open.

My Data is a list of files that you could be currently working on.

My Recent Data is a list of files that you have recently worked on.

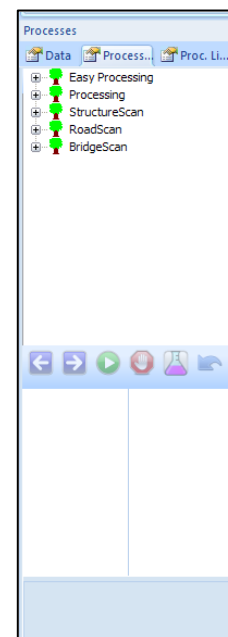
GSSI Example Data is a list of files that are examples of data.

Note: You may right click on any of the options to Create New Categories, Add New files and folders, Import or Export your data files. You may also drag a file from one category to another category.



Processes Tab

The Processes Tab is the same options as the options located in the Ribbons. This allows you to open a menu option by click on the “+” and select which steps to process. Selecting these steps will open a dialog for you to process your data.



Proc. List Tab

A macro is a series of steps and/or options put together as one option. GSSI has created these commonly used processes as macros.

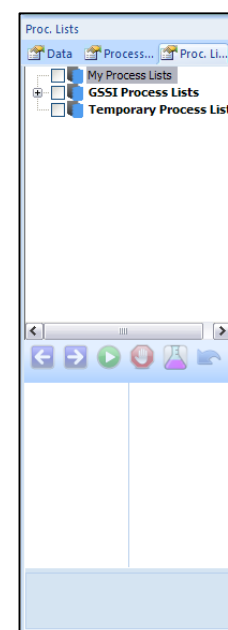
With the Process Icon turned on in the Ribbon, you may select the following macros:

My Process List: A list of processes (macros) you create for future use.

GSSI Process List: A list of processes created by GSSI for the more commonly used processes.

Temporary Process List: A list of processes created for temporary use.

Note: You may right click on any of the options to create a New Category or Process List, Import or Export Macros.



When you select process list, you may test the macro by clicking on the Test Icon, Reset the Test, Stop the Processing, or click Go to run and process the macro.



Go Icon



Stop Icon



Test Icon



Reset Icon

Properties Pane

The Properties Pane will display one of four options.

Global Parameters

Before any files are open, you may configure RADAN so that opening, processing, displaying and saving files can “default” to your desire.

Auto Save File:

- **True:** Automatically save a processed file (GSSI naming convention). See Appendix A for the GSSI name convention.
- **False:** User is prompted to save a processed file and you name the file.

Source Directory: Select the folder where the source (original data is located).

Output Directory: Select the folder where all processed data will be located.

Display Channel: Select the number of channels you would like to display/process.

Marker Style: Select None, Long, or Short for the type of User/Distance marks to be displayed.

Horiz. Scale:

- **Show:** Display horizontal scale.
- **Hide:** Do not display horizontal scale.

Grid Lines:

- **Show:** Display grid lines.
- **Hide:** Do not display grid lines.

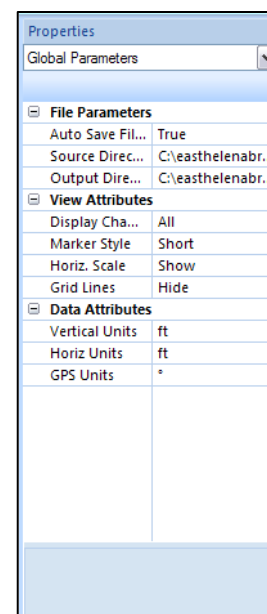
Vertical Units: Units for the vertical scale.

Horiz Units: Units for the horizontal scale.

Note: When a file is opened, the scale will default to whatever units were saved with the file.

GPS Units: Units for GPS horizontal scale.

Note: Once you open a file, you cannot change any of the Global Parameters. You must close all files to make changes.



Header Parameters

Once a file is open, Header information about the file will be displayed.

Header File Parameters

- Original File: Name of the original file. It will display the name of the original file even if a processed file is opened.
- Created: Date original file was created.
- Modified: Date file was last modified.
- Number of Channels: Number of channels files contains.

Horizontal Parameters

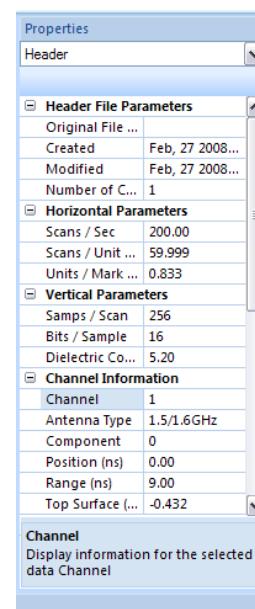
- Scans/Sec: Number of scans per second file was collected.
- Scans/unit: Number of scans per unit file was collected.
- Units/Mark: Number of units per mark. This can be modified.

Vertical Parameters

- Samples/Scan: Number of samples per scan. This can be modified.
- Bits/Sample: Number of bits per sample. This can be modified.
- Dielectric Constant: Dielectric value when data was collected. This can be modified.

Channel Information

- Channel: Which of channel to display in Header Information.
- Antenna Type: Antenna used to collect the data.
- Component: Component used to collect the data.
- Position (ns): Position used when collecting the data.
- Range (ns): Range of the data (depth) when data was collected.
- Top Surface: Surface information when data was collected.
- Depth: Depth of the data when data was collected.
- Processing History: Any processing done to the current file is recorded here.
 - Range Gain: Any Gain modifications.
 - Position Correction: Time Zero processing.
 - IIR Filters: IIR filters applied to the data.
 - FIR Filter: FIR filters applied to the data.
 - Background Removal: Background Removal applied.



Display Parameters

Once a file is open, you may view and change how the information is displayed.

View Attributes

- Display Channel: Display All or a specific channel.
- Marker Style: None, Short, or Long.
- Grid Lines: Hide or Show.

Data Attributes

- Use Surface: Yes or No.
- Start Scan: Starting scan to display data.
- End Scan: Ending scan to display data.

Oscope Parameters

- Fill: None, Positive, or Negative.
- Fill Level: Enter Fill Level Number.
- Stack- Enter Stack Number.
- Line Width- Enter Line Width.
- Color- Select color.

Wiggle Parameters

- Chop: Show Both, Show Positive, or Show Negative.
- Fill: Fill in peaks, positive, negative, or none.
- Fill Level: Fill upper n% of the peaks.
- Scale: Enter Scale.
- Skip: Skip intervals of n scans.
- Stack: Stack n scans per wiggle.
- Space: Enter space number.
- Line Width: Line width of traces in pixels.
- Color: Color of the traces.

Properties	
Display Parameters	
View Attributes	
Display Cha...	All
Marker Style	Short
Grid Lines	Hide
Data Attributes	
Use Surface	Yes
Start Scan	0
End Scan	0
Oscope Parameters	
Fill	None
Fill Level	0
Stack	1
Line Width	1
Color	000000
Wiggle Parameters	
Chop	Show Both
Fill	None
Fill Level	0
Scale	16
Skip	0

3D Display Parameters

In 3D View, you may view and change how the 3D cube is displayed.

XYZ Slicing

- X Position: Move the X Slice.
- Y Position: Move the Y Slice.
- Z Position: Move the X Slice.
- Thickness As: Select type of units as Depth, Sample or Time.
- Thickness: Thickness of the Z Slice.
- Z Map Width: Width of the Z Slice.
- 3D Z Slice Mode: Toggle between On/Off to give the Z Slice a “3D” look.

XYZ Visibility

- Section: Create a Blocked Cube image.
- Transparency: Only show the positive amplitude.
- Snaps: None, To Profile, or To Bookmarks.
- X, Y, Z: Toggle On/Off X, Y, Z slices.
- X, Y Data: Toggle On/Off X,Y Data.

3D Data Processing

- Max RMS: Max or RMS.
- Bkg Removal: Toggle between On/Off.
- Median Filter: None, 1, 2, or 3.

3D Toggles

- Show or do not show Points, Lines, Waypoints, Contour or Surfaces.

3D Style

- Grid: Toggle On/Off Grid.
- Minor Grid X, Y, Z: Toggle On/Off Grid lines.
- Mesh: Hide Mesh.
- North Arrow: Show/Hide North Arrow.
- Background: Toggle between Black and White.
- Lock Aspect Ratio: Enter Lock Aspect Ratio.
- Open Shape Files: Open a file that saves your target information.

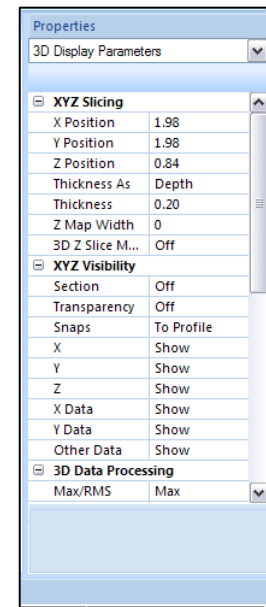


Table Pane

This pane will display data base information about the file. Some of the tabs will not be displayed, depending on the type of application being processed.

Tables									
	Scan	User Mark	Dist. Mark	Name	Dist.(ft)	Lat(°)	Long(°)	Elev(ft)	Time
1	0	☐	☐		0.000				
2	1	☐	☐						
3	2	☐	☐						
4	3	☐	☐						

Way Points Notes Channel Positions Profiles Free Draw Points Free Draw Groups

Way Points: Computer generated or User generated marks. You may add and delete any of these marks.

Notes: Enter any notes about this file.

Channel Positions: For multi-channel files, enter the offset information for each channel.

Profiles: For 3D files, display all the profiles associated with the 3D file. You may edit the position of each profile by clicking on Edit 3D Position.

Free Draw Points: Enter Target information.

Free Draw Groups: Enter Group information for entering target information.

Note: There are other tabs depending upon the application of the data. These tabs will be discussed in detail processing specific data in Section 4.

Section 4: Basic Processing/Tutorials

This section will give you a mouse click by mouse click tutorial for some basic processing. For details on what each option does, please refer to Section 3.

Creating a Manual Grid

This will allow you to combine individual data files (.dzt) that were collected in a grid format and combine them together to create a 3D Grid file (.m3d). Files collected can be collected in the X direction only, Y direction only, or both X and Y direction.

Note: It is highly recommended that files collected in the X direction be put in a folder separate from files collected in the Y direction.

- 1 Click the GSSI Button, then Assemble, then 3D File, enter a filename for the Grid you want to create. Click Save to save and continue.

3D Filename: This is the name of the single .DZT file that RADAN will construct from the individual profiles, and the location where it will be stored. Clicking on this button will allow you to change either the name or the storage directory.

Files in X/Y Direction: Both of these tabs MUST be filled even if you collected the grid in one direction. If you collected the same grid twice, but with perpendicular transects, these tabs allow you to define different input parameters for each direction. For example, if you collected the X transects in zig-zag, but the Y as unidirectional lines, you can input the different line order here. Also if you collect data in the Y direction as with some other geophysical instruments (Geoscan RM15 or FM36, or Geometrics 858), you can just check the Files in Y-Direction tab.

Starting Coords (units): This is the coordinate of the bottom left corner of your grid. If your area is tied into a larger site grid, you can input those coordinates here so that the axis of the resultant 3D file matches with the larger area.

X-Length/Y-Length: These are the maximum coordinates of your grid. For example, if your grid is 100 inches $\times$ 100 inches, you would put those values in here. If your profile lines are not all the same length, you should put in the measurement of the longest one. **This must be filled in for BOTH the Files in X-Direction and Files in Y-Direction tabs.**

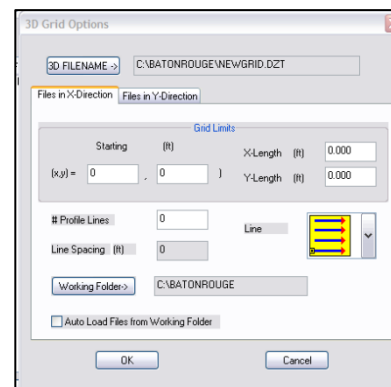
Profile Lines: Total number of lines in either the X or the Y direction.

Line Spacing: This is the distance between each survey transect. The software figures out this number by dividing the grid size by the number of profile lines. You should use this as error checking. If you collected data with transects placed one foot apart, and you have the correct number of transects for the grid size, then this number should be 1. Anything else and you have a positioning error.

Helpful Hint: Do not forget to count the “0” transect. If you are scanning a 10 $\times$ 10 foot area with profiles every one foot, and your first profile is at 0 and your last is at 10, you will have 11 profiles.

Line Order: This is a pull down menu. Visualize your site grid and the order that your files were collected, and choose the orientation that matches your collection method.

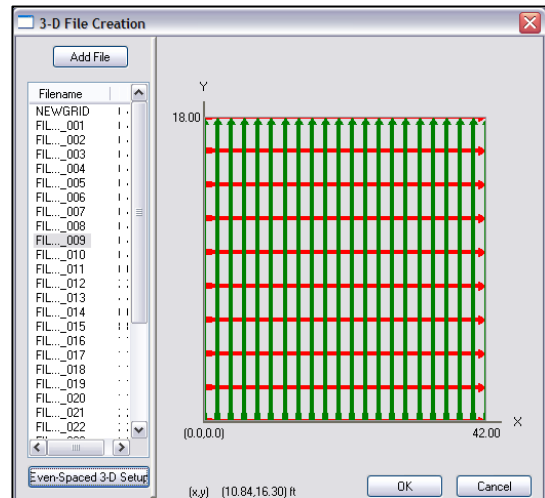
Working Folder: This is where your data is stored. Clicking this button will open a browser so you can select a different directory.



Auto Load Files: If you check this box, RADAN will go to the working directory and automatically input the data files in alpha-numerical order. This is the same order that is shown when you sort the data files by name in Windows Explorer (by clicking on the “Name” column header). If the files are not in the correct naming convention, you may find it easier to rename them in Windows Explorer.

After you have worked with all the selections in the dialog, click OK to open the 3-D File Creation window.

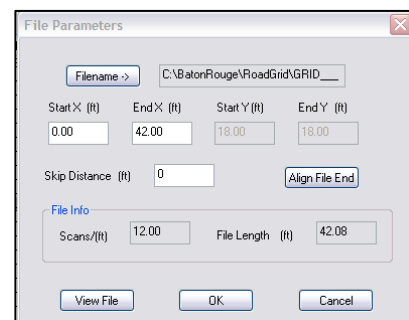
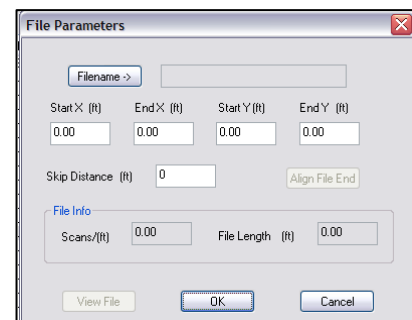
- 2** The window shows the actual locations and orientations of your data profiles. If you clicked “Auto Load Files...” on the previous the left pane will show a list of file names with starting and ending coordinates.
- 3** Existing filenames and coordinates can be edited two ways. The first way is click on a filename located in the left pane. Then double-click that filename to edit the coordinates. The second way to edit a file is to move the mouse cursor to a line located in the right pane and click on it.
- 4** You may delete a line by clicking on the file in the left pane and pressing the delete key.



- 5** When you are satisfied with the look of the grid, click OK and the software will combine all the files and create a single grid file.

Other 3D File Creation Options

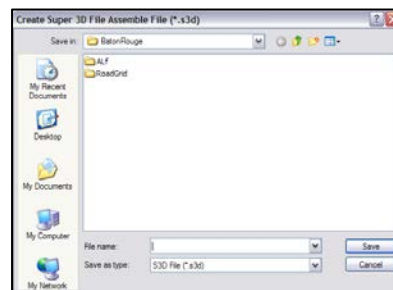
- 6** You may, if necessary, add files to your grid. Click Add File. Click on Filename to browse and select the appropriate file. Then enter the X and Y starting and ending coordinates for that line. Click OK to save and continue.
- 7** The Skip Distance option permits you to skip a certain distance from the start of the file when writing to the 3D output file. This is particularly beneficial for files that were mistakenly started with the antenna in back of the starting point for the grid.
- 8** If your files were collected in Zig-Zag, you will see the align file button: to the right of the Skip Distance window. This button will adjust the file so that the last scan is aligned with the end of the grid. This option is typically used in cases where the user is more confident in the ending position of the profile than the starting position. This option is only available for evenly spaced x- or y-directed files.
- 9** If you did select a line to edit, you may change the coordinates of this line. Click OK when finished.



Creating a Super Grid

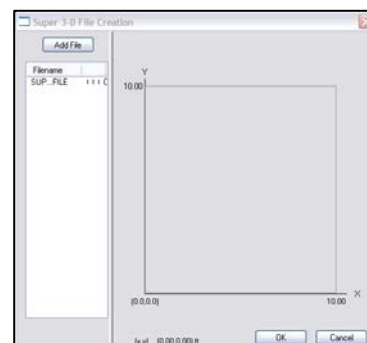
You may combine multiple processed 3D Grids to create one “super” 3D Grid. This will create a new file with the extension .s3d.

- 1** Click the GSSI Button, then click Assemble. After you select Super 3D File, Enter a filename for the newly created Super 3D file (.s3d). Click Save to save and continue.



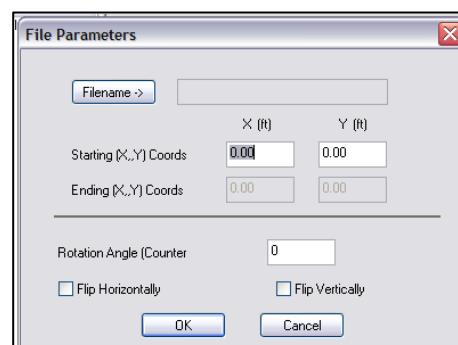
Follow these next set of steps:

- 2** **Add File:** Click Add File to retrieve a grid.



- 3** **Populate:** Populate the File Parameters window.

- a)** Filename: Click Filename >> to browse and retrieve a grid.
- b)** Starting (X,Y) Coords: Enter the X,Y position for this grid.
 - If this is the first grid you are adding, the Starting (X,Y) coordinate is more likely 0,0.
 - If this is NOT the first grid you are adding, the Starting (X,Y) coordinate is relative to the first grid you added.
- c)** If necessary, you can Rotate, Flip Horizontally, and/or Flip Vertically, depending on how the grid was collected relative to the coordinates of the first grid you added.
- d)** Click OK when you are done.



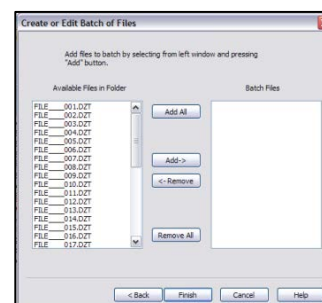
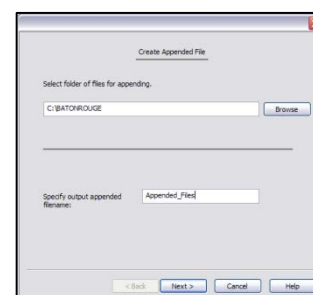
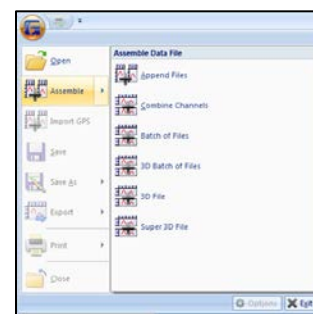
- 4** Repeat: If you have more grids to add, continue back to step 1.

- 5** Once you have completed adding all the grids, click OK and the system will combine all the grids and create a Super Grid file.

Appending Files

This option will allow you to append files to create one file.

- 1 Click the GSSI Button, then click Assemble. After Append Files is selected, browse (if necessary) to the folder where the files you wish to append are located. Enter a name of the file for the newly created file. Click Next to continue or Cancel to cancel the process.
- 2 There will be two windows: Available Files in Folder are files available to append; and Batch Files are the files that will be appended.
 - a) Adding Files to append together:
 - **Add All:** Click here to add all files from the left window to the right window.
 - Click on a file and click Add to add a file to the right window
 - You may add multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to add. Then click Add.
 - If the files you want to add are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Add.
 - b) Removing Files:
 - **Remove All:** Click here to remove all files from the right window to the left window.
 - Click on a file and click Remove to remove a file from the right window.
 - You may remove multiple files at once by clicking on one file, then while holding the CTRL key, click on the other files to remove. Then click Remove.
 - If the files you want to remove are grouped together, you may click on the first file, then while holding down the SHIFT key, click on the last file in that group. Then click Remove.
- 3 Click Back to return to the previous screen.
- 4 Click Finish to complete the process.
- 5 Click Cancel to cancel the process.



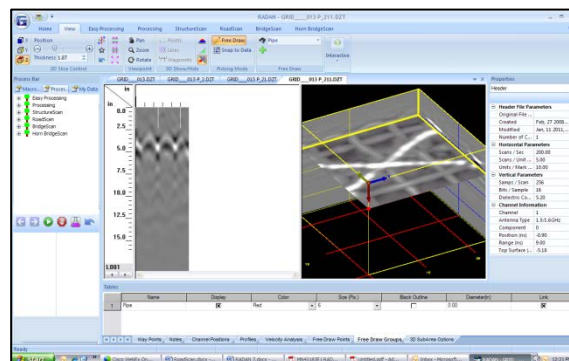
Basic 3D Grid Processes

When viewing data in the 3D mode (Home Ribbon->3-D View), there are many options available to you to assist in interpretation of the data and provide improved images for documentation purposes.

- 1 To begin, open a 3D file and process the file as needed (Time Zero Correction, Filtering, Migration and so on). Then, click on 3-D View from the Home Ribbon. When this is done, you will automatically be switched to the View Ribbon.

The left pane of the display window is the 2D image, one profile at a time. To view other profiles, simply click the left and right arrows located at the bottom of the left pane.

The right pane of the display window is the 3D image.



Right Click Options

- Right Click on the 3D Grid.
- Click Display Gain and select the Gain that will best display your data.
- Click Absolute to display all targets as a positive amplitude.
- Click Negate to display all targets as a negative amplitude.
- Click None to display all targets as their original amplitude.

Zooming In/Out the 3D Image

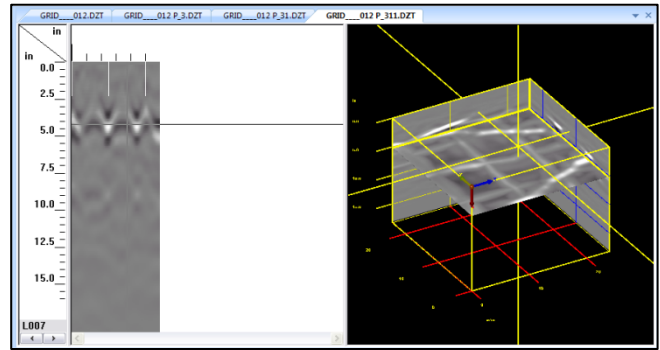
- Point your mouse on the image.
- Hold down the [Ctrl] key on your keyboard and press the up and down arrows on your keyboard. If you have a wheel on your mouse, you can zoom in and out on the image using that as well.

Resizing the Display

- Point to the vertical bar located between the 2D image and 3D image.
- Hold your left mouse button down and move the bar left or right.
- Note: You may resize any pane by “grabbing” any horizontal or vertical bar and “dragging” the bar up and down or left and right.

Corresponding 2D Location with the 3D Location

- Hold your left mouse button down while pointing at the 2D image in the left pane. You will see cross hairs appear in the 3D image that corresponds with the location of your mouse.
- The left pane will display one profile at a time. To view other profiles, click the left and right arrows located at the bottom of the left pane.



Slicing Through the 3D Image

- You may move the X, Y, and Z slices by:
 - Point to the X, Y, or Z slice with your mouse.
 - Hold down the [Ctrl] key on your keyboard.
 - Hold down the left mouse button, and move your mouse.
- You can use the options on the View Ribbon as well to do this.
 - If you wish to move the X Slice, click the “X” icon and move the slider bar.
 - If you wish to move the Y Slice, click the “Y” icon and move the slider bar.
 - If you wish to move the Z Slice, click the “Z” icon and move the slider bar.



Drawing Pipes

You may add pipes and other images to your 3D Grid that will give you other ways to present your information.

1 Click Free Draw in the View Ribbon.

2 To draw a pipe:

a) Double-click on one end of a target.

b) While holding down the [Space Bar] on your keyboard, double-click the other end of the target.

3 To add additional pipes, click the Add Target icon located in the View Ribbon. Then continue the process of drawing a pipe.



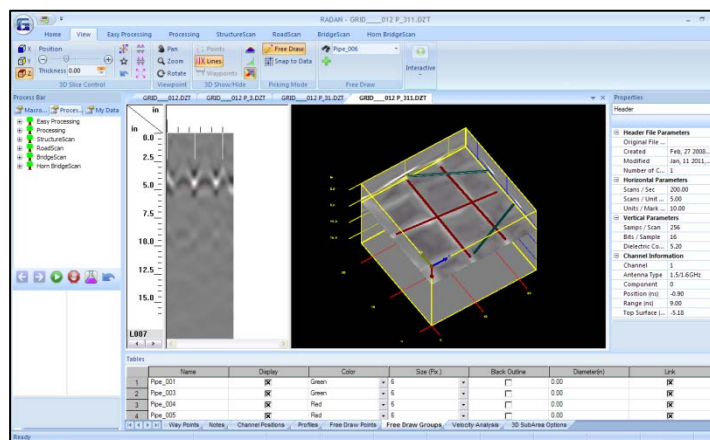
4 As you add new pipes, note the pipe information located in the Free Draw Group Tab in Table Pane located at the bottom of the screen. Here you may:

- Rename the target.
- Choose to display the target.
- Choose the color of the target.
- Choose the pixel size of the target.
- Choose whether or not the target will display a black outline.
- Choose the diameter of the target.
- Choose to link the two dots.

5 Right click on a pipe and select Delete to delete a pipe.

6 To save an image as a picture to use for documentation or export targets for AutoCAD applications.

- Click the GSSI Button.
- Click Export.
- Click Save as JPG Image or Save as AutoCAD format.



Tables						
	Name	Display	Color	Size (Px)	Back Outline	Diameter(in)
1	Pipe_001	☒	Green	5	☐	0.00
2	Pipe_002	☒	Green	5	☐	0.00
3	Pipe_004	☒	Red	5	☐	0.00
4	Pipe_005	☒	Red	5	☐	0.00

Properties Pane

Using the 3D Display Parameters information located in the Properties Pane, you may select other display options.

XYZ Slicing

- X Position: Move the X Slice.
- Y Position: Move the Y Slice.
- Z Position: Move the X Slice.
- Thickness As: Select type of units as Depth, Sample or Time.
- Thickness: Thickness of the Z Slice.
- Z Map Width: Width of the Z Slice.
- 3D Z Slice Mode: Toggle between On/Off to give the Z Slice a “3D” look.

XYZ Visibility

- Section: Create a Blocked Cube image.
- Transparency: Only show the positive amplitude.
- Snaps: None, To Profile, or To Bookmarks.
- X, Y, Z: Toggle On/Off X, Y, Z slices.
- X, Y Data: Toggle On/Off X,Y Data.

3D Data Processing

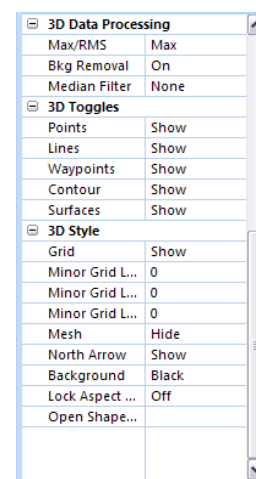
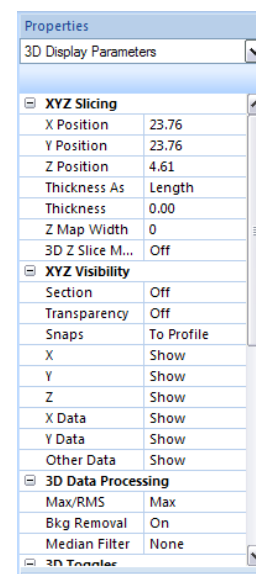
- Max RMS: Max or RMS.
- Bkg Removal: Toggle between On/Off.
- Median Filter: None, 1, 2, or 3.

3D Toggles

- Show or do not show Points, Lines, Waypoints, Contour or Surfaces.

3D Style

- Grid: Toggle On/Off Grid.
- Minor Grid X, Y, Z: Toggle On/Off Grid lines.
- Mesh: Hide Mesh.
- North Arrow: Show/Hide North Arrow.
- Background: Toggle between Black and White.
- Lock Aspect Ratio: Enter Lock Aspect Ratio.
- Open Shape Files: Open a file that saves your target information.



Interactive Processing

This section will describe the basic processing for picking layers and/or targets in your data. Used primarily for RoadScan and BridgeScan, this can be used for any application. This will allow you to produce and output a comma delimited ASCII file for use in any third party software. If GPS was used with your data, you may export the data to a KML file for use with Google Earth.

Click Interactive in the Home Ribbon, RoadScan Ribbon, or BridgeScan Ribbon.



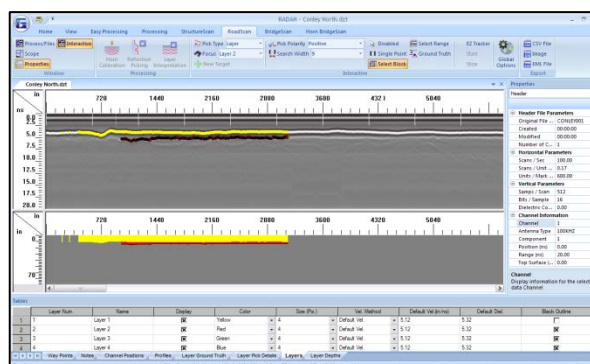
Pick Type: Pick between Layer or Target.

Focus: Select the type of layer or target you will be picking.

New Target: Add more targets to pick.

Pick Polarity: Select whether the picks are Positive, Negative, Absolute, or None.

Search Width: Enter the number of pixels your cursor will be when adding or deleting picks.



Picking Tools

These are the modes in which the picks are selected.

Single Point: Layer picks are added each time the user presses the left mouse button and deleted each time the user presses the right mouse button. Only one pick can be added at a time with this option, but multiple picks can be deleted depending on the Search Width.

Adding Picks in Single Point Mode

- 1 Position the mouse cursor over the area where you would like a pick to be added.
- 2 Click the left mouse button. A search will be performed on all of the scans between the left and right inside edges of the mouse cursor to locate the maximum amplitude. If the search is successful, a pick will appear on the data. The type of search used to locate the new pick can be customized in the Layer Options menu choice (Layer Tab of the Tables Pane).

A pick will only be successfully added if a reflection can be located over the cursor search width. If no pick is added after pressing the left mouse button, reposition the mouse cursor and click the left mouse button again.

For cases where the reflection peak is reversed polarity, the user would first need to select Neg. Peak for the Layer Properties under Layer Options in the Interactive Interpretation main menu, then place the mouse cursor over a negative polarity reflection.

Deleting Picks in Single Point Mode

- 1 To delete poor picks, a procedure similar to adding picks is followed. Position the mouse cursor so that the poor pick is located within the search area of the cursor.

- Click the RIGHT mouse button in Delete Picks mode.
- Click the LEFT mouse button in Add Picks mode.

2 Then, click the right mouse button. If a portion of the pick circle bisects the vertical midpoint of the cursor, then the pick will be deleted from the database and from the screen.

Select Block and Select Range

The Select Block and Select Range picking tools are designed to operate over a large number of scans.

Select Block

When Select Block is activated, a multi-colored, translucent square will appear over the data when the user clicks the left mouse button. The select block contains tiny squares on each face and corner. These squares act as handles that can be used to resize the select block. Select Block is only available if you have purchased the Structure Identification module for RADAN.

To Resize Select Block

- 1** Position the mouse cursor over one of the handles and press the left mouse button.
- 2** Move the mouse cursor to the desired location with the left mouse button still pressed down.
- 3** When the desired size is reached, simply release the left mouse button.

To Move Select Block

- 1** Place the mouse cursor within the select block and press the left mouse button down.
- 2** Move the mouse cursor with the left button still pressed down until the block is in the desired position. Finally, release the left mouse button.

Select Range

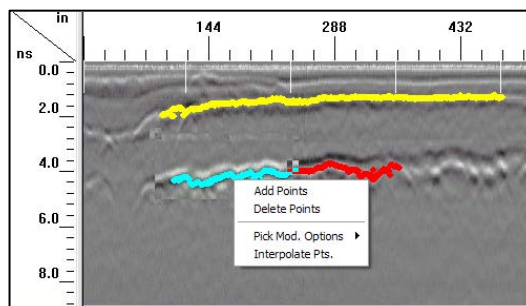
When Select Range is activated, a multicolored translucent overlay appears over the data, extending from the beginning to the end of the file. It operates similarly to the Select Block, except that all operations performed using the Select Range picking tool are performed within the time interval (slice width) of the selected area on *all* of the scans in the file.

The slice width can be changed by clicking the left mouse button on one of the handles (located at the top and bottom at the horizontal midpoint of the slice) and with the left mouse button still pressed, dragging the handle to the desired location.

Adding and deleting points are performed within the Select Block and Select Range areas. These options are activated by pressing the right mouse button down while the mouse cursor is positioned within the multicolored Select Block or Select Range area. This action activates a pop-up menu.

Add Points: Clicking on Add Points will activate the program to begin a smart search for reflection peaks within the selected region. Circles will overlay the data where reflection peaks are identified by the search algorithm.

Delete Points: Clicking on Delete Points will activate the program to start deleting the picks of the Current Layer located within the selected region.



Layer Modification Options:

Change Velocity changes the velocity of the currently selected layer points located in the selected region. Clicking on Change Velocity opens a dialog box for entering the desired velocity. The new velocity is used for all of the currently selected layer picks in the selected region.

Change Pick ID allows the user to change the layer number assigned to the picks located within the selected region. For example, the user desires to change the layer # of a group of layer 3 picks to layer 2. The user must select layer 3 as the Current Layer, position the Select Block (or Select Range) over the group of points and click the right mouse button to access the pick modification options shown in. Clicking on Change Pick ID will open a dialog box for entering the desired layer number. After changing the layer number, the color of the selected picks will change to the color of the new layer.

Note: A selected point is not changed to the new layer if a pick from the new layer is already present in the scan, or if the new layer overlaps another layer pick (i.e., changing a layer 3 pick to a layer 1 pick with a layer 2 pick present in the scan).

Interpolate Points: Will interpolate layer picks (add new picks between existing ones) using the interpolation method (Linear or Nearest Neighbor) specified in the Properties dialog under Layer Options.

Ground Truth

Selecting the Ground Truth menu item from the Interactive Interpretation main menu (see next section) opens a spreadsheet that allows the user to enter ground truth data that can automatically be used to calculate layer velocities.

Ground truth data can be entered on the left half of the spreadsheet and the calculated velocity and nearest scan location are automatically updated and shown in the right side of the spreadsheet.

For example, let's say a core was drilled 104.5 m from the starting location of the file and the measured depths of the layer interfaces were 12 cm, 22 cm, and 33 cm. Each of the first three columns in the Ground Truth spreadsheet should be filled with this information.

The 5th, 6th and 7th columns of the spreadsheet were automatically calculated by RADAN. The Data X shows the location of the nearest scan containing the layer pick, Time is its arrival time, and V is the calculated velocity for the layer. If no picks are located within 20 scans of the core location, then zeros will appear in the last 3 columns.

Once core data have been entered, the spreadsheet can be closed or left open. When the user saves the data as an ASCII *.lay file, the core data are stored in a separate ASCII file with the same name as the *.dzt file and the extension .gtr. Once this file has been created it will automatically open and its contents read by RADAN the next time the user opens the ASCII file providing it is located in the same directory as the ASCII file. Alternatively, if the user saves the pick data in a *.bii file, all of the ground truth information will be stored with the picks in the *.bii file.

The velocities calculated from core data can be used to calculate layer depths. To change the velocities of existing picks, use the select block or select range picking tool and move the selected region over the picks that need to be changed. Then, click the right mouse button to bring up the pick options and select the Change Pick Velocity option. Proceed to change the selected pick velocities as directed in the dialog.

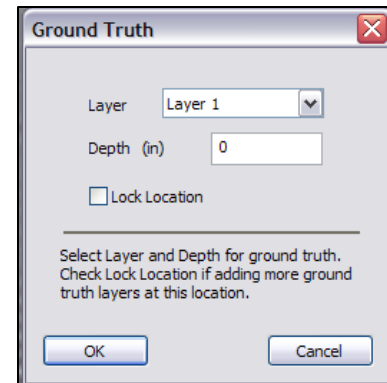
To specify all new layer picks to have velocities calculated from the ground truth data, the velocity option must be changed from the Layer Properties Dialog page accessed through the Layer Options menu item.

When ground truth data from more than one core are entered in the Ground Truth spreadsheet and the user selects core data as the velocity option for picks, the velocity used for each pick is obtained from the nearest core.

Any changes in core data are automatically updated in the picks that have core data calculated velocities.

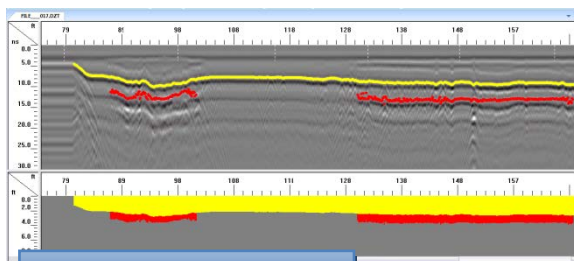
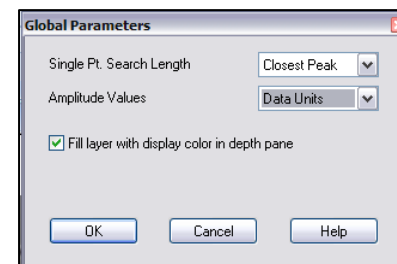
Ground Truth: Enter Ground Truth Information.

- Click in the data the spot where the ground truth was collected.
- Layer: Select Layer for ground truth.
- Depth (): Enter Depth information.
- Lock Location: Input a check mark if adding more ground truth layers at this location.
- Click OK or Cancel.

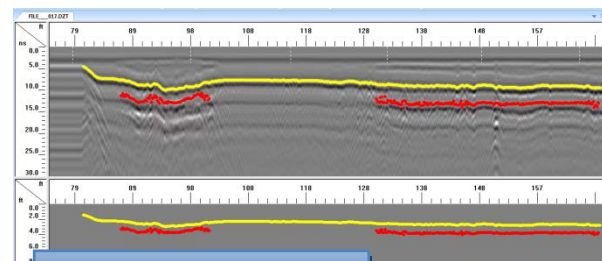


Global Parameters

- Single Pt Search Length: When using Single Point Picking Tool, to enter a "Dot" at the closest peak or use the cursor length.
- Amplitude Values: Use Data Unit, dB, or Normalized dB.
- Check to fill layer with display color in depth pane.



Filled Layers



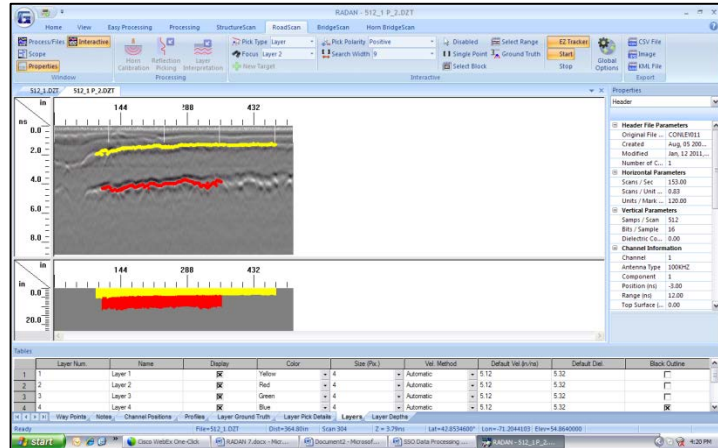
Non-Filled Layers

EZ Tracker (available only with RoadScan Application)

EZ Tracker will attempt to trace a layer by following similar conditions in successive scans. You should note the desired layer's phase information and select Positive, Negative, or Both under layer options. To use EZ Tracker, make sure that the current focus is set to the layer that you want to track and click Start. Then click on the layer. Move the cursor further down the layer and click again. EZ Tracker will attempt to trace that layer between the two points. A closer click spacing will produce better results. If you make a mistake, you can right click to undo the previous section. You should click Save every few picks to save changes. If you come to a gap in the layer, click Stop and then Start to restart the layer at the other side of the gap.

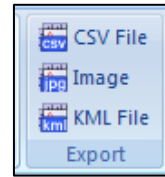
Start by clicking on EZ Tracker. Then click Start. Click, in the Data Area, the beginning of a layer. Then at small intervals, click along the layer. The software will fill the layer in between the dots.

If you wish to begin at a new area of the layer, simply click Stop, then Start again. You may Right Click the mouse to delete the previous click.



Export Data

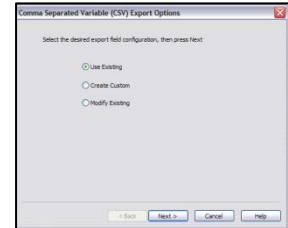
You may export your picks to a comma delimited ASCII (CSV) file, save your image to a JPG file, or export the layer information to a KML file (if GPS was used with your data) for Google Earth



CSV File

Click on CSV file to export your picks to a comma delimited file. You may select pick what fields you would like to export to the CSV file. You may:

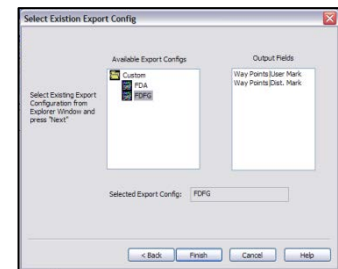
- **Use Existing:** Use the existing custom fields what you created in Create Custom.
- **Create Custom:** You may pick and choose which fields you want.
- **Modify Existing:** Modify an existing field list to create a new one.



Use Existing: Select an existing customize field list you created from Create Custom.

Create Custom: Select from Categories and a list of fields from each of the Categories you want for you CSV file. You will give this Category a user-friendly name for future use.

Modify Existing: Modify an existing category to create a new category.



Image

Save the image as a JPG file.

KML File

Export each layer to a KML file for Google Earth.

A KML file is a file type used to display geographic data in an earth browser, such as Google Earth and Google Maps. KML files have a tag-based structure with names and attributes used for specific display purposes. In RADAN you can export certain kinds of information as a KML file and easily view that information in Google Earth. This section discusses exporting layers identified in Interactive Interpretation.

Note: In order to create a KML file, you must have collected the data with GPS.

Layer: Selects the Layer that you want to export. If you have multiple layers, you must export them one at a time.

Channel: Selects the data channel to export. You can export different channels if you have collected a multi-channel file.

Output Parameter: This can be pick depth, amplitude, velocity, or time depending on the type of data that you want to depict.

Distance Placemarks/Distance Interval: This option outputs a notation at a prescribed distance.

Line width: The KML output line width in pixels.

Minimum/Maximum: This sets the range of output values specified in the Output Parameter option.

Number of Intervals: The number of evenly spaced breakpoints for the output data.

Range/Colors: The range will be set by the number of output intervals. You can decide to code different intervals with colors.

The output of this is a KML file with the same root name as the ASCII pick LAY file. To import into Google Earth, simply drag this file onto Planet Earth and it will zoom to the given location of the data.

Layer Options

In the Layers Tab located in the Table Pane, you can define each layer, how they are displayed and how they behave.

Tables										
	Layer Num.	Name	Display	Color	Size (Pix.)	Vel. Method	Default Vel (in/ns)	Default Diel.	Black Outline	
1	1	Layer 1	☒	Yellow	4	Default Vel	5.12	5.32	☐	
2	2	Layer 2	☒	Red	4	Default Vel	5.12	5.32	☒	
3	3	Layer 3	☒	Green	4	Default Vel	5.12	5.32	☒	
4	4	Layer 4	☒	Blue	4	Default Vel	5.12	5.32	☒	
5	5	Layer 5	☒	Brown	4	Default Vel	5.12	5.32	☒	
6	6	Layer 6	☒	Black	4	Default Vel	5.12	5.32	☒	
7	7	Layer 7	☒	Black	4	Default Vel	5.12	5.32	☒	

Name: The Layer Names page permits the user to specify a name for each layer. For example, for a typical road structure, the user may want to rename Layer 1 as Asphalt, Layer 2 as Base and Layer 3 as Subbase. Layer name changes are shown in all dialogs and will be output to the ASCII file when the user performs the next ASCII file save

Display: Display the layer or not.

Color: Pick the color for the layer.

Size: Select the pixel size of each pick.

Velocity Method: Sets the propagation velocity used to calculate the thickness of each layer. If the Auto box is checked, the velocity will be automatically calculated for each radar scan. The other options are: (1) User-specified, (2) Core Data, and (3) Velocity Analysis File. The Velocity analysis option is most commonly used for geological investigations where a common-midpoint file has been collected and processed using the Velocity Analysis Option in RADAN. The output file from the Velocity Analysis must be entered in the Global page before this option is activated here.

Default Vel (ins/ns): The velocity of RADAR at the interface of that layer.

Default Diel: The dielectric of the layer.

Black Outline: Toggle On/Off to have a black outline for each pick.

Section 5: Processing Specific Applications

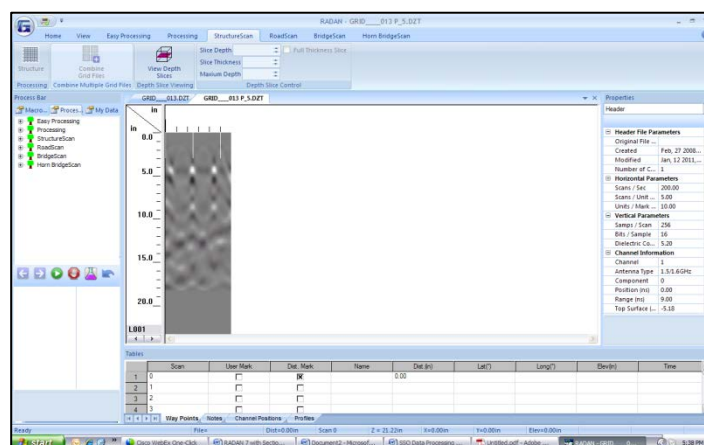
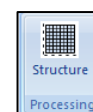
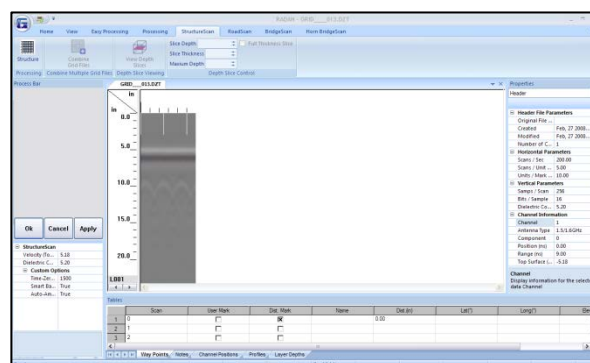
This section will provide you a step by step processing guide for specific applications. These are simply suggestions for basic processing. For detailed information of each step, please refer to Section 3 and Section 4.

StructureScan

This is for data collected in a grid format using the 1500/1600 MHz antennas or data collected using the StructureScan Mini.

Processing the Data

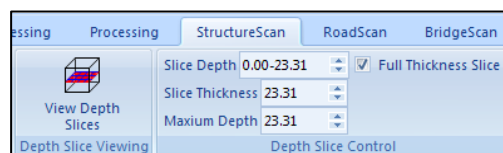
- 1** Before opening any file, configure the software using the Global Parameters located in the Right Pane.
 - a)** Source Folder.
 - b)** Output Folder.
 - c)** Vertical and Horizontal Units.
- 2** Click the GSSI Button and Open a Grid file (.b3d).
- 3** This will display the first profile of the grid. You may click the Left and Right arrows located in the lower left corner of the Display Pane to view other profiles.
- 4** If necessary, right click in the data and adjust the display gain to your desire.
- 5** Click the StructureScan Tab in the Ribbon Pane.
- 6** Click the Structure Icon located on the left side of the Ribbon. This will load a GSSI created macro in the Process Bar Pane.
- 7** Click Apply and it will automatically test three processes:
 - a)** Time Zero Adjustment
 - b)** Smart Background Removal
 - c)** Auto Amplitude Correction (Migration)
- 8** If necessary, click Reset to make any adjustments to the three processes and click Apply again. Continue this until you have the desired results.
- 9** Click OK once you have the desired results. (Note the surface is at the top, noise has been removed, and migration was done).



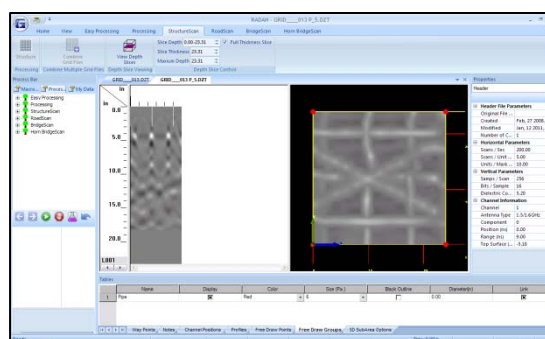
View Depth Slices

If you like, you make View Depth Slices. This view will allow you to view the data from the top and also give you the ability slice through the data.

- 1** Click the View Depth Slices Icon.
- 2** This will give you the top-down view of your data.
- 3** You may make adjustments as follows:

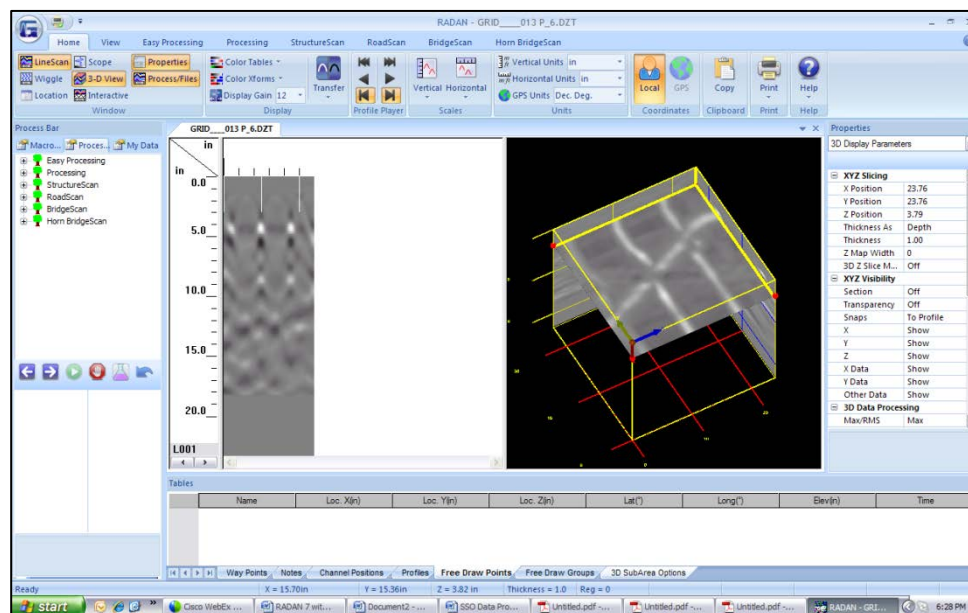


- a)** Slice Depth: Slice the Z slice up and down. You will need to uncheck Full Thickness Slice to do this.
 - b)** Slice Thickness: Adjust the thickness of the slice
 - c)** Maximum Depth: Adjust the Depth of the data
 - d)** When viewing from the very top of the data, you may have a Full Thickness Slice, or turn it off.
- 4** You can right-click the Depth Slices View:
 - a)** Adjust the display gain.
 - b)** Drill a hole.
- 5** You may print this image, or save it as a jpeg file.
 - a)** From the Home Ribbon, click Print to print.
 - b)** Click the GSSI Button, then click Export, and click Save as a JPG file to save as a picture file.



3-D View

Simply clicking on the Home Ribbon tab will allow you to view and manipulate the view as a 3-D cube. Please refer to Section 4, Basic 3D Grid Processing for options in this view.

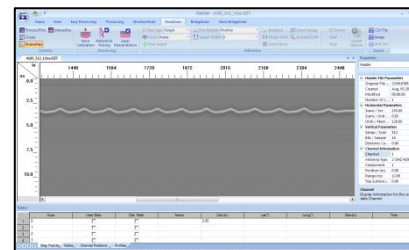


RoadScan

This is for data collected specific to Road data.

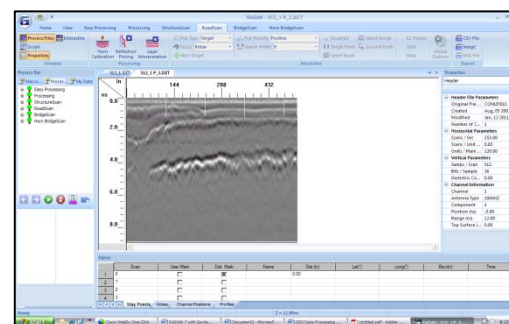
Creating the Calibration File

- 1** Before opening any file, configure the software using the Global Parameters located in the Right Pane.
 - a)** Source Folder.
 - b)** Output Folder.
 - c)** Vertical and Horizontal Units.
- 2** Click the GSSI Button and Open a bumper jump file.
- 3** If necessary, right click in the data and adjust the display gain to your desire.
- 4** Click the RoadScan Tab in the Ribbon Pane.
- 5** Click the Horn Calibration Icon. This will open the Horn Calibration process in the Process Bar Pane.
- 6** Click OK. This will create a calibration file (.czt) for the road data collected.
- 7** Close all files.



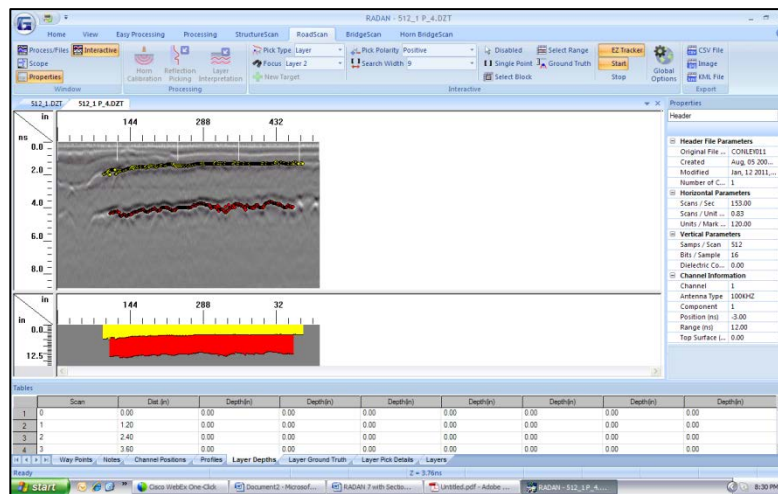
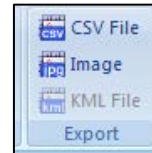
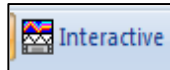
Calibrating the Road File

- 1** Click the GSSI Button and open a Road File.
- 2** Click the Reflection Picking Icon.
 
- 3** The Reflection Picking Process will display in the Process Bar Pane.
 - a)** Auto Load Calibration File: True or False.
 - b)** Current Folder: If Auto Load Calibration File is True, this will use the Calibration File located in the Output Folder. If Auto Load Calibration file is False, you must browse to the location of the Calibration File.
 - c)** Automatic Mode: Leave True.
- 4** Click OK to apply the Calibration File to the Road File.



Layer/Target Picking

- 1 Click the Interactive Icon.
- 2 Using the Interactive Pane of the RoadScan Ribbon, you may begin to enter your picks for each layer. Please refer to Section 3, RoadScan Ribbon or Section 4, Interactive for detailed information on this process.
- 3 When complete, save the comma delimited ASCII file for use in other 3rd party software. Click CSV file to do this.

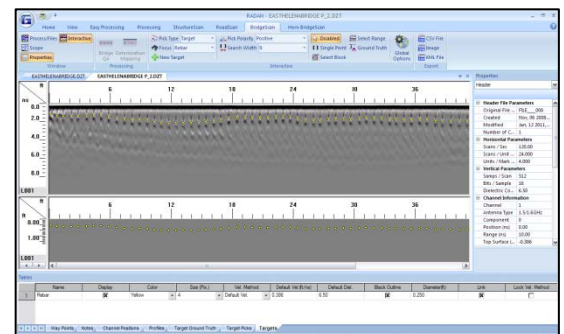
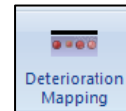
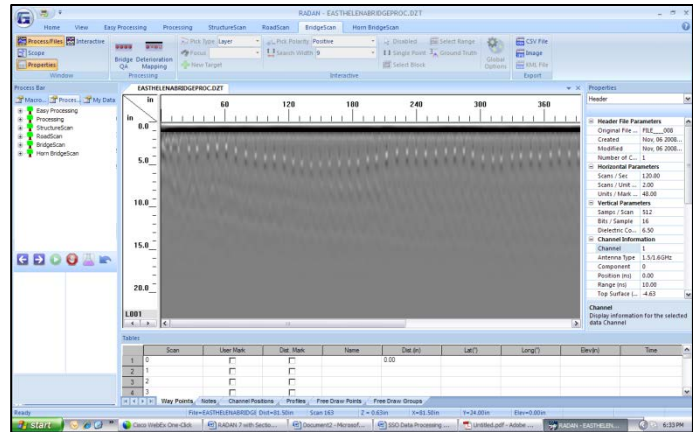


BridgeScan

This is for data collected specific to determine bridge deterioration.

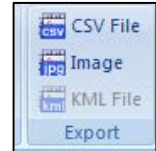
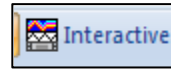
Preparing the Data

- 1** Before opening any file, configure the software using the Global Parameters located in the Right Pane.
 - a)** Source Folder.
 - b)** Output Folder.
 - c)** Vertical and Horizontal Units.
- 2** Click the GSSI Button and Open a Bridge file.
- 3** This will display the first profile of the bridge. You may click the Left and Right arrows located in the lower left corner of the Display Pane to view other profiles.
- 4** If necessary, right click in the data and adjust the display gain to your desire.
- 5** Click the BridgeScan Tab in the Ribbon Pane.
- 6** Click the Deterioration Mapping Icon. This process will correct the time zero of the data as well, apply picks to the rebar, and create a comma delimited ASCII file for export. This will open the Deterioration Mapping process in the Process Bar Pane.
- 7** Select:
 - a)** Concrete Overlay or Asphalt.
 - b)** Time Zero Amplitude Threshold.
 - c)** Velocity (Top Layer/Ft).
- 8** Click Apply and it will automatically test this process.
- 9** If necessary, click Reset to make any adjustments to the three values and click Apply again. Continue this until you have the desired results.
- 10** Click OK once you have the desired results.



Target Picking

- 1** Once this process is complete, click the Interactive Icon.
- 2** Using the Interactive Pane of the BridgeScan Ribbon, you may make any adjustment to the picks the Deterioration Mapping process applied. Please refer to Section 3, BridgeScan Ribbon or Section 4, Interactive Processing for detailed information on this process.
- 3** When complete, save the comma delimited ASCII file for use in other 3rd party software. Click CSV file to do this.



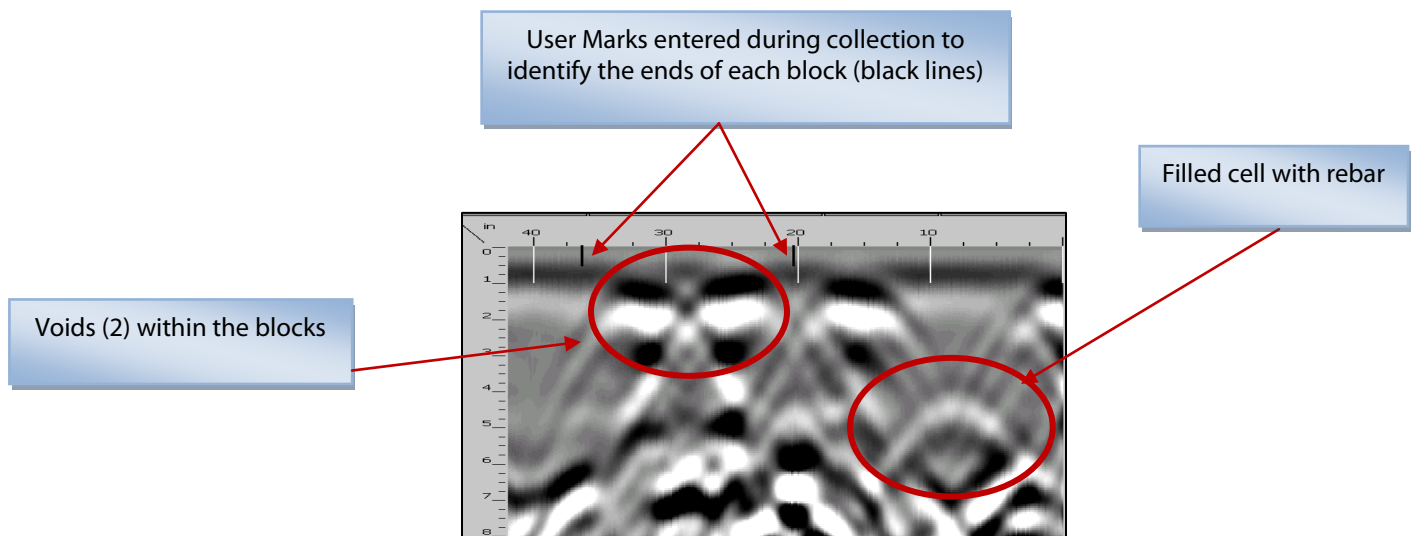
Appendix A: Sample Data

These examples are presented for informational use only. Conditions may be different at your site that may cause the images to look different from the data shown here.

Examples 1 – 4 are data collected with a 1600 MHz antenna.

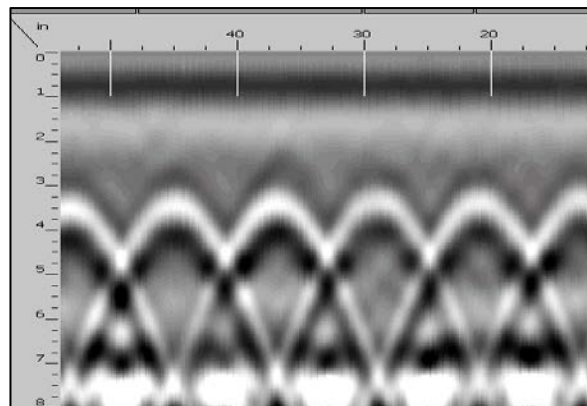
Examples 5 – 8 are data collected with a 400 MHz antenna.

Example 1: Concrete Block Data



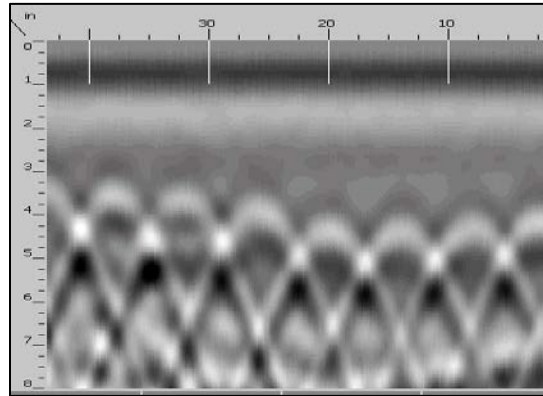
Example 2: Rebar Data

This is an example of rebar. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 8 inches. Also note the approximate depth, the brightness of the hyperbolas, and they tend to be at a constant depth over the short term.



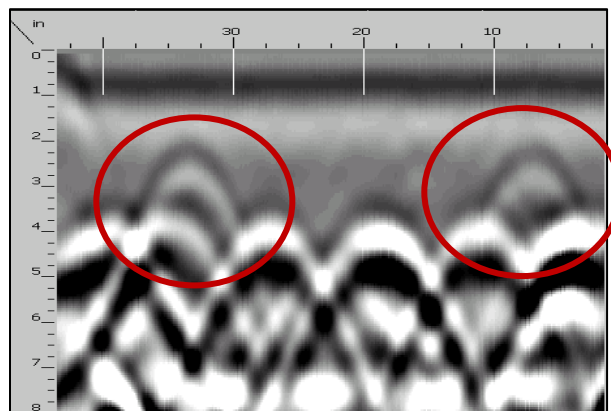
Example 3: Mesh Data

This is an example of mesh. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 6 inches. Also note the approximate depth, the brightness of the hyperbolas, and mesh may be more uneven than rebar.



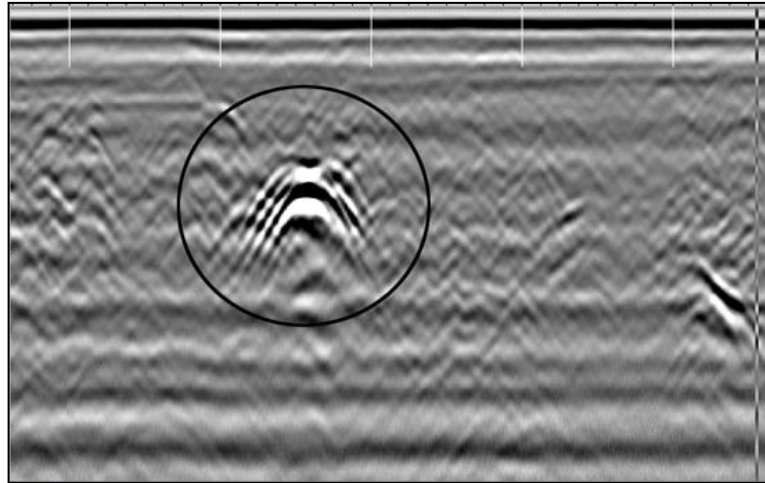
Example 4: PVC

This is an example of two PVC pipes. Note they tend to be dimmer than rebar.



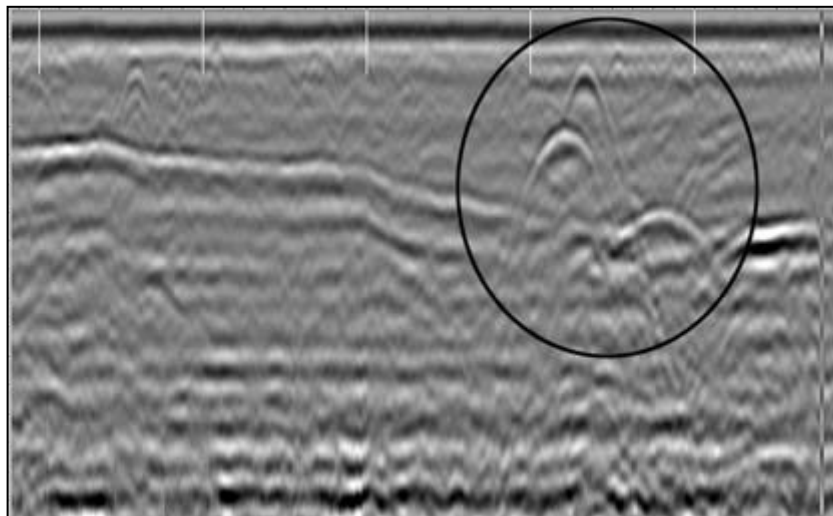
Example 5: Bank of Conduits

This piece of data shows a bank of conduits feeding into an industrial building. Note the dipping edge of a trench cut to the upper left of the conduits.



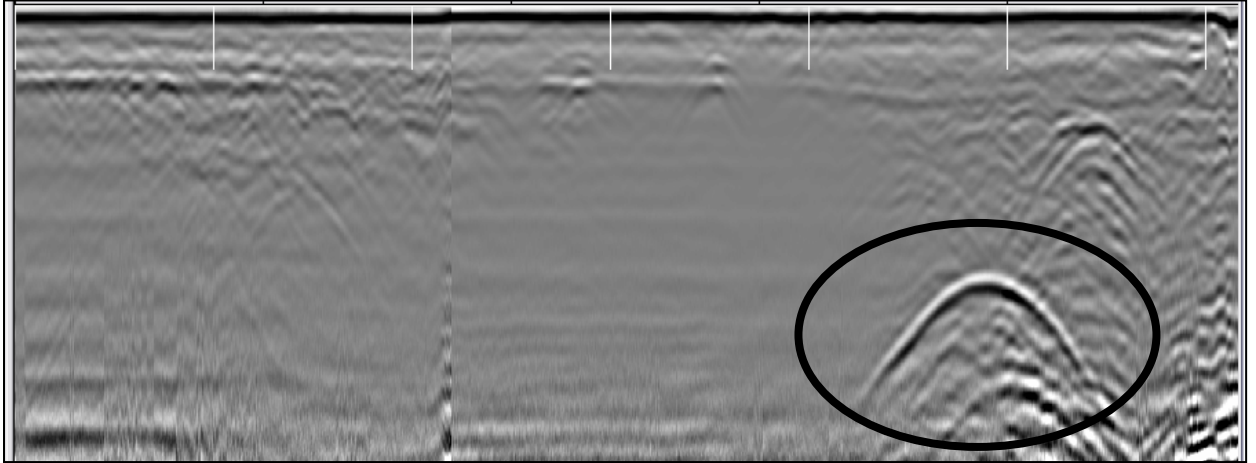
Example 6: Empty PVC Pipe

This image shows three drain lines. The one in the middle (depth-wise) is an 8" PVC pipe (empty).



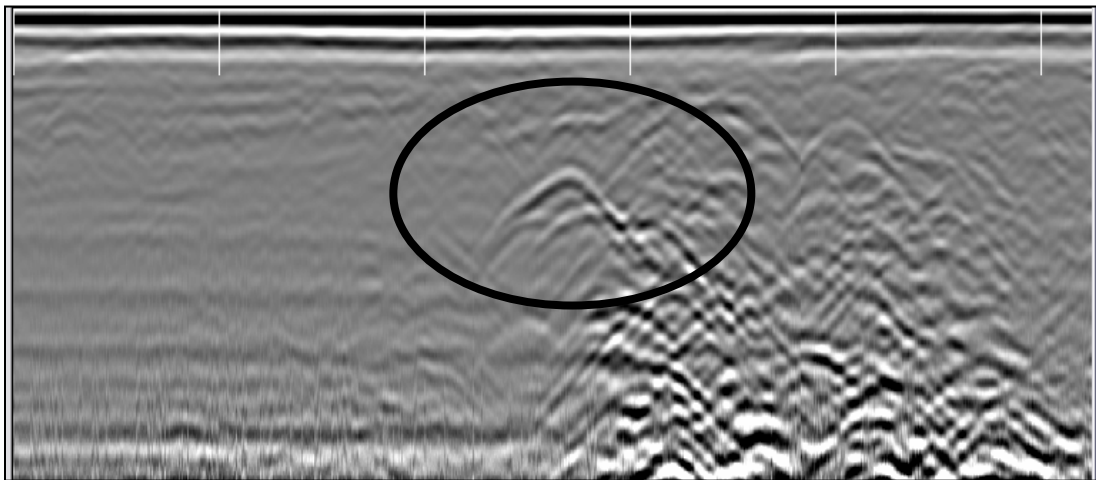
Example 7: Water Main

This image shows data collected across a road, looking for a water main.



Example 8: Culvert

This image shows data collected along a driveway next to a business. This shows a concrete culvert.



Appendix B: The GSSI Naming Convention

If you set “Auto Save” to True, GSSI will automatically name your processed files and place them in the PROC folder inside your source folder. The naming convention is as follows:

The original file is represented by <FileName>.dzt. If you process this file, a new file is created in the PROC folder and named <FileName> P1.dzt.

If you continue to process the ORIGINAL file, then each processed file will increment by 1.

However, if you process a processed file, another digit will be added to the name of the processed file to create a new file in the PROC folder. Example: If you process a processed file <FileName> P1.dzt, then the new file is named <FileName> P11.dzt.

The figure below diagrams the GSSI naming convention:

A. Original File	B. Processed Files from Original (Column A)	C. Processed Files from Processed Files (Column B)	D. Processed Files from Processed Files (Column C)
<FileName>.dzt	<FileName> P1.dzt	<FileName> P11.dzt	<FileName> P111.dzt
			<FileName> P112.dzt
			<FileName> p113.dzt
		<FileName> P12.dzt	<FileName> P121.dzt
			<FileName> P122.dzt
			<FileName> p123.dzt
		<FileName> P13.dzt	<FileName> P131.dzt
			<FileName> P132.dzt
			<FileName> p133.dzt
	<FileName> P2.dzt	<FileName> P21.dzt	<FileName> P211.dzt
			<FileName> P212.dzt
			<FileName> p213.dzt
		<FileName> P22.dzt	<FileName> P221.dzt
			<FileName> P222.dzt
			<FileName> p223.dzt
		<FileName> P23.dzt	<FileName> P231.dzt
			<FileName> P232.dzt
			<FileName> p233.dzt
	<FileName> P3.dzt	<FileName> P31.dzt	<FileName> P311.dzt
			<FileName> P312.dzt
			<FileName> p313.dzt
		<FileName> P32.dzt	<FileName> P321.dzt
			<FileName> P322.dzt
			<FileName> p323.dzt
		<FileName> P33.dzt	<FileName> P331.dzt
			<FileName> P332.dzt
			<FileName> p333.dzt

Appendix C: Dielectric Constants

The relative dielectric permittivity is a dimensionless measure of the capacity of a material to store a charge when an electric field is applied. The dielectric constant is the real part of dielectric permittivity, as it is normalized to air. Dielectric constant values vary from material to material. In dry materials the dry bulk density primarily influences the dielectric constant. In partially saturated or saturated soils the dielectric constant is primarily determined by the water content. Radar energy is reflected at boundaries of electrically dissimilar materials where there is a contrast in the dielectric constants (i.e., where there is electrical impedance). These boundaries typically occur at stratigraphic boundaries, but may occur at the water table and within stratigraphic units where changes in electrical properties occur.

The chosen value of the dielectric constant defines the relationship between time and depth (i.e., it is directly related to velocity) for the file. In resistive soils, those with conductivities less than 20 millimhos per meter, the average soil velocity may be approximated by:

$$V_m C E_r =$$

where: V_m is the average soil velocity (m/s)

C is the speed of light or 2.998×10^8 m/s

E_r is the dielectric constant or real dielectric permittivity normalized to air (dimensionless).

Note that this equation is not valid in conductive soils (such as clays) or soils with conductive pore fluids (such as brackish and conductive groundwater).

Note: When you input a dielectric constant value in the header, RADAN makes the assumption that this value is valid throughout the entire file, which may not be representative.

The dielectric constant may be changed to evaluate the effect of different velocity assumptions of the location of features.

The dielectric constant may be determined from:

- 1** On-site calibrations over targets of known depth.
- 2** Common depth point or midpoint (CDP or CMP) calibrations using bistatic antennas over a reflector of known depth (such as the water table) or,
- 3** Estimated velocities or dielectric constants observed at similar sites.

Values of dielectric constants (at 100 MHz) for common materials follows:

Air	1
Glacial ice	3.6
PVC	3
Asphalt	3 - 5
Concrete	4 - 11
Granite	4 - 7
Sandstone	6
Shale	5 - 15
Limestone	4 - 8
Basalt	8 - 9
Water saturated sands (20% porosity)	19 - 24
Soils & sediments	4 - 30
Water	81

Appendix D: Glossary of Terms

Antenna: A component of an impulse radar system designed to radiate radio waves (electromagnetic radiation) from Applied voltage impulses (transmitting antenna), or conversely, to intercept radio waves and convert them back into electrical impulses (receiving antenna). Antennas radiate or receive electromagnetic energy.

Antenna Radiation Pattern: A plot of the intensity of the radiation received at a given radial distance from an antenna versus angle, relative to a given reference axis. The pattern is a three dimensional measure of the energy at a fixed radial distance from the antenna.

Attenuation: A measure of the loss of radiated signal amplitude or signal energy as it progresses through a lossy medium. The loss can be due to a spreading loss as the wave expands out into the medium and also due to an ohmic loss, due to the finite conductivity of the medium.

Background Removal: A digital signal processing function that filters by subtracting an average of a large number of scans from each individual scan. The result is horizontal changes in the data are accentuated while linear features (background) are suppressed.

Backscatter: A portion of a radar's transmitted energy that is intercepted by a target, or other object and reflected (scattered) back in the radar's direction.

Bandwidth: The band of frequencies occupied by the central lobe of the spectrum of an electromagnetic signal. Bandwidth is usually defined so that it includes most, but not all of the signal power. Generally, it includes the portion lying between the points at which the power has dropped to half that at the center of the band.

Beamwidth: The angular width of a slice through the main lobe of the radiation pattern of an antenna.

Bias: The amount by which the average of a set of values departs from a reference value.

Bistatic: The survey method that utilizes two separate antennas at a constant distance. One antenna has a transmitter and the other contains a receiver sampler. With this method it is possible to transmit a higher power signal and receive reflections with a greater time delay.

Common Depth Point (CDP): Also known as Common Midpoint (CMP). Having the same midpoint between source and detector.

Common Depth Point Method: A survey method that can be performed to calculate the electromagnetic velocity of a material. This is conducted by transmitting from one antenna and receiving from a second antenna at several known offsets (surface horizontal distance).

Clutter: Unwanted reflections from the ground, from within the ground or from above the ground. In the case of ground penetrating radar (GPR), clutter may be produced by boulders, soil interfaces and other scatterers that are not of interest. Clutter is also produced within the radar system.

Conductivity: The electrical conductivity of a dielectric material is a measure of the ease with which an electrical current can be made to flow through it. In the MDS system, the unit of conductivity is the Siemens per meter (S/m). Conductivity is the reciprocal of resistivity. The higher the conductivity of the subsurface materials, the greater the attenuation of the radar signal.

Control Unit (C/U): An electronic instrument that interfaces a transducer (s) to recorders, processors, displays, survey wheel, power supply, etc. It also has controls to allow radar functions such as range, gain and filtering to be adjusted. A C/U can be analog, digital, or hybrid.

Data Channel: A software channel on the control unit that displays and records a received signal. It is possible to have one transducer and four data channels with the same or varied processes.

Decibel (db): A unit of measure for gain. A logarithmic unit used to express power ratios. One decibel equals 20 LOG (P2/P1). Decibels are also used to express voltage ratios.

Deconvolution: A digital signal processing function designed to attenuate multiples and improve the recognition and resolution of reflected events. A process that restores a waveshape to the form it had before it underwent a linear filtering action (convolution).

Delay Time: The amount of time the radar wave propagates through a material, reflects off an interface and returns to the receiver. Time lag introduced by either hardware (cable length, etc.) or software (filter averaging, etc.).

Depth of Penetration: In any medium, the radar wave is attenuated as it progresses due to losses that occur. At radar frequencies in a conductive material (sea water, metallic materials, clay soils, etc.) the rate of attenuation is very great and the wave may penetrate only a short distance (<1m) before being reduced to a negligibly small value. In a resistive dielectric earth material (fresh water, granite, quartz sand, etc.), where the losses are low, the depth of penetration can be quite great (>30m).

Dielectric Constant: See Dielectric Permittivity.

Dielectric Permittivity: Dielectric Permittivity is a property of an electrical insulating material (a dielectric) equal to the ratio of the capacitance of a capacitor filled with the given material to the capacitance of the identical capacitor filled with air. Earth materials are classified generally as conductors, semiconductors and insulators (dielectrics). A dielectric material is a poor conductor of electric current. The specific capacitance of a vacuum is $\epsilon_0 = 8.85 \times 10^{-12}$ Farads per meter. The relative dielectric constant, ϵ_r for air is 1 and is approximately 81 for fresh water.

Dielectric Interface: A place in the subsurface of a dielectric material where the dielectric permittivity changes.

Diffraction: The phenomenon that causes electromagnetic waves in the beam of a directional antenna to spread out. The bending of wave energy around obstacles without obeying Snell's Law. An event that occurs at the termination of curved topped, or steeply dipping reflectors that is characterized by a distinctive curved alignment.

Diffuse: To break up and distribute the energy in an incident electromagnetic wave in many directions.

Dipole: A simple antenna having two elements driven from the center of the antenna by a balanced source.

Directivity: Ability of an antenna to concentrate transmitted energy in a given direction and to emphasize the returned energy received from that direction.

Dynamic Range: The spread between the minimum signal at the input of a system, which produces a discernible change in the output and the maximum input that the system can handle without saturating. Measured in decibels (db).

Echoes: Radar energy reflected from a given target or object.

Electromagnetic Wave: A wave that is propagated by the mutual interaction of electric and magnetic fields. Radiant heat, light and radio waves, are electromagnetic waves.

Finite Impulse Response (FIR) Filter: A digital signal processing function that convolves a finite length function (boxcar, triangle) with the data. Each data value is multiplied by the corresponding filter value and added together. FIR filters are digital filters and have no time delay.

Frequency: The number of positive and negative voltage amplitude cycles that a pure unmodulated sine wave completes per second. The unit of frequency is Hertz (Hz).

Frequency Domain: Mathematical realm in which the amplitudes of signals are expressed as functions of frequency, rather than time. The frequency spectrum of a time varying signal is obtained by translating the expression for the signal from the time domain to the frequency domain.

Frequency Domain (FK) Filter: Also called a velocity filter. A digital signal processing function that discriminates on the basis of apparent velocity. Coherent arrivals with certain apparent velocities are attenuated.

Fresnel Zone: The portion of a reflector from which reflected energy can reach a detector within one-half wavelength of the first reflected energy.

Gain: A change in signal amplitude or power from one point in a circuit or system to another, often from system input to output. See Range Gain.

Gigahertz: A unit of frequency. One GHz equals 10⁹ Hz.

GPR: Acronym for Ground Penetration Radar.

GPS: Acronym for Global Positioning System.

Hertz (Hz): A unit of frequency. One Hertz equals one cycle per second.

High Pass Filter: A filter that passes without significant attenuation frequencies above some cutoff frequency while attenuating lower frequencies. The same as low-cut filter.

Hilbert Transform: A digital signal processing function that determines the magnitude envelope, instantaneous phase and instantaneous frequency of a received signal.

Horizontal Filter: A digital signal processing function that attenuates signals outside the filter function across adjacent scans.

Infinite Impulse Response (IIR) Filter: A digital signal processing function that emulates an analog filter function. An IIR filter is a filter function that offsets the data in time.

Interface: The common surface separating two different media in contact. A location in the subsurface where the dielectric constant changes.

Isotropic Radiator: An antenna that radiates equally (both in amplitude and in phase) in all directions. The imaginary source of the radiation used as a reference for the gain of a directional antenna.

Lateral Resolution: The ability of the system to resolve in a horizontal direction the smallest discernible target. This is a function of antenna frequency, scan rate and speed of travel.

Low Pass Filter: A filter that passes frequencies below some cutoff frequency while substantially attenuating higher frequencies. Same as a high-cut filter.

Magnetic Permeability: The magnetic permeability of a material is a measure of the difficulty of magnetization of the material in an external field. The magnetic permeability of earth materials is taken to be that of free space, $\mu_0 = 4 \times 10^{-7}$ Henrys per meter. Therefore, the relative permeability of earth materials is $\mu_r = 1$.

Megahertz: A unit of frequency. One MHz equals 10⁶ Hertz.

Migration: A digital signal processing function that rearranges data so that reflections and diffractions are plotted at the locations of the reflectors and diffracting points rather than with respect to observation points on the profile. Migration by computer is accomplished by integration along diffraction curves

(Kirchhoff migration), by numerical finite-difference downward-continuation of the wave equation and other algorithms.

Monostatic: A survey method that utilizes a single transducer with either a single antenna with transceiver or a dual antenna transducer with separate electronics.

Multiple: Also known as ringing. Wave energy that has been reflected more than once. In radar data multiples may occur when there is a large change in the dielectric permittivity or conductivity of the medium. A multiple can be identified by signals that have the same time delay as between the surface and the first reflector.

Nanosecond: A unit of time. One nanosecond equals 10^{-9} seconds; one billionth of a second.

Noise: Unwanted, usually random, electrical or electromagnetic energy that interferes with the detection of wanted signals. The term is also applied to any unwanted random variations in the measured value of any quantity.

PC: Acronym for Personal Computer.

Permittivity: See Dielectric Permittivity.

Phase: Degree of coincidence in time between a repetitive signal, such as a sine wave and a reference signal, having the same frequency. The angle of lag or lead of a sine wave with respect to a reference. Generally expressed in degrees. 360 degrees corresponds to the period of the signal.

Point Reflector: A subsurface feature with electromagnetic properties different from its surroundings, whose dimensions are approximately the same as the fresnel zone of the radar wave.

Polarization: The orientation of the electric and magnetic fields of an electromagnetic wave, such as a radio wave. By convention, the polarization of the wave is the direction of the electric field. If the polarization does not change as the wave propagates, the polarization is said to be linear.

Power: A measure of the quantity of electric energy, commonly expressed in Watts. One watt equals one Joule per second.

Profile: A graph showing the depth measurements as a vertical cross section of the medium along a horizontal line.

PRF (Pulse Repetition Frequency): The number of pulses per second transmitted by a pulsed radar.

Propagation: The outward spreading, or travel, of an electromagnetic wave, such as a radio wave.

Radar: Radio Detection And Ranging. An electronic system that transmits electromagnetic energy and detects the location of reflected energy.

Radar Cross Section: A factor relating the power of the radio waves that a radar target scatters back in the direction of the radar, to the power density of the radar's transmitted waves at the target's range. Takes account of the cross sectional area of the target, as viewed by the radar, the target's reflectivity and its directivity.

Radiation: Energy in the form of an electromagnetic wave emitted by an antenna, in which free electrons are accelerated. Radiant heat, light and radio waves are electromagnetic radiation. They differ only in wavelengths.

Range: The radial distance from a radar to a target or other object. A user adjustable setting on a control unit to determine the amount of time displayed and recorded (0 - thousands of nanoseconds).

Range Gain: Also known as time gain control or time varying gain. Control for varying the amplification or attenuation of an amplifier, used to compensate for variations in input signal strength over time.

Receiver: The portion of the antenna used, to intercept the radio waves reflected from the subsurface and convert them back into electrical impulses (receiving antenna).

Reflection: The degree to which an object returns incident radio waves.

Reflection Coefficient: A description of the reflected field strength from an infinite interface between two media 1 and 2. The reflection coefficient r is defined by:

$$r = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

where Z is the impedance of the respective media.

Refraction: The bending of an electromagnetic wave that occurs when the wave passes obliquely from one medium into another whose dielectric constant is different from that of the first medium. The bending results from the speed of the propagation being different in one medium than in the other. Refraction may also occur in a single medium whose dielectric constant gradually changes in a direction normal to the wave's direction of propagation.

Resistivity: The reciprocal of the electrical conductivity is the electrical resistivity. The electrical resistivity of a substance is a measure of the difficulty an electrical current can have flowing through it. In the MKS system the unit of resistivity is the ohm-meter ($\Omega \cdot m$). The symbol is Ω

Resolution: The ability to separate two features that are very close together. The minimum separation of two bodies before their individual identities are impossible to interpret. The smallest change in input that will produce a detectable change in output.

Sampler: A circuit whose output is a series of discrete values representative of the values of the input at a series of points in time.

Scan: One discrete sequence of events such as a sampling at all time points of the amplitudes at a receiver.

Scatter: The irregular and diffuse dispersion of energy caused by inhomogeneities in the medium through which the energy is traveling.

Signal Position: The relative delay between the time when the Radar system sends out a transmit pulse and when the Radar systems sends out a receiver pulse.

Signal-to-Noise Ratio: Ratio of the power or energy of a received signal to the power or energy of the accompanying noise.

SIR: Acronym for Subsurface Interface Radar.

Spectrum: The distribution of the power or energy of a signal over the range of possible frequencies is commonly represented by a plot of amplitude versus frequency. If the amplitude is a voltage, a plot of the square of the amplitude is the power spectrum. The area under the power spectrum corresponds to the signal's energy.

Specular Reflection: Mirror-like reflection occurring when an electromagnetic wave strikes a flat surface, the irregularities (roughness) in which are small compared to the wavelength of the incident wave.

Running Average: A digital signal processing function that averages a finite number of scans with a user designated function to produce a single output scan. This method can be operated in continuous or static modes.

Static Correction: Corrections applied to data to compensate for the effects of variations in elevation, weathering thickness, weathering velocity, or reference to a datum.

Time Domain: Mathematical realm in which the amplitudes of electromagnetic signals are expressed as functions of time.

Time Varying Gain: See Range Gain.

Top Surface Normalization: Correction of data for the effects of surface elevation changes by time (depth) shifting the data. The result is to present the data as if all measurements had been made on a flat plane.

Transducer: An antenna or antennae with built-in or plug-in transmitter and receiver electronics.

Transducer Input Connector: The connector on the control unit that connects the cable from the transducer/antenna.

Trans-illumination: This a method of surveying where a transmitting antenna transmits one way through a material to a receiving antenna. Example; crosshole investigations.

Transect: The line along the surface that a profile is acquired.

Transmit Pulse: The impulse of radar energy from the transmitting antenna as seen at the receiving antenna. This is shown on the recorded data at the top of the display. When the transducer is moved on the ground surface the first arrival of the transmit pulse is interpreted as the surface.

Transmitter: The electronics, which after receiving a trigger pulse from the control unit, sends an impulse of electromagnetic energy to the attached antenna.

Travel Time: The amount of time (nanoseconds) that the radar signal takes to travel from the transmitting antenna to a target or receiving antenna. This is used for transillumination methods (also called Transit Time).

Trigger Pulse: Pulse generated in the control unit that is sent through the cable to the transmitting antenna.

Two-way Travel Time: The amount of time (nanoseconds) that the radar signal takes to travel from the transmitting antenna, reflect off a target and return to the receiving antenna. This is used for most standard GPR field methods.

Velocity: The speed at which electromagnetic signals propagate. In air or free space, electromagnetic energy propagates at the speed of light. In dielectric materials, the velocity of propagation is slower by the square root of the dielectric constant.

Vertical Filter: A digital signal processing function that attenuates signals outside of the filter function for each individual scan

Vertical Resolution: The ability to separate two feature within one scan that are very close together. The minimum separation of two bodies before their individual identities are lost on the resultant map or cross-section. A function of transducer frequency, sampling interval and range.

Some of the above definitions are from the following sources:

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